Highlights

A coupled Farquhar-Jarvis whole-plant simulation model: modular formulation, solver verification and global sensitivity analysis

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- Whole-plant simulator couples Farquhar-Jarvis gas exchange to eleven state equations.
- Analytic C_i solution replaces a fixed point that cycles under water stress.
- Sobol' estimator verified against a closed-form benchmark before use on the model.
- Carbon budget closes to relative residuals below 1.3×10^{-14} over all photoperiods.
- Out-of-sample photoperiod tests replace an earlier, calibration-bound RGR agreement.

A coupled Farquhar-Jarvis whole-plant simulation model: modular formulation, solver verification and global sensitivity analysis

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ABSTRACT

Whole-plant growth simulation requires environmental forcing, leaf gas exchange, carbohydrate storage, nutrient uptake and biomass allocation to be advanced together over the diel cycle, yet many whole-plant and crop simulators prescribe the photosynthetic carbon input instead of computing it from the gas exchange on which the rest of the model depends. We reformulate a whole-plant, Tedone-based physiological framework as a modular coupled simulation model in which Farquhar-von Caemmerer-Berry photosynthesis and Jarvis stomatal conductance compute the carbon input, propagated into an eleven-state system of cytosolic and vacuolar sucrose, transient starch, nitrogen and phosphorus pools, and leaf and root biomass. The implicit gas-exchange constraint is solved in closed form: the coupled supply-demand system is a quadratic in the intercellular CO_2 concentration. The fixed-point iteration it replaces is retained as a switchable alternative and verified against it, revealing a period-two orbit below a leaf water potential of -1.15 MPa in which the sign of the assimilation rate depends on the iteration cap, and non-convergence in 15.2% of sensitivity-design draws. The carbohydrate budget, integrated alongside the state equations, closes to a relative residual below 1.3×10^{-14}. For *Arabidopsis thaliana* the model captures the diel starch cycle within a normalised-RMSE threshold of 0.25 across 4-12 h photoperiods; the per-photoperiod relative growth rate (RGR) fit is saturated, and a held-out photoperiod test over 14 cross-validated fits gives a mean absolute percentage error of 28.5% for photoperiod-independent coefficients, falling to 7.3% once one coefficient is allowed to vary linearly with photoperiod — reported here as the predictive figure of merit, and locating the model's photoperiod dependence in a single calibrated growth coefficient rather than in the coupled dynamics. An ablation study shows that computing rather than prescribing the carbon input removes a systematic RGR underestimation of up to 23%. A Sobol' analysis, its estimator first verified against a function with closed-form indices, attributes the majority of growth variance to parameter interactions — the interaction remainder exceeds the first-order index in 44 of the 52 influential parameter-photoperiod cells — and shows control shifting from nutrient uptake and respiration under short days to photosynthetic and light-supply capacity under long days. Alongside the simulation core we report verification tests that need no observational data and that found four defects invisible at the calibrated operating point, including an adaptive integrator step that stalls without diverging — a failure mode invisible to any check formulated on the values the integrator produces.

1. Introduction

Simulating the growth of a whole plant requires several physical and biological subsystems to be updated together over time: environmental forcing must drive leaf gas exchange, the resulting carbon must be partitioned between transient reserves and mobile sugars, sugars must support nutrient uptake and maintenance, and the remaining assimilate must be allocated to new biomass. Each of these subsystems has an established modelling literature, but their behaviour is only biologically meaningful when they are coupled: the carbon available for growth at any instant is the outcome of the environment, the photosynthetic machinery and the internal carbohydrate and nutrient state acting together Farquhar, von Caemmerer and Berry (1980); Thornley and Johnson (1990). For simulation, the main difficulty is therefore not any single process equation but the construction of a stable numerical workflow in which these processes exchange fluxes consistently over the day-night cycle. Photoperiod makes this problem clear: changing the day length alters the ratio of the light phase, during which carbon is assimilated and stored, to the dark phase, during which growth depends entirely on the mobilisation of stored starch Scialdone, Mugford, Feike, Skeffington, Borrill, Graf, Smith and

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Howard (2013); Graf, Schlereth, Stitt and Smith (2010), so a correctly coupled model should reproduce the shape of this shifting day-night balance across photoperiods from a single mechanism. We state this as the aspiration it is rather than as an achieved property: the model reported here requires two coefficients to be recalibrated per photoperiod, and Section 6 quantifies how much of the photoperiod response that recalibration carries.

The individual physiological components needed for such a model are well established. The Farquhar-von Caemmerer-Berry biochemical model Farquhar et al. (1980) and Jarvis-type stomatal conductance Jarvis (1976) describe leaf-level gas exchange, whole-plant and crop simulators such as APSIM, DSSAT and WOFOST integrate growth over field scales Holzworth, Huth, deVoil, Zurcher, Herrmann, McLean, Chenu, van Oosterom, Snow, Murphy, Moore, Brown, Whish, Verrall, Fainges, Bell, Peake, Poulton, Hochman, Thorburn, Gaydon, Dalgliesh, Rodriguez, Cox, Chapman, Doherty, Teixeira, Sharp, Cichota, Vogeler, Li, Wang, Hammer, Robertson, Dimes, Whitbread, Hunt, van Rees, McClelland, Carberry, Hargreaves, MacLeod, McDonald, Harsdorf, Wedgwood and Keating (2014); Jones, Hoogenboom, Porter, Boote, Batchelor, Hunt, Wilkens, Singh, Gijsman and Ritchie (2003); de Wit, Boogaard, Fumagalli, Janssen, Knapen, van Kraalingen, Supit, van der Wijngaart and van Diepen (2019), and day-night starch turnover in *Arabidopsis thaliana* has been characterised in detail Scialdone et al. (2013); Stitt and Zeeman (2012). In many of these whole-plant and crop simulators, however, the carbon input is still prescribed or strongly simplified, supplied as a light-use-efficiency term, a look-up table or an externally imposed assimilation rate, rather than computed from the simulated gas exchange that the rest of the model depends on Jones et al. (2003); de Wit et al. (2019). When the carbon source is decoupled from the environment and from the plant's internal state in this way, the simulator cannot represent the feedbacks between stomatal behaviour, intercellular CO_2, carbohydrate status and growth that give the day-night cycle its structure. What is missing is not another process equation but a transparent, reproducible workflow in which a mechanistic gas-exchange layer is coupled to downstream carbohydrate, nutrient and biomass dynamics, with clearly declared states, inputs, parameters and outputs.

This paper addresses that gap by reformulating the whole-plant physiological framework of Tedone (2020) as a coupled dynamical simulation model in which the photosynthetic carbon input is computed rather than prescribed, and by documenting the model as a modular simulator whose submodels, interfaces and numerical procedure are stated explicitly. A Farquhar-Jarvis gas-exchange layer supplies the carbon input, which is propagated into an eleven-state description of cytosolic and vacuolar sucrose, transient starch, nitrogen and phosphorus pools, and leaf and root biomass. We calibrate the model per photoperiod against literature-derived relative growth rate targets for *Arabidopsis thaliana*, and we use an ablation study and global and local sensitivity analyses to test whether the coupling, rather than any individual component, is responsible for the model's behaviour. The intended contribution is thus not a new process equation but a documented, reproducible simulation core in which mechanistic gas exchange is transparently coupled to carbohydrate, nutrient and biomass dynamics.

Although the application is a plant, the problems the model raises are those of simulation practice, and they are the aspects of the work we expect to transfer. Four are addressed explicitly. First, *architecture*: the simulator is decomposed into six submodels with declared inputs, outputs, parameters and state ownership, so that a component such as the stomatal-conductance function can be substituted without touching the integrator or the downstream pools; the ablation study in Section 5 is possible only because the interfaces are explicit. Second, *multi-rate dynamics*: the coupled system mixes an algebraic gas-exchange loop that equilibrates within a time step with carbohydrate pools that turn over in hours and biomass that accumulates over weeks. This is a separation of time scales rather than stiffness in the sense that would motivate an implicit method: the fast subsystem is algebraic and is eliminated exactly, so the differential system that remains is not stiff, and an explicit adaptive method integrates it without difficulty. What the separation does govern is where the discontinuities of the diel forcing force the step-size controller to contract, and how the algebraic layer must be embedded in the derivative so that the error estimate remains meaningful. Third, *solver verification as distinct from validation*: agreement with measurements does not establish that the numerical procedures inside a simulator are correct, and we show that a fixed-point iteration which behaves well over the calibrated operating range fails, silently and with a sign error in the reported flux, once a forcing variable crosses a threshold; we therefore verify each numerical component against an analytic solution before comparing the model with data. Fourth, *estimator verification in sensitivity analysis*: variance-based sensitivity indices are themselves the output of a Monte Carlo estimator whose sampling design can be wrong in ways that produce plausible but meaningless numbers, and we demonstrate that testing the estimator against a function with closed-form indices exposes a defect that no amount of resampling of the model would have revealed. These are methodological rather than physiological contributions, and each is reported together with the diagnostic that detected the problem, so that the same checks can be applied to other modular simulators of coupled biophysical systems.

The remainder of the paper is organised as follows. Section 2 reviews mechanistic plant-growth models, gas-exchange formulations and the validation, ablation and sensitivity methods on which the evaluation builds. Section 3 presents the coupled model formulation as six linked submodels with declared states, inputs, parameters and assumptions. Section 4 describes the numerical implementation and the steps taken to make the simulations reproducible. Section 5 defines the photoperiod scenarios, validation metrics, ablation variants and sensitivity protocol, and Section 6 reports the corresponding results. Section 7 interprets the findings and states the limitations of the model, and Section 8 concludes and outlines future work.

2. Related Work

This section summarises the main lines of research on plant simulation models and on the biophysical modules relevant to the approach proposed here. The focus is on mechanistic growth models, photosynthesis and stomatal-conductance models, carbohydrate dynamics, nutrient processes, and on approaches that couple these components into a single simulation framework.

2.1. Mechanistic plant growth simulation models

Plant simulation models provide the basis for linking physiological processes, microclimatic conditions and biomass change over time. The spatial and temporal scales a model covers determine where and how it can be applied. Spatially, models may span the leaf, organ, whole-plant, canopy or field level; temporally, they range from minutes and hours, for microclimatic and physiological processes, through days, for carbon growth and allocation, to whole seasons, for yield estimation. Reviews of functional-structural plant models show that models combining plant architecture with physiological modules can integrate spatial structure, phenotypic information and dynamic growth processes Louarn and Song (2020); Sievänen, Godin, De Jong and Nikinmaa (2014). The architecture of modern plant simulation models is typically modular: the photosynthetic component may use the biochemical Farquhar model, stomatal conductance may be described through the Ball-Berry model, the Medlyn optimality approach or Jarvis-type response functions, and the water balance is often coupled through a Penman-Monteith formulation. This structure allows each module to be validated individually and photosynthesis, stomatal conductance, transpiration, carbon allocation and growth to be linked flexibly Farquhar et al. (1980); Ball, Woodrow and Berry (1987); Medlyn, Duursma, Eamus, Ellsworth, Prentice, Barton, Crous, Angelis, Freeman and Wingate (2011); Jarvis (1976); Allen, Pereira, Raes and Smith (1998); Kim and Lieth (2003). Platforms such as APSIM, DSSAT and WOFOST are well-established tools for simulating crop growth, water and nitrogen dynamics and climate scenarios Holzworth et al. (2014); Jones et al. (2003); de Wit et al. (2019), while functional-structural approaches such as GreenLab add an explicit coupling between plant architecture and physiological processes de Reffye, Hu, Kang, Letort and Jaeger (2020). Such models demonstrate that modular organisation is particularly advantageous when several processes acting on different temporal and spatial scales must be combined.

Within plant simulation models several methodological families can be distinguished. Mechanistic, process-based models rely on biophysical and biochemical equations to describe photosynthesis, transpiration, growth and water and nitrogen balances through explicitly defined mechanisms; recent studies show that, unlike purely empirical approaches, they respond better to seasonal variation in climatic conditions Mamassi, Lang, Tychon, Lahlou, Wellens, El Gharous and Marrou (2024). Newer examples such as T&C-CROP integrate complex agronomic management, including crop rotation, irrigation and fertilisation, and achieve a high level of validation against field data Paules, Fatichi, Warring and Paschalis (2025). A second family models water transport through the soil-plant system; a one-dimensional model coupling plant xylem to the soil, for instance, can be as effective as more complex three-dimensional formulations while substantially reducing computational cost Janott, Gayler, Gessler, Javaux, Klier and Priesack (2011). Discrete plant-development models, such as DEM-based approaches, are used to analyse root-soil interactions, including physical contacts between seeds, roots and soil particles Tian, Zeng and Xing (2024). Optimality models assume that plants allocate resources so as to maximise a given criterion such as yield, growth or resource-use efficiency; Dynamic Energy Budget theory is one framework for describing resource allocation under variable environmental conditions Russo, Ledder, Muller and Nisbet (2022). Finally, continuous models describe plant dynamics through systems of ordinary or partial differential equations, enabling formal mathematical analysis and representing biomass, water or vegetation dynamics as smooth time-varying quantities Meron, Bennett, Fernandez-Oto, Tzuk, Zelnik and Grafi (2019).

2.2. Photosynthesis and stomatal conductance modelling

Photosynthesis is one of the central processes in mechanistic growth models because it sets the primary carbon input that is subsequently partitioned to carbohydrates, respiration, maintenance and growth. For C_3 plants the most widely used biochemical description is the Farquhar-von Caemmerer-Berry model, which relates net assimilation to the intercellular CO_2 concentration, Rubisco-limited carboxylation and RuBP regeneration Farquhar et al. (1980). This model is broadly used as the basis for leaf photosynthetic response and is commonly parametrised through quantities such as V_{cmax}, J, Γ^*, K_c, K_o and R_d Sharkey, Bernacchi, Farquhar and Singsaas (2007); Dubois, Fiscus, Booker, Flowers and Reid (2007).

The photosynthetic response depends not only on the biochemical capacity of the leaf but also on the diffusive availability of CO_2, which is governed by stomatal conductance. Stomatal models therefore form the necessary link between microclimatic conditions and carbon assimilation. Common formulations include the empirical Jarvis approach, the Ball-Berry model and the Medlyn optimality model of stomatal regulation Jarvis (1976); Ball et al. (1987); Medlyn et al. (2011). The Jarvis approach describes stomatal conductance as a product of response functions to external factors such as irradiance, vapour-pressure deficit, temperature, CO_2 concentration and plant water status Jarvis (1976); Yu, Cai, Zheng, Li and Wang (2017); Yu, Guo, Liu, Wang, Ma, Li and Liu (2021). This formulation is well suited to simulation models because it allows time-varying inputs from scenarios or measurements to be incorporated directly.

Coupling photosynthesis and stomatal conductance introduces a feedback between the biochemical consumption of CO_2 and the diffusive flux through the stomata. The intercellular CO_2 concentration sets the assimilation rate in the Farquhar model, while stomatal conductance determines how much CO_2 can diffuse from the surrounding air into the leaf interior. The combination of the Farquhar biochemical model with Jarvis-type stomatal regulation is therefore a natural basis for simulating gas exchange under variable microclimatic and photoperiodic conditions.

In the context of this work, the Farquhar and Jarvis equations are not new physiological contributions on their own but are inherited model components integrated into a broader dynamic simulation framework. The contribution lies in coupling them to the carbohydrate, nutrient and biomass state variables, so that the photosynthetic input is not treated as a prescribed external forcing but emerges from simulated gas exchange and environmental conditions.

2.3. Carbohydrate and photoperiod-dependent growth in *Arabidopsis thaliana*

Carbohydrate dynamics form the key link between photosynthesis and plant growth. During the light phase, assimilated carbon is partitioned between mobile sugars, chiefly sucrose, and transient reserves, chiefly starch. During the dark phase, when photosynthetic input ceases, starch is degraded to provide a continuous carbon supply for respiration, maintenance and growth Stitt and Zeeman (2012); Smith and Stitt (2007). Models that describe only instantaneous photosynthesis, without explicit starch and sucrose dynamics, therefore have a limited ability to represent the day-night balance of growth.

In *Arabidopsis thaliana*, a range of studies shows that nocturnal starch degradation is tightly regulated and that starch reserves are typically consumed almost exactly by dawn, reducing the risk of nocturnal carbon starvation Scialdone et al. (2013); Stitt and Zeeman (2012); Webb and Satake (2015). Mathematical models of this process link circadian regulation, sucrose availability, enzyme activity and sugar signalling to the dynamics of growth and carbon partitioning Feugier and Satake (2013); Seki, Ohara, Hearn, Frank, Silva, Caldana, Webb and Satake (2017). These properties make *Arabidopsis thaliana* a suitable system for validating models that explicitly describe photoperiod-dependent carbohydrate dynamics.

Photoperiod strongly affects the balance between daytime carbon accumulation and nocturnal reserve consumption. A shorter day limits the time available for photosynthetic assimilation and lengthens the period during which growth depends on starch mobilisation, whereas a longer day extends the period of direct carbon input. Experimental data for *Arabidopsis thaliana* show that the plant coordinates diurnal carbon-allocation regulation and growth across a wide range of photoperiods Sulpice, Flis, Ivakov, Apelt, Krohn, Encke, Abel, Feil, Lunn and Stitt (2014). Photoperiod scenarios are therefore a suitable validation framework for checking whether a simulation model reproduces the expected changes in relative growth rate, biomass and carbohydrate variables.

Beyond starch and sucrose, growth also depends on the nutrient status of the plant. Nitrogen and phosphorus affect photosynthetic capacity, metabolism and biomass allocation, and mechanistic models often introduce them as internal states or limiting factors Gastal and Lemaire (2002); Xu, Fisher, Wullschleger, Wilson, Cai and McDowell (2012); Barber (1995); Shen, Yuan, Zhang, Li, Bai, Chen, Zhang and Zhang (2011); Carstensen, Herdean, Schmidt, Sharma, Spetea, Pribil and Husted (2018). Carbon partitioning between leaf and root can be viewed through source-sink

relations, in which assimilate availability, internal nutrient demand and root status govern resource allocation Marcelis (1996); Feller, Favre, Janka, Zeeman, Gabriel and Reinhardt (2015); Louarn and Song (2020). In the present work these processes are treated as part of a coupled dynamical system linking gas exchange, carbohydrate reserves, nutrient fluxes and biomass growth.

2.4. Simulation validation, ablation, and sensitivity analysis

In dynamical plant-growth models, validation usually compares simulated and reference values for biomass, relative growth rate, water or nutrient fluxes, and other physiological variables when available Wallach, Makowski, Jones and Brun (2018); Holzworth et al. (2014); Jones et al. (2003); de Wit et al. (2019). For a model validated across photoperiods, the most informative indicators are the relative growth rate, the final biomass and the starch and sucrose dynamics.

Beyond standard validation, an analysis of model behaviour should also expose the contribution of individual components. Ablation experiments compare model variants in which specific parts are removed, simplified or replaced by alternative formulations. Here it is relevant to compare a baseline model with prescribed or simplified photosynthetic input, a model with Farquhar photosynthesis, a model with stomatal regulation, and the full variant in which gas exchange, carbohydrates, nutrients and biomass are coupled into a single dynamical system. Such a comparison helps to show that the contribution is not merely the implementation of known equations but their functional coupling within a simulation workflow.

Sensitivity analysis is a further step in assessing model robustness. It identifies which parameters most strongly influence the model outputs, such as final biomass, relative growth rate, net assimilation, starch and sucrose. For a model that includes gas exchange, the parameters linked to light, temperature, ambient CO_2 concentration, relative humidity or vapour-pressure deficit are particularly important, together with the biochemical parameters V_{cmax} and J. For the downstream processes, the relevant parameters are those governing starch degradation, sucrose allocation, nutrient limitation and initial carbohydrate reserves.

In this work, validation, ablation and sensitivity analysis are used as complementary elements of evaluation. Validation checks agreement with literature targets, the ablation study assesses the contribution of coupling gas exchange to the rest of the model, and the sensitivity analysis examines the stability and dominant sources of variability in the simulation behaviour.

2.5. Open problems and scope of this paper

Read together, the literature reviewed above leaves four issues unresolved, and they are not independent of one another. First, the two bodies of work sit at different scales and are rarely joined: leaf-level gas-exchange models are formulated and tested as steady-state algebraic systems at a fixed carbohydrate status Farquhar et al. (1980); Jarvis (1976); von Caemmerer (2000), whereas whole-plant and crop simulators integrate growth over days to seasons and, for reasons of robustness and runtime, obtain their carbon input from a light-use-efficiency term, a look-up table or an externally prescribed assimilation rate Jones et al. (2003); de Wit et al. (2019); Holzworth et al. (2014). The result is that the feedback loop closing stomatal behaviour, intercellular CO_2, carbohydrate status and growth is normally opened somewhere, and its consequences for simulated behaviour are not quantified. Second, where the carbohydrate literature does resolve the diel cycle in detail Scialdone et al. (2013); Graf et al. (2010); Stitt and Zeeman (2012), the starch and sucrose dynamics are usually described by dedicated models that take assimilation as given, so the day-night balance and the gas exchange that produces it are not solved as one system. Third, the numerical procedures that make such a coupling possible are seldom documented or verified. Coupling a demand function to a supply function introduces an implicit algebraic constraint that has to be solved at every time step, yet the solver, its convergence criterion and its behaviour outside the calibrated operating range are typically left unstated, and agreement with data is reported as if it also established numerical correctness. Fourth, evaluation practice is uneven: sensitivity analyses of plant models are widely reported, but the sampling designs and estimators used to compute the indices are rarely verified against cases with known answers, and calibration and validation are often performed against the same small set of aggregate observations, so it is not always possible to tell a predictive error from a calibration residual.

This paper addresses a specific part of that space. We do not propose new process equations for photosynthesis, stomatal conductance or starch turnover; we take those from the literature. What we contribute is the coupled formulation and its verification: an eleven-state whole-plant system in which the Farquhar-Jarvis layer computes the carbon input at every step rather than receiving it, documented submodel by submodel with declared states, inputs, parameters and interfaces (Section 3); an explicit treatment of the implicit gas-exchange constraint, including an

analytic solution of the intercellular CO_2 concentration and a verification study of the fixed-point alternative over the full forcing range (Sections 3.6 and 4.3); a closed carbon budget integrated with the state equations, which tests conservation independently of any comparison with data (Section 6); a sensitivity protocol whose estimator is first verified against a function with closed-form indices before being applied to the model (Section 5.5); and an out-of-sample photoperiod test that separates predictive accuracy from calibration fit (Section 5). Four defects found by these checks, and the tests that would have found them earlier, are reported in Section 7.3. Questions of canopy geometry, water transport, temperature acclimation and multi-species parameterisation lie outside this scope and are returned to in Section 7.5.

Table 1 places the present model against the closest published alternatives on the axes that matter for this argument. The point of the comparison is not that any individual capability is unique — every column is occupied by several models — but that the specific combination in the last row is not, and in particular that the two columns most relevant to a simulation journal, the explicit treatment of the fast–slow coupling and the reporting of numerical verification, are the ones most often left empty.

Table 1
The present model against the closest related models. FvCB: Farquhar–von Caemmerer–Berry biochemistry. "Carbon input" distinguishes models that compute assimilation from gas exchange at every step (*computed*) from those that prescribe it via light-use efficiency, a radiation-use coefficient or a look-up table (*prescribed*). "Fast–slow" marks an explicit numerical separation of the algebraic gas-exchange subsystem from the differential growth subsystem. "Num. verif." marks published convergence, conservation or estimator-verification evidence as distinct from agreement with data. Entries reflect the standard formulations as described in the cited references; individual modules and versions vary.

Model	FvCB	Stom. model	Starch/ sucrose	N/P	Root alloc.	Carbon input	Fast–slow	Num. verif.
APSIM Holzworth et al. (2014)	–	–	–	N	yes	presc.	–	–
DSSAT Jones et al. (2003)	–	–	–	N	yes	presc.	–	–
WOFOST de Wit et al. (2019)	–	–	–	N	yes	presc.	–	–
GreenLab de Reffye et al. (2020)	–	–	–	–	yes	presc.	–	–
T&C-CROP Paules et al. (2025)	yes	B–B	–	N	yes	comp.	–	–
Scialdone et al. Scialdone et al. (2013)	–	–	yes	–	–	presc.	–	–
Graf et al. Graf et al. (2010)	–	–	yes	–	–	presc.	–	–
Weraduwage et al. Weraduwage, Chen, Anozie, Morales, Weise and Sharkey (2015)	yes	–	part.	–	yes	comp.	–	–
This work	yes	Jarvis	yes	N, P	yes	comp.	yes	yes

3. Model Formulation

This section develops the model in the sequence conventionally followed in a modelling-and-simulation study (Cellier, 1991; Zeigler, Praehofer and Kim, 2000): we first state the model concept (the system boundary, its decomposition into interacting submodels, and the simplifying assumptions), then the mathematical model (the state-space equations and constitutive relations of each submodel), and finally the numerical model (the numerical scheme and simulation algorithm through which the equations are advanced in time). Presenting the three levels separately is intended to make the modelling decisions, their mathematical effects, and their software implementation clear and easier to check. All symbols are collected in Table 2.

Figure 1 gives the structure of the section as a whole and should be read before the individual subsections. Section 3.1 fixes the system boundary and the decomposition into the six submodels S1–S6, and Section 3.2 collects the eleven state variables and their governing equations in one place. The remaining subsections then take the submodels in the order in which information flows through one time step: environmental forcing (Section 3.3, S1) supplies radiation, temperature, humidity and CO_2; the stomatal-conductance supply function (Section 3.4, S2) and the photosynthetic demand function (Section 3.5, S3) are reconciled at their common operating point by the gas-exchange closure (Section 3.6, S4); the resulting assimilation rate is scaled to a whole-plant carbon input (Section 3.7); that input drives the carbohydrate and nutrient economy (Sections 3.8 and 3.9, S5); and the sucrose available after maintenance and uptake costs is allocated to leaf and root biomass (Section 3.10, S6), whose leaf mass gates the carbon input on the following evaluation. Section 3.11 states the numerical model that advances this cycle in time. The grouping in Figure 1 separates the three algebraic submodels S2–S4, which are evaluated in closed form within a derivative evaluation, from the differential submodels S5–S6, which carry the state forward; that separation is the organising principle of both the formulation and the implementation.

3.1. Conceptual model and system decomposition

The system studied is a single vegetative Arabidopsis plant together with its immediate physical environment. The system boundary encloses the plant; radiation, temperature, atmospheric humidity and CO_2, and soil water and nutrient status cross the boundary as external inputs, while carbon and nutrient acquisition, storage, and allocation to leaf and root biomass are the internal states. The plant is represented as a two-compartment (leaf/root) whole-organism system rather than a spatial one, so that transport within the plant is represented by aggregate pools rather than by explicit geometry. This lumped, functional-balance decomposition and the underlying carbon-nutrient allocation states follow the whole-plant growth framework of Tedone (2020), which the present work extends by replacing the prescribed carbon input with a computed Farquhar-Jarvis gas-exchange layer.

The system is decomposed into six submodels that exchange a small number of defined signals (Fig. 2; the block diagram is described here in text and its interfaces are those listed below):

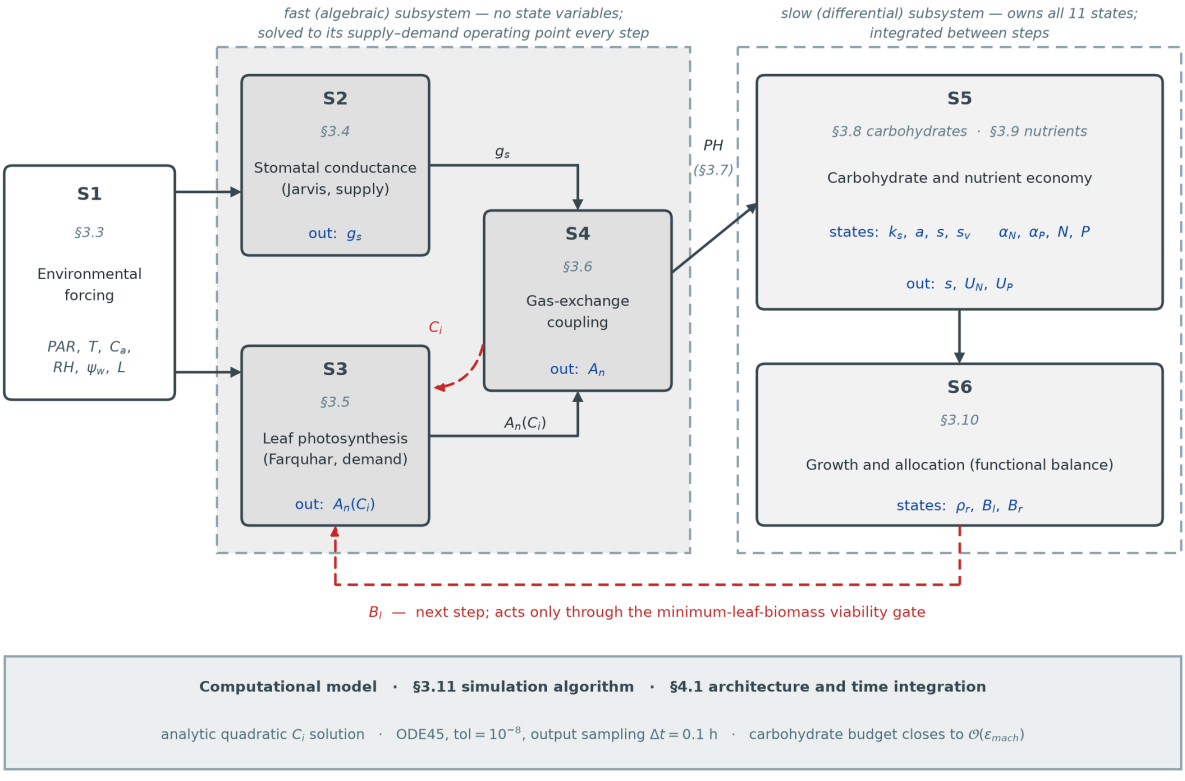


Figure 1: Structure of the model formulation and road map of Section 3. Boxes are the six submodels S1–S6 of Section 3.1, annotated with the subsection in which each is specified, the state variables it owns and the signals it exports. Solid arrows carry fluxes and algebraic signals within a single derivative evaluation. The single dashed return arrow carries leaf biomass B_l from S6 back to the gas-exchange layer, where it acts as a viability gate only ($\Phi_c = 0$ when $B_l < B_{l,\min}$; Eq. (24)): there is no feedback of photosynthetic capacity in the model as implemented, and the Farquhar capacities $V_{c\max}$ and $J_{\max}$ are fixed parameters throughout. Background tint distinguishes the fast algebraic gas-exchange subsystem S2–S4, whose supply-demand operating point is obtained in closed form (Eq. (22)) at every derivative evaluation, from the slow differential subsystem S5–S6; the two are labelled in the figure. The slow states are advanced by an adaptive Dormand–Prince `ODE45` embedded pair under local error control, not by a fixed-step scheme, so the gas-exchange layer is re-evaluated at every Runge–Kutta stage rather than once per step (Section 3.11). Figure 2 shows the same decomposition at the level of individual interface signals.

- **S1** *Environmental forcing* – maps clock time to the photoperiod switch and the instantaneous radiation and microclimate. *In:* time t, scenario parameters. *Out:* $PAR, T, C_a, RH, \psi_w, L$.
- **S2** *Stomatal conductance (supply)* – the Jarvis multiplicative model. *In:* PAR, T, C_a, RH, ψ_w. *Out:* g_s.
- **S3** *Leaf photosynthesis (demand)* – the Farquhar C_3 biochemical model. *In:* PAR, C_i, and the fixed capacity parameters $V_{c\max}, J_{\max}$. *Out:* $A_n(C_i)$.
- **S4** *Gas-exchange coupling* – closes supply against demand for the operating point. *In:* g_s, $A_n(C_i)$, C_a. *Out:* A_n, and after scaling the carbon input Φ_c.
- **S5** *Carbohydrate and nutrient economy* – diel starch/sucrose partitioning and sucrose-limited N, P uptake. *In:* Φ_c, soil N_s, P_s. *Out:* sucrose s, uptakes U_N, U_P.

S6 *Growth and allocation* – converts sucrose into leaf and root biomass under functional-balance control. *In:* s, U_N, U_P. *Out:* B_l, B_r and the allocation/investment states.

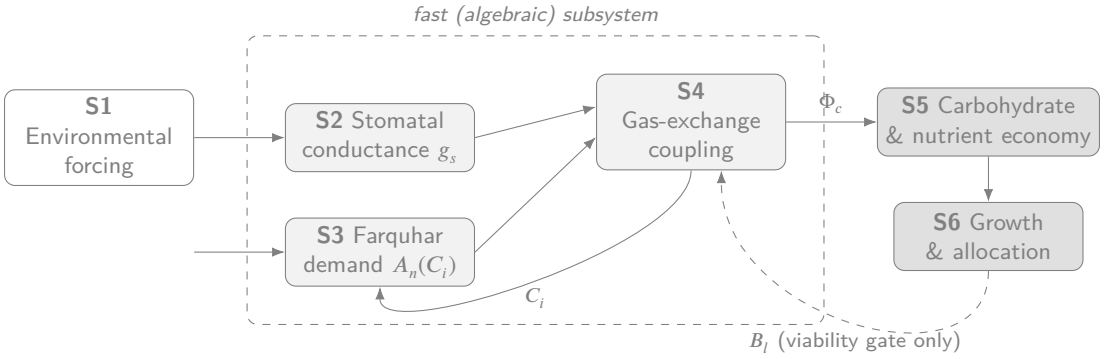


Figure 2: Decomposition of the whole-plant model into six submodels and their interface signals. S2–S4 form the fast algebraic gas-exchange subsystem (dashed box), whose supply-demand operating point is obtained in closed form at every derivative evaluation; its scaled output Φ_c drives the slow carbohydrate/nutrient and growth states S5–S6. The only return path from the slow to the fast subsystem is leaf biomass B_l, and it enters solely as the viability gate on the carbon input (Eq. (24)); the photosynthetic capacities are fixed parameters and are not fed back.

Submodels S2–S4 form a *fast* algebraic subsystem: gas exchange equilibrates on a timescale of seconds and is treated as instantaneously relaxed relative to the *slow* states in S5-S6, which evolve over the day-night cycle. This fast-slow separation is exploited directly by the numerical model of Section 3.11, where the fast subsystem enters as an algebraic constitutive layer inside the derivative of the slow states, evaluated in closed form and therefore without an inner iteration (Cellier, 1991).

The formulation rests on the following simplifying assumptions.

(A1) The plant is spatially lumped into leaf and root compartments; no within-organ gradients or explicit canopy architecture are resolved.

(A2) Leaf gas exchange is at steady state at every instant, so the intercellular CO_2 concentration is the algebraic supply-demand balance rather than a differential state (Section 3.6).

(A3) Stomatal conductance responds to the environment multiplicatively and independently of the assimilation rate (Jarvis-Stewart closure), so the supply function contains no explicit feedback from A_n (Jarvis, 1976; Damour, Simonneau, Cochard and Urban, 2010).

(A4) Transitory starch is degraded at a near-constant nocturnal rate calibrated to be almost exhausted at dawn, consistent with the observed circadian control of starch turnover (Graf et al., 2010; Scialdone et al., 2013).

(A5) Active nutrient uptake is energetically limited by cytosolic sucrose, so carbon and nutrient acquisition are explicitly co-limiting (Smith and Stitt, 2007).

(A6) Biomass partitioning between shoot and root follows functional-balance theory: investment shifts toward the resource that most limits growth (Thornley, 1972; Thornley and Johnson, 1990).

3.2. State-space representation

The model state is the vector $X(t) \in \mathbb{R}^n$, whose evolution is governed by the ordinary differential system with fixed parameters

$$\frac{dX(t)}{dt} = f(X(t), U(t), \boldsymbol{\theta}, t), \tag{1}$$

where $U(t)$ is the exogenous environmental forcing, $\boldsymbol{\theta}$ is the time-invariant parameter set, and the time dependence enters only through the diel forcing. The state vector comprises $n = 11$ components,

$$X(t) = \left[k_s,\ a,\ s,\ \alpha_N,\ \alpha_P,\ N,\ P,\ \rho_r,\ B_l,\ B_r,\ s_v\right]^{\mathsf{T}}, \tag{2}$$

with the following interpretation: k_s is the starch partition coefficient (dimensionless), i.e. the fraction of gross assimilate directed to starch; a is the transitory starch pool and s the cytosolic sucrose pool (μmol gFW^{-1}); α_N, α_P are the nitrogen and phosphorus investment (affinity) states; N, P are the internal nitrogen and phosphorus pools (μmol gFW^{-1}); ρ_r is the sucrose-to-root allocation state; B_l, B_r are leaf and root biomass (gFW); and s_v is a vacuolar sucrose pool (μmol gFW^{-1}) exchanging with the cytosolic pool. All five pool states are specific quantities on a whole-plant fresh-weight basis; Section 3.8 states precisely what that basis is and what it implies for the balance equations. The exogenous forcing is

$$U(t) = \left[PAR(t),\ T(t),\ C_a(t),\ RH(t),\ VPD(t),\ \psi_w(t),\ L(t)\right]^{\mathsf{T}}, \tag{3}$$

collecting photosynthetically active radiation PAR (μmol m^{-2} s^{-1}), leaf temperature T (°C), ambient CO_2 concentration C_a (μmol mol^{-1}), relative humidity RH (%), vapour pressure deficit VPD (kPa), leaf water potential ψ_w (MPa), and the binary photoperiod signal $L(t)$. As indicated in Eqs. (12)-(14), VPD is a diagnostic function of T and RH and is therefore not an independent forcing; it is listed in $U(t)$ because it enters the stomatal response directly.

3.3. Submodel S1: environmental forcing

Let $h(t) = t \bmod 24$ denote the hour of day and f the photoperiod length (h). The light/dark switch is

$$L(t) = \begin{cases} 1, & 0 \le h(t) < f, \\ 0, & f \le h(t) < 24, \end{cases} \tag{4}$$

and the incident radiation follows a half-sinusoidal daytime profile that is zero at night,

$$PAR(t) = \begin{cases} PAR_{\max}\ \sin\!\left(\pi\,\dfrac{h(t)}{f}\right), & 0 \le h(t) < f, \\ 0, & f \le h(t) < 24, \end{cases} \tag{5}$$

where $PAR_{\max}$ is the scenario-specific peak irradiance. The sinusoidal envelope is a smooth idealisation of the daily light course that reaches its maximum at solar noon ($h = f/2$); it is the driver of both the demand (through electron transport, Eq. (18)) and the supply (through the Jarvis light response, Eq. (7)) side of gas exchange, and thereby imposes the day-night rhythm on the entire system.

3.4. Submodel S2: stomatal conductance (supply)

This submodel takes the micro-climate from S1 and returns the stomatal conductance g_s; it is the supply side of gas exchange and carries no dependence on the assimilation rate (assumption A3). Stomatal conductance is represented by the multiplicative (Jarvis-Stewart) form (Jarvis, 1976; Damour et al., 2010), in which a maximal conductance is reduced by a product of independent, dimensionless environmental modifiers, each bounded in $[0, 1]$:

$$g_s = g_{s,\max}\, f_{PAR}(PAR)\, f_{VPD}(VPD)\, f_T(T)\, f_{CO_2}(C_a)\, f_w(\psi_w). \tag{6}$$

The multiplicative closure assumes that the environmental controls act independently and non-synergistically; it remains the most widely used empirical conductance model and requires no iterative optimisation (Damour et al., 2010; Buckley, 2017). The maximal conductance $g_{s,\max}$ is carried as an explicit parameter separate from the shape parameters of the modifiers. This separation is not cosmetic: it keeps the magnitude of the conductance in one place, so that each f_i can be required to satisfy $f_i \in [0, 1]$ with a value of unity at its own non-limiting reference condition, and so that a correction to the shape of one modifier does not silently rescale the whole conductance. The individual response functions are specified as follows.

Light response. A saturating (rectangular-hyperbola) response with a small residual (cuticular/night) conductance b_{10},

$$f_{PAR}(PAR) = \begin{cases} b_{10}, & PAR \le q, \\ \dfrac{b_1 b_2 (PAR - q)}{b_1 + b_2 (PAR - q)}, & PAR > q, \end{cases} \qquad q = \frac{b_{10}}{b_1}. \tag{7}$$

The two branches do not meet. With the values of Table 3 the breakpoint is $q = b_{10}/b_1 = 0.0833$ μmol m^{-2} s^{-1}, where the first branch gives $f_{PAR} = b_{10} = 0.05$ and the second gives $f_{PAR} \to 0$: the modifier drops by b_{10} as PAR crosses q from below, and does not regain 0.05 until $PAR = 27.36$ μmol m^{-2} s^{-1}. The intent of the construction is evidently a residual conductance floor, $f_{PAR} = \max(b_{10}, \text{hyperbola})$, which would place the crossover at 27.36 and be continuous there; the implemented form instead switches at q and is not. We report the behaviour as implemented, since it is what produced every number in this paper, but the jump is a defect rather than a modelling choice and the floor form is the appropriate correction.

Its practical consequence is small and bounded, for a reason worth stating because it is not the reason one would guess. The interval affected, $0.0833 < PAR < 27.36$ μmol m^{-2} s^{-1}, lies almost entirely below the light compensation point of the gas-exchange system (13.4 μmol m^{-2} s^{-1} at the default parameters, and higher under the reduced conductance the jump itself produces), where the carbon input is zero regardless of g_s. The discontinuity is therefore mostly invisible in Φ_c; where it is not, it acts by lowering conductance in the first minutes after dawn and the last before dusk. It does, however, remain a discontinuity in a quantity the integrator differences, and it is one of the reasons the step-size controller contracts sharply at the light–dark transitions (Section 3.11). The half-sinusoidal forcing of Eq. (5) traverses this interval in 1.7 to 5.2 minutes of simulated time at the four photoperiods studied here (rising and again falling), which bounds the exposure.

Vapour-pressure-deficit response. A linear decline with atmospheric demand, floored at zero,

$$f_{VPD}(VPD) = \max\bigl(0,\ 1 - b_5\, VPD\bigr)\,. \tag{8}$$

Temperature response. A skewed parabolic optimum (Jarvis, 1976) defined between a lower limit T_1 and an upper limit T_h with optimum near T_0,

$$f_T(T) = b_3\,(T - T_1)\,(T_h - T)^{b_4}, \qquad b_4 = \frac{T_h - T_0}{T_0 - T_1}, \quad b_3 = \frac{1}{(T_0 - T_1)\,(T_h - T_0)^{b_4}}, \tag{9}$$

so that $f_T(T_1) = f_T(T_h) = 0$ and $f_T(T_0) = 1$ with the maximum located *at* T_0; with $T_1 = 5$, $T_0 = 25$ and $T_h = 45$°C this gives $b_4 = 1$ and a single-peaked curve requiring no clipping. Both constants differ from those used in the originally submitted implementation, and Section 4.3 documents the correction and its consequences.

CO_2 response. A piecewise-linear reduction of conductance with increasing ambient CO_2,

$$f_{CO_2}(C_a) = \begin{cases} 1, & C_a < 100, \\ 1 - b_7\,(C_a - 100), & 100 \le C_a < 1000, \\ b_8, & C_a \ge 1000, \end{cases} \qquad b_7 = \frac{1 - b_8}{900}, \tag{10}$$

with C_a in μmol mol^{-1}.

Water-status response. A saturating (exponential) recovery of conductance above a minimum leaf water potential $\psi_{\min}$,

$$f_w(\psi_w) = 1 - \exp\bigl(-b_6\,(\psi_w - \psi_{\min})\bigr)\,, \tag{11}$$

clipped to [0, 1]. The vapour pressure deficit driving Eq. (8) is obtained from leaf temperature and relative humidity through the Tetens saturation relation (Monteith, 1965; Nobel, 2009),

$$e_s = 0.61078\ \exp\!\Big(\frac{17.27\,T}{T + 237.3}\Big), \tag{12}$$

$$e_a = e_s\,\frac{RH}{100}, \tag{13}$$

$$VPD = e_s - e_a, \tag{14}$$

with e_s, e_a, VPD in kPa. The Jarvis model thus returns an environment-dependent conductance that modulates the CO_2 supply to the mesophyll without any explicit dependence on the assimilation rate, providing the supply constraint that is closed against the biochemical demand in Section 3.6.

Table 2 lists symbols and units. Because a reader reproducing the model needs the numerical values as well, and needs to know which of them were fitted, Table 3 gives the complete parameter set with nominal value, provenance and calibration status. Every value is the one used in the reported runs, read from `config/default.config` or from the source where a constant is compiled in; the four coefficients marked as calibrated are the only fitted quantities in the model.

3.5. Submodel S3: leaf photosynthesis (demand)

This submodel returns the net assimilation A_n as a function of the intercellular CO_2 concentration C_i; it is the demand side of gas exchange and is closed against the S2 supply in Section 3.6. Gross CO_2 demand follows the Farquhar-von Caemmerer-Berry biochemical model (Farquhar et al., 1980; von Caemmerer, 2000), in which assimilation is the minimum of a Rubisco-limited and an electron-transport-limited (RuBP regeneration) rate. The Rubisco-limited rate is

$$A_c = V_{c\max}\,\frac{C_i - \Gamma^*}{C_i + K_c\big(1 + O/K_o\big)}, \tag{15}$$

where C_i is the intercellular CO_2 concentration, Γ^* the CO_2 compensation point in the absence of dark respiration, $V_{c\max}$ the maximum carboxylation rate, K_c and K_o the Michaelis constants of Rubisco for CO_2 and O_2, and O the oxygen concentration. The electron-transport-limited rate is

$$A_j = J\,\frac{C_i - \Gamma^*}{4.5\,C_i + 10.5\,\Gamma^*}, \tag{16}$$

in which the denominator coefficients {4.5, 10.5} encode the electron requirement per carboxylation and per oxygenation under the standard stoichiometry of RuBP regeneration (von Caemmerer and Farquhar, 1981; von Caemmerer, 2000). Net assimilation is the limiting process less dark respiration R_d,

$$A_n = \min(A_c,\ A_j) - R_d. \tag{17}$$

The potential electron transport rate J is obtained from PAR through the non-rectangular hyperbola (Farquhar et al., 1980; Farquhar and Wong, 1984),

$$\theta\,J^2 - \big(J_{\max} + \phi\,PAR\big)\,J + J_{\max}\,\phi\,PAR = 0, \tag{18}$$

whose lower root is taken,

$$J = \frac{\big(J_{\max} + \phi\,PAR\big) - \sqrt{\big(J_{\max} + \phi\,PAR\big)^2 - 4\,\theta\,J_{\max}\,\phi\,PAR}}{2\,\theta}, \tag{19}$$

Table 2
Principal symbols. Rate/flux units for the carbohydrate pools are μmol gFW^{-1} h^{-1}; leaf-level assimilation is in μmol CO_2 m^{-2} s^{-1}.

Symbol	Meaning	Units
State variables $X(t)$		
k_s	Starch partition coefficient	–
a	Transitory starch pool	μmol gFW^{-1}
s	Cytosolic sucrose pool	μmol gFW^{-1}
s_v	Vacuolar sucrose pool	μmol gFW^{-1}
α_N, α_P	N, P investment (affinity) states	–
N, P	Internal nitrogen, phosphorus content	μmol gFW^{-1}
ρ_r	Sucrose-to-root allocation state	–
B_l, B_r	Leaf, root biomass	gFW
Environmental forcing $U(t)$		
PAR	Photosynthetically active radiation	μmol m^{-2} s^{-1}
T	Leaf temperature	°C
C_a	Ambient CO_2 concentration	μmol mol^{-1}
RH	Relative humidity	%
VPD	Vapour pressure deficit	kPa
ψ_w	Leaf water potential	MPa
L	Photoperiod (light/dark) switch	–
Gas exchange		
g_s, g_c	Stomatal conductance to H_2O, CO_2	mol m^{-2} s^{-1}
C_i	Intercellular CO_2 concentration	μmol mol^{-1}
A_c, A_j, A_n	Rubisco-, RuBP-limited, net assimilation	μmol m^{-2} s^{-1}
$V_{c\max}$	Maximum carboxylation rate	μmol m^{-2} s^{-1}
$J, J_{\max}$	Actual, maximum electron transport rate	μmol m^{-2} s^{-1}
Γ^*	CO_2 compensation point ($R_d = 0$)	μmol mol^{-1}
K_c, K_o	Michaelis constants (CO_2, O_2)	μmol mol^{-1}; mmol mol^{-1}
O	Oxygen concentration	mmol mol^{-1}
R_d	Dark (day) respiration rate	μmol m^{-2} s^{-1}
ϕ, θ	Quantum yield, curvature (Eq. (19))	–; –
Φ_c	Whole-plant gross carbon input	μmol gFW^{-1} h^{-1}
SLA	Specific leaf area	m^2 gFW^{-1}
Nutrients, growth, and parameters		
$U_N^{\text{pot}}, U_P^{\text{pot}}$	Potential N, P uptake	μmol gFW^{-1} h^{-1}
U_N, U_P	Realised (sucrose-limited) N, P uptake	μmol gFW^{-1} h^{-1}
$I_{N,\max}, I_{P,\max}$	Maximal uptake rates	μmol gFW^{-1} h^{-1}
K_N, K_P	Uptake half-saturation constants	μmol (soil)
c_N, c_P	Sucrose cost per unit N, P acquired	μmol (sucrose) μmol^{-1}
τ	Reference time scale of the uptake limitation (Eqs. (40)-(41)), = 1 h	h
D_a	Starch degradation flux	μmol gFW^{-1} h^{-1}
μ_l, μ_r	Leaf, root mortality rates	h^{-1}
γ_l, γ_r	Self-competition coefficients	gFW^{-1} h^{-1}
$\boldsymbol{\theta}$	Time-invariant parameter set	–
ϵ	Numerical regularisation constant	–

with ϕ the apparent quantum yield for electron transport (including leaf absorptance) and θ the curvature of the light response.

The submodel is evaluated whenever $PAR > 0$ and leaf biomass exceeds the viability minimum $B_{l,\min}$; in darkness, or below that biomass, the carbon input is set to zero. No irradiance threshold is applied. The originally submitted version of the model treated the leaf as non-assimilating below $PAR = 20$ μmol m^{-2} s^{-1}, which was removed for two reasons. First, it was a discontinuity in the carbon input at a point every diel trajectory crosses twice a day: the light

Table 3
Complete parameter set. "Status" distinguishes parameters fixed from the literature or by construction (F) from the four calibrated coefficients (C); "Range" gives the sensitivity-analysis interval where the parameter was included in the global sensitivity design (see Table 17 for sampling scales and per-parameter provenance). Parameters compiled in as constants rather than exposed in the configuration file are marked †.

Symbol	Meaning	Value	Units	St.	Range
Jarvis stomatal response (Eq. 6), all †					
$g_{s,\max}$	Unstressed maximum conductance	0.7071068	$\mathrm{mol\,m^{-2}\,s^{-1}}$	F	–
b_1	PAR response asymptote	0.6	–	F	–
b_2	PAR response initial slope	0.002	$\mathrm{m^2\,s\,\mu mol^{-1}}$	F	–
b_{10}	PAR response floor	0.05	–	F	–
q	PAR breakpoint, b_{10}/b_1	0.0833	$\mu\mathrm{mol\,m^{-2}\,s^{-1}}$	F	–
b_5	VPD sensitivity	0.2	$\mathrm{kPa^{-1}}$	F	–
T_0	Temperature optimum	25	°C	F	–
T_1, T_h	Lower, upper temperature zeros	5, 45	°C	F	–
b_4	$(T_h - T_0)/(T_0 - T_1)$	1.0	–	F	–
b_3	$1/((T_0 - T_1)(T_h - T_0)^{b_4})$	2.5×10^{-3}	–	F	–
b_7, b_8	CO_2 response slope, floor	8.9×10^{-4}, 0.2	–	F	–
b_6	Water-potential response rate	3.0	$\mathrm{MPa^{-1}}$	F	–
$\psi_{\min}$	Water potential at stomatal closure	−1.5	MPa	F	–
Farquhar biochemistry (Eqs. 15–17), all †					
$V_{c\max}$	Maximum carboxylation rate	45.0	$\mu\mathrm{mol\,m^{-2}\,s^{-1}}$	F	[20, 120]
$J_{\max}$	Maximum electron transport rate	90.0	$\mu\mathrm{mol\,m^{-2}\,s^{-1}}$	F	[50, 200]
Γ^*	CO_2 compensation point	42.0	$\mu\mathrm{mol\,mol^{-1}}$	F	–
K_c	Michaelis constant, CO_2	404.9	$\mu\mathrm{mol\,mol^{-1}}$	F	–
K_o	Michaelis constant, O_2	278.4	$\mathrm{mmol\,mol^{-1}}$	F	–
O	Oxygen concentration	210.0	$\mathrm{mmol\,mol^{-1}}$	F	–
R_d	Day respiration rate	1.5	$\mu\mathrm{mol\,m^{-2}\,s^{-1}}$	F	–
ϕ	Quantum yield	0.385	$\mathrm{mol\,mol^{-1}}$	F	–
θ	Curvature of the J response	0.7	–	F	–
Carbon-input conversion (Eqs. 24–25)					
κ'	Carbon-input coefficient	1.222×10^{-2}	–	C†	–
SLA	Specific leaf area	0.025	$\mathrm{m^2\,gFW^{-1}}$	F	–
$B_{l,\min}$	Leaf-biomass viability floor	0.001	gFW	F	–
$\Phi_{c,\max}$	Prescribed max. rate (M0 only)	12.7	$\mu\mathrm{mol\,gFW^{-1}\,h^{-1}}$	F	–
Carbohydrate dynamics					
λ_{sdi}	Daytime starch synthesis rate	0.05	$\mathrm{h^{-1}}$	C	–
λ_{sdr}	Starch degradation rate coeff.	0.25	$\mathrm{h^{-1}}$	F	–
λ_{sni}	Nocturnal starch partition	0.16/0.15/0.13/0.08 (4/6/8/12 h)		F	excl.
λ_c	Starch feedback coefficient	0.82/0.79/0.67/0.62 (4/6/8/12 h)		F	–
λ_g	Night-efficiency coefficient	0.65	–	F	–
k_s^0	Initial starch partition	0.6	–	F	–
$D_{a,\max}$	Max. starch degradation rate	6.0	$\mu\mathrm{mol\,gFW^{-1}\,h^{-1}}$	F	–
$s_{\max}$	Sucrose reference scale	2.0	$\mu\mathrm{mol\,gFW^{-1}}$	F	–
k_{in}	Cytosol-to-vacuole rate	0.10	$\mathrm{h^{-1}}$	F	[0.005, 0.40]
k_{out}	Vacuole-to-cytosol rate	0.08	$\mathrm{h^{-1}}$	F	–
r_m	Maintenance respiration freq.	0.79	$\mathrm{h^{-1}}$	F	–
ϕ_{load}	Sucrose loading frequency	1.98	$\mathrm{h^{-1}}$	F	–
c_{tr}	Sucrose cost of transport	0.0035	–	F	–
Nutrients and growth					
λ_{sb}	Growth-rate coefficient	calibrated per photoperiod, ≈ 3.8–5.9×10^{-3}		C	–
$I_{N,\max}$	Maximal N uptake rate	see Table 17		F	incl.
$I_{P,\max}$	Maximal P uptake rate	see Table 17		F	incl.
λ_{csn}	Sucrose cost of N assimilation	0.0267	–	F	–
λ_k	Uptake half-saturation scale	50.0	μmol	F	–
r^*	Optimal N:P stoichiometry	10.0	$\mathrm{mol\,mol^{-1}}$	F	–
ρ_r^0	Initial root allocation	0.5	–	F	–
μ_l, μ_r	Leaf, root mortality rates	5×10^{-7}	$\mathrm{h^{-1}}$	F	–
γ_l, γ_r	Self-competition coeff.	5×10^{-7}	$\mathrm{gFW^{-1}\,h^{-1}}$	F	–
$B_l(0), B_r(0)$	Initial leaf, root biomass	0.02, 0.02	gFW	F	incl.

modifier at that irradiance is $f_{PAR}(20) = 0.0374 \neq 0$, so the assimilation rate jumped rather than passing through the threshold, which is a hazard for the error-controlled integrator of Section 3.11. Second, it is unnecessary: with the closed-form operating point of Section 3.6 the net rate passes smoothly through zero at the light compensation point — $PAR = 13.4\ \mu\mathrm{mol\,m^{-2}\,s^{-1}}$ under the default parameterisation — and the terminal $\max(0, \cdot)$ applied to the carbon input returns exactly zero below it, continuously and without a special case. Sub-compensation irradiance is therefore

handled by the physics rather than by a gate. Note that A_n itself remains negative there, and it is the whole-plant carbon input, not A_n, that is clamped; the respiratory efflux is not treated as an assimilatory gain.

3.6. Submodel S4: gas-exchange coupling

Equations (15)-(17) express biochemical *demand* as a function of the intercellular CO_2 that is itself set, through Fickian *supply*, by the diffusion of CO_2 across the stomata:

$$C_i = C_a - \frac{A_n}{g_c}, \qquad g_c = \frac{g_s}{1.6}, \tag{20}$$

where the factor 1.6 converts the water-vapour conductance g_s to a CO_2 conductance g_c, reflecting the lower molecular diffusivity of CO_2 in air (von Caemmerer, 2000; Nobel, 2009). Because A_n depends on C_i and C_i depends on A_n, the operating point is the mutual solution of Eqs. (17) and (20). We obtain it by fixed-point iteration (Press, Teukolsky, Vetterling and Flannery, 2007): starting from $C_i^{(0)} = \eta\, C_a$ with $\eta = 0.7$, the demand Eqs. (15)-(17) are evaluated at the current $C_i^{(k)}$, the supply Eq. (20) returns an updated $C_i^{(k+1)} = \max\!\left(0,\ C_a - A_n^{(k)}/g_c\right)$, and the iteration proceeds until both the concentration and the assimilation rate are stable,

$$\left|C_i^{(k+1)} - C_i^{(k)}\right| < \varepsilon \quad \text{and} \quad \left|A_n^{(k)} - A_n^{(k-1)}\right| < \varepsilon, \tag{21}$$

with separate tolerances ε_{C_i} and ε_{A_n} for the two criteria — the two residuals have different dimensions and differ by more than an order of magnitude in typical size, so a single absolute tolerance is not defensible — and an iteration cap $k_{\max}$. The conductance g_s is held fixed at its Jarvis value (Eq. (6)) during the inner iteration, so the procedure resolves a genuine runtime coupling between the Jarvis supply function and the Farquhar demand function at every time step. The scheme is summarised in Algorithm 1.

The coupled system does not, however, require iteration at all, and the production simulator does not use it. Substituting the supply relation Eq. (20) into either Farquhar branch yields a *quadratic* in C_i that can be solved in closed form. Writing each branch of Eqs. (15)-(16) as a rational function $A = \alpha(C_i - \Gamma^*)/(C_i + \beta)$ — with $(\alpha, \beta) = \left(V_{c\max},\ K_c(1 + O/K_o)\right)$ for the Rubisco-limited branch and $(\alpha, \beta) = \left(J/4.5,\ (10.5/4.5)\,\Gamma^*\right)$ for the electron-transport-limited branch — and equating supply and demand, $g_c(C_a - C_i) = \alpha(C_i - \Gamma^*)/(C_i + \beta) - R_d$, gives

$$g_c\, C_i^2 + \left[\alpha - R_d + g_c(\beta - C_a)\right] C_i - \left[\alpha\Gamma^* + R_d\,\beta + g_c\, C_a\,\beta\right] = 0, \tag{22}$$

whose physically admissible root is the larger one, since the product of the roots is negative and exactly one root is positive. The operating point is the branch-wise solution with the smaller assimilation rate, consistent with the $\min(A_c, A_j)$ co-limitation of Eq. (17). The sign of R_d in the linear coefficient is worth stating explicitly, because the originally submitted manuscript printed it as $+R_d$. That was a transcription error in the manuscript only: the implementation has always formed the coefficient as $\alpha - R_d + g_c(\beta - C_a)$ — in the source, as $g_c(d_1 C_a - d_0) - \alpha_{\text{num}} + R_d d_1$ with the leading coefficient $-g_c d_1$, which is the same quadratic multiplied by -1 — so every number reported in this paper and in the earlier submission was computed from the correct expression. We record the correction here rather than silently amending the equation, since a reader checking the algebra against the code would otherwise find a discrepancy that does not exist. We thank the reviewer who caught it.

The admissibility rule applied to the two roots is $C_i \geq 0$ alone. A root exceeding the ambient concentration is retained rather than rejected: by Eq. (20), $C_i > C_a$ is exactly what the supply relation returns whenever $A_n < 0$, that is, whenever the leaf is a net source below the light compensation point, and it is physically correct there.

When the conductance vanishes the quadratic degenerates and no operating point exists, so the implementation applies a boundary condition rather than solving: for $g_c \leq 0$ it sets

$$C_i = \Gamma^*, \qquad A_n = -R_d, \tag{23}$$

that is, a genuine efflux of respiratory CO_2 and not zero assimilation. Equation (23) is a stated convention of the implementation at the boundary of the admissible conductance range, not the $g_s \to 0$ limit of the coupled system; the

limit itself is approached from the interior and is not evaluated. We flag the distinction because a boundary convention and a limit are different objects, and only the former is what the code executes. Equation (22) is the default solver in all results reported here. The iterative scheme of Eq. (21) and Algorithm 1 is retained behind a run-time switch, and is used in Section 4.3 to characterise its own failure modes against the closed form; presenting both is what makes that verification possible.

Algorithm 1 Farquhar-Jarvis gas-exchange coupling: fixed-point solution of the intercellular CO_2 concentration.

Require: C_a, g_s, Γ^*, $V_{c\max}$, K_c, K_o, O, J, R_d; tolerance ε, cap $k_{\max}$
Ensure: A_n, converged C_i

$C_i \leftarrow 0.7\,C_a$
$g_c \leftarrow g_s/1.6$; $\quad A_n^{\text{prev}} \leftarrow 0$
for $k \leftarrow 1$ **to** $k_{\max}$ **do**
  $A_c \leftarrow V_{c\max}\,(C_i - \Gamma^*)/\big(C_i + K_c(1 + O/K_o)\big)$
  $A_j \leftarrow J\,(C_i - \Gamma^*)/(4.5\,C_i + 10.5\,\Gamma^*)$
  $A_n \leftarrow \min(A_c, A_j) - R_d$
  $C_i^{\text{new}} \leftarrow \max\big(0,\ C_a - A_n/g_c\big)$
  if $|C_i^{\text{new}} - C_i| < \varepsilon$ **and** $|A_n - A_n^{\text{prev}}| < \varepsilon$ **then**
    break
  end if
  $C_i \leftarrow C_i^{\text{new}}$; $\quad A_n^{\text{prev}} \leftarrow A_n$
end for
return $A_n,\ C_i$

3.7. Conversion from assimilation to model carbon input

The leaf-level net assimilation A_n (μmol CO_2 m^{-2} s^{-1}) is converted to the whole-plant gross carbon input Φ_c that drives the carbohydrate pools (μmol gFW^{-1} h^{-1}):

$$\Phi_c = \underbrace{\kappa'}_{\text{dimensionless}}\ \underbrace{3600}_{\text{s h}^{-1}}\ \underbrace{\text{SLA}}_{\text{m}^2\,\text{gFW}^{-1}}\ \underbrace{A_n}_{\mu\text{mol m}^{-2}\,\text{s}^{-1}}, \qquad L(t) = 1, \tag{24}$$

where SLA is the specific leaf area and the two explicit factors carry the entire unit conversion: SLA converts the per-area flux to a per-gram-fresh-weight basis and the factor 3600 $s\,h^{-1}$ converts seconds to hours, so that Φ_c is in μmol gFW^{-1} h^{-1} as stated. Only κ' is dimensionless, and it is a calibration coefficient, not a unit factor. The earlier submission wrote this relation as $\Phi_c = \kappa A_n\,\text{SLA}/\text{SLA}_{\text{ref}}$ with a dimensionless κ said to absorb the unit conversion. That form is dimensionally inconsistent, as a dimensionless coefficient cannot convert $m^{-2}\,s^{-1}$ into $gFW^{-1}\,h^{-1}$; we thank the reviewer for identifying it. The implementation is unchanged — it applies a single constant 1.10 at the reference SLA of 0.025 m^2 gFW^{-1} — so the two forms agree numerically with

$$\kappa' = \frac{1.10}{3600\,\text{SLA}_{\text{ref}}} = \frac{1.10}{90} = 1.222 \times 10^{-2}. \tag{25}$$

Writing the conversion this way makes the size of the fitted coefficient visible: the coupled model retains about 1.2% of the gross per-area assimilation as carbon entering the whole-plant carbohydrate pools. That is not a small correction to a physical conversion but the dominant empirical scaling in the carbon-input pathway, and it is reported here as such rather than concealed inside a symbol described as a unit factor. Its magnitude is consistent with the model's treatment of A_n as an upper-canopy, light-saturated leaf rate while Φ_c is a whole-rosette average over self-shaded leaves of mixed age, but the coefficient is calibrated, not derived, and κ' is therefore listed as a fitted parameter in Table 2. Leaf biomass B_l enters this conversion only through the on/off viability guard ($\Phi_c = 0$ when $B_l < B_{l,\min}$) and not as a continuous multiplier. In darkness there is no gross fixation,

$$\Phi_c = 0, \qquad L(t) = 0. \tag{26}$$

3.8. Submodel S5: carbohydrate and nutrient economy

Before the balance equations, the meaning of the pool states must be fixed precisely, because the units μmol gFW^{-1} admit two readings that lead to different equations. The states a, s, s_v (and likewise N, P) are *specific* pools: amount of substance per unit of whole-plant fresh weight, i.e. intensive quantities, not extensive contents of the growing plant. They are stated per gram of fresh weight so that Φ_c, which Eq. (24) delivers on the same basis, can drive them directly without carrying $B_l + B_r$ through the carbohydrate subsystem.

One consequence must be stated explicitly. For an intensive quantity in a growing system, the material balance carries a dilution term: if A is the extensive amount and $B = B_l + B_r$ the fresh weight, then $a = A/B$ obeys

$$\frac{da}{dt} = \frac{1}{B}\frac{dA}{dt} - \frac{a}{B}\frac{dB}{dt}, \tag{27}$$

and the second term — growth dilution — is *not* present in Eqs. (28), (30) or (31), nor in Eqs. (42)-(43). The pool equations as implemented are therefore per-gram balances of the fluxes only, and are exact for an intensive pool only in the limit $B^{-1}\,dB/dt \to 0$. Over the horizons simulated here that relative growth rate is 0.069 to 0.261 d^{-1}, i.e. 2.9×10^{-3} to 1.1×10^{-2} h^{-1}, against diel turnover of the sucrose pool on the order of 10^{-1} h^{-1}; the omitted term is thus one to two orders of magnitude below the retained ones, which is why the diel trajectories are qualitatively unaffected. It is not, however, negligible in the sense of being absent from the model's intent: the calibrated coefficients λ_{sb} and λ_{sdi} absorb its time-averaged effect, which is one more reason the calibration residuals of Section 6 cannot be read as predictive errors. We flag this as a formulation limitation rather than correct it here, since correcting it would change every calibrated value in the paper.

The alternative reading — that the pools are normalised state variables on a nominal per-gram basis, with no claim to be concentrations of a growing body — is self-consistent and requires no dilution term. It is the reading under which the implementation is correct as written, and it is the one we adopt. What it costs is the direct interpretation of a and s as measurable tissue concentrations: the comparison against measured carbohydrate contents in Section 6 is then a comparison of shape and timing, and the absolute levels carry the calibration in them. The figures are read that way throughout.

Fixed carbon is partitioned between transitory starch and cytosolic sucrose. The starch pool accumulates a fraction k_s of the carbon input during the day and is remobilised at night,

$$\frac{da}{dt} = k_s\,\Phi_c - D_a, \tag{28}$$

where D_a is the starch degradation flux. Consistent with the experimental observation that transitory starch is degraded at an approximately constant rate calibrated to be nearly exhausted at dawn (Graf et al., 2010; Scialdone et al., 2013; Stitt and Zeeman, 2012), D_a is active only in darkness ($L = 0$) and is scaled by the remaining reserve and the time left until dawn. Explicitly, with $a^\dagger$ the starch content latched at the onset of darkness, $t_d = \mathrm{mod}(t, 24)$ the hour of day and $\Delta_n = 24 - P$ the night length,

$$D_a = \begin{cases} 0, & a^\dagger < a_{\min}, \\ (1-L)\,\min\!\left(D_{a,\max},\ \left[\dfrac{t_d}{24} + \left(1-\dfrac{t_d}{24}\right)\left(1-\dfrac{s}{s+s_{\max}}\right)\right]\dfrac{a^\dagger - a_{\min}}{\Delta_n}\right), & a^\dagger \geq a_{\min}, \end{cases} \tag{29}$$

where $D_{a,\max}$ caps the instantaneous mobilisation rate. The first branch is a guard rather than a modelling choice: if the reserve latched at dusk already lies below the residual pool $a_{\min}$ there is nothing mobilisable, and prescribing a positive flux would drive the starch state negative. The bracket interpolates between a purely clock-driven rate and a sucrose-status-driven one, so that mobilisation accelerates when cytosolic sucrose is depleted, and the divisor Δ_n paces the reserve to be almost exhausted at dawn. Because $a^\dagger$ is held constant through the night, D_a has no explicit dependence on the current starch state a during a night, which is what makes the near-linear nocturnal decline observed by Graf et al. (2010) an emergent rather than an imposed property. The cytosolic sucrose pool receives the complementary fraction of the carbon input plus the starch-derived carbon, and is drained by nutrient-uptake, transport, respiration, and phloem-loading costs, and by net exchange with the vacuole:

$$\frac{ds}{dt} = (1 - k_s)\,\Phi_c + D_a - C_{\text{uptake}} - C_{\text{transport}} - C_{\text{resp}} - C_{\text{load}} - k_{\text{in}}\,s + k_{\text{out}}\,s_v, \tag{30}$$

$$\frac{ds_v}{dt} = k_{\text{in}}\,s - k_{\text{out}}\,s_v. \tag{31}$$

The four sink terms in Eq. (30) were not defined in the original formulation and are given here in full. Respiration is first-order in the cytosolic pool, and phloem loading is proportional to a night-efficiency factor E_{night} that represents the fraction of mobile sucrose usable for export under the current light and partitioning state:

$$C_{\text{resp}} = \phi_r\,s, \tag{32}$$

$$C_{\text{load}} = \phi_{\text{load}}\,E_{\text{night}}, \tag{33}$$

$$E_{\text{night}} = \min(s, s_{\max})\left[\lambda_g + (1 - \lambda_g)\big(1 - L + L\,(1 - k_s)\big)\right], \tag{34}$$

$$C_{\text{uptake}} = c_N\,U_N + c_P\,U_P, \tag{35}$$

$$C_{\text{transport}} = c_{\text{tr}}\left(\phi_r\,s + \phi_{\text{load}}\,E_{\text{night}}\right) = c_{\text{tr}}\left(C_{\text{resp}} + C_{\text{load}}\right), \tag{36}$$

where ϕ_r is the respiration frequency, ϕ_{load} the sucrose-loading frequency, $\lambda_g \in (0, 1]$ the floor of the night-efficiency factor, c_N, c_P the sucrose cost per unit nitrogen and phosphorus uptake (Eqs. (40)-(41)), and c_{tr} a dimensionless transport surcharge. Two features are worth making explicit. The bracket in Eq. (34) equals unity in darkness ($L = 0$) and $\lambda_g + (1 - \lambda_g)(1 - k_s)$ in the light, so export efficiency falls as a larger fraction of new assimilate is diverted to starch. And because $C_{\text{transport}}$ is proportional to $C_{\text{resp}} + C_{\text{load}}$ rather than being an independent flux, it is a surcharge on those two costs and not a third pathway; the combined drain is $(1 + c_{\text{tr}})(C_{\text{resp}} + C_{\text{load}})$.

Equations (30)-(31) represent the mobile sucrose as a two-compartment (cytosol-vacuole) system with linear exchange rate constants $k_{\text{in}}, k_{\text{out}}$ (Lohaus, Hussmann, Pennewiss, Schneider, Zhu and Sattelmacher, 2000; Nadwodnik and Lohaus, 2008). The exchange conserves total sucrose. At equilibrium ($k_{\text{in}}\,s = k_{\text{out}}\,s_v$) the two transfer terms cancel and the cytosolic steady state coincides with that of a single-pool description, but under diel forcing the pools are never at equilibrium: the vacuole fills while the cytosolic concentration is high during the light period and returns sucrose as the cytosolic pool is drawn down at night, so its dynamic role is to damp the amplitude of the cytosolic oscillation and to extend the period over which export and uptake costs can be met. The compartment therefore alters the trajectory while leaving the integrated carbon balance unchanged, which is the behaviour reported for vacuolar sugar storage in *Arabidopsis* (Lohaus et al., 2000; Nadwodnik and Lohaus, 2008).

The partition coefficient k_s is not a constant but the first of six dynamic states whose governing equations complete the system. Its balance equation drives the day/night division of new assimilate between starch and sucrose according to the cytosolic sucrose status. Writing $\sigma^- = s_{\min}/(s_{\min} + s)$ for a scarcity signal and $\sigma^+ = s/(s + s_{\max})$ for a sufficiency signal,

$$\frac{dk_s}{dt} = L\left[-k_s\,\lambda_{sdr}\,\sigma^- + (1 - k_s)\,\lambda_{sdi}\,\sigma^+\right] + (1 - L)\,(1 - k_s)\,\lambda_{sni}\,\sigma^-, \tag{37}$$

with rate coefficients λ_{sdr} (daytime decrease), λ_{sdi} (daytime increase) and λ_{sni} (nocturnal increase). In the light, storage is suppressed when sucrose is scarce ($\sigma^- \to 1$) and promoted when it is abundant ($\sigma^+ \to 1$); in darkness the coefficient relaxes towards $k_s = 1$ at a rate gated by scarcity, which pre-positions the partitioning state for the following photoperiod. The factors $(1 - k_s)$ and k_s confine the state to $[0, 1]$ without an external clip. Of the three rate coefficients, λ_{sni} was excluded from the sensitivity analysis (Appendix A) because its screening effect was indistinguishable from zero; the exclusion is a documented result, not an omission.

3.9. Submodel S5 (continued): nutrient uptake and costs

Potential nutrient uptake follows saturating Michaelis-Menten kinetics in the soil concentration (Epstein and Hagen, 1952; Tinker and Nye, 2000), weighted by the plant's investment state and by the fraction of biomass allocated to roots. For nitrogen and phosphorus,

$$U_N^{\mathrm{pot}} = I_{N,\max}\,\frac{N_s}{N_s + K_N}\,\alpha_N\,\frac{B_r}{B_l + B_r}, \tag{38}$$

$$U_P^{\mathrm{pot}} = I_{P,\max}\,\frac{P_s}{P_s + K_P}\,\alpha_P\,\frac{B_r}{B_l + B_r}, \tag{39}$$

where N_s, P_s are soil nutrient concentrations, $I_{N,\max}, I_{P,\max}$ maximal uptake rates, K_N, K_P half-saturation constants, and α_N, α_P the investment (affinity) states. Because active uptake is energetically coupled to the carbon economy, realised uptake is limited by sucrose availability through a saturating factor,

$$U_N = U_N^{\mathrm{pot}}\,\frac{s}{s + \tau\, c_N\, U_N^{\mathrm{pot}} + \epsilon}, \tag{40}$$

$$U_P = U_P^{\mathrm{pot}}\,\frac{s}{s + \tau\, c_P\, U_P^{\mathrm{pot}} + \epsilon}, \tag{41}$$

where c_N, c_P are the sucrose costs per unit nutrient acquired (μmol sucrose (μmol N)$^{-1}$ and μmol sucrose (μmol P)$^{-1}$; numerically 0.68 and 0.308), τ is a reference time scale, and ϵ is a small regularisation constant preventing division by zero. The consumed sucrose appears as the uptake cost $C_{\mathrm{uptake}} = c_N U_N + c_P U_P$ in Eq. (30), so that nutrient acquisition and carbon supply are explicitly co-limiting (Smith and Stitt, 2007). Two distinct saturations act in sequence and should not be conflated: the Michaelis-Menten term in Eqs. (38)-(39) saturates uptake in the external soil concentration, while the term in Eqs. (40)-(41) saturates it in the internal energy supply. A plant in nutrient-rich soil with depleted sucrose is limited by the second and not the first.

The factor τ requires comment, because the equations as printed in the original submission omitted it and were dimensionally inconsistent as a result. The state s is a pool (μmol gFW^{-1}) while U_N^{pot} is a rate (μmol gFW^{-1} h^{-1}), so s and $c_N U_N^{\mathrm{pot}}$ cannot be added: the product $c_N U_N^{\mathrm{pot}}$ is a sucrose consumption *rate*, and converting it to the pool-commensurable quantity that competes with s requires multiplication by a time. In the implementation that time is present, as an explicit unit factor: the denominator is formed as `sucrose + c_N * pot_uptake * 1.0`. Writing it as $\tau = 1$ h makes the equation dimensionally homogeneous and leaves every computed value unchanged, which is why no result in this paper is affected. We state it explicitly rather than absorbing it into c_N, because absorbing it would give c_N units of hours in Eqs. (40)-(41) and dimensionless mole-ratio units in C_{uptake} — the same symbol with two dimensions, which is what produced the inconsistency in the first place. The interpretation of τ is that the limitation compares the sucrose pool against the cost of one hour of uptake at the potential rate; its value is not independently identified by the calibration and is not treated as a free parameter.

The nutrient contents are then the balance of realised uptake against the nitrogen and phosphorus committed to maintenance, export and the photosynthetic apparatus:

$$\frac{dN}{dt} = U_N - \Big[\phi_r\,(s + a) + \phi_{\mathrm{load}}\,E_{\mathrm{night}}\Big]\,c_N^{\mathrm{as}}\,\frac{B_l}{B_l + B_r + \epsilon} - \frac{\Phi_c}{\Phi_{c,\max}}\,N_{\mathrm{min}}^{\mathrm{ph}}\,\kappa_n, \tag{42}$$

$$\frac{dP}{dt} = U_P - \Big[\phi_r\,(s + a) + \phi_{\mathrm{load}}\,E_{\mathrm{night}}\Big]\,c_P^{\mathrm{as}}\,\frac{B_l}{B_l + B_r + \epsilon} - \frac{\Phi_c}{\Phi_{c,\max}}\,P_{\mathrm{min}}^{\mathrm{ph}}\,\kappa_n, \tag{43}$$

where c_N^{as} and $c_P^{\mathrm{as}} = c_N^{\mathrm{as}}/q^*$ are the nutrient costs of assimilation, related through the optimal N:P stoichiometric ratio q^*; $N_{\mathrm{min}}^{\mathrm{ph}}, P_{\mathrm{min}}^{\mathrm{ph}}$ are the nutrient requirements of the photosynthetic machinery; $\Phi_{c,\max}$ is the maximum carbon input; and κ_n is a conversion coefficient. The first sink term scales with the *total* carbohydrate turnover, sucrose plus starch, because both pools are respired, and is weighted by the leaf fraction of total biomass, since the assimilatory

demand is a shoot property. The second sink term makes the nutrient cost of maintaining photosynthetic capacity proportional to the capacity actually being used.

The investment states $\alpha_N, \alpha_P \in [0, 1]$ govern how strongly the root system is committed to acquiring each nutrient. Both follow the same increase-minus-decrease form, in which the increase is driven by unmet demand and the decrease by demand already satisfied. For nitrogen, with λ_k an adjustment-rate coefficient,

$$\frac{d\alpha_N}{dt} = (1-\alpha_N)\underbrace{\left[\left(1-\frac{U_N}{U_N+C_N+\epsilon}\right)\frac{N_{\max}}{N+N_{\max}}+\frac{\Phi_c}{\Phi_{c,\max}}\right]}_{\text{unmet demand}} \tag{44}$$

$$-\underbrace{\alpha_N\,\lambda_k\,\frac{N-q^*P}{N+q^*P+\epsilon}}_{\text{stoichiometric imbalance}} \underbrace{-\alpha_N\left[\frac{N}{N+N_{\min}}+\frac{c_N U_N}{c_N U_N+\Phi_c(1-k_s)+D_a+\epsilon}\right]}_{\text{satisfied demand and carbon burden}}, \tag{45}$$

and identically for phosphorus under $N \leftrightarrow P$ with the stoichiometric term reversed in sign, since an excess of nitrogen relative to q^*P is a phosphorus deficit:

$$\frac{d\alpha_P}{dt} = (1-\alpha_P)\left[\left(1-\frac{U_P}{U_P+C_P+\epsilon}\right)\frac{P_{\max}}{P+P_{\max}}+\frac{\Phi_c}{\Phi_{c,\max}}\right] \tag{46}$$

$$-\alpha_P\,\lambda_k\,\frac{q^*P-N}{N+q^*P+\epsilon}-\alpha_P\left[\frac{P}{P+P_{\min}}+\frac{c_P U_P}{c_P U_P+\Phi_c(1-k_s)+D_a+\epsilon}\right]. \tag{47}$$

Here C_N, C_P are the total nutrient costs incurred per unit time (the two sink terms of Eqs. (42)-(43)), and $N_{\min}, N_{\max}, P_{\min}, P_{\max}$ are the minimum and maximum nutrient contents, the minima being derived from the maintenance and photosynthetic requirements at saturating carbohydrate content and the maxima set at four times the minima. Three signals therefore compete in each equation. Uptake that falls short of cost, combined with a content below $N_{\max}$, argues for greater investment; so does a high carbon input, which raises nutrient demand. A content comfortably above $N_{\min}$, or an uptake cost that is large relative to the sucrose currently available for growth ($\Phi_c(1-k_s)+D_a$), argues for less. The stoichiometric term redirects investment toward whichever nutrient is scarce relative to q^*, and is the mechanism by which the two nutrient submodels interact rather than operating independently.

3.10. Submodel S6: biomass growth and allocation

Leaf and root biomass grow in proportion to the sucrose delivered for structural synthesis, partitioned by the allocation state ρ_r, and are reduced by first-order mortality and a quadratic self-competition term:

$$\frac{dB_l}{dt} = G_l(s,\rho_r,\theta) - \mu_l B_l - \gamma_l B_l^2, \tag{48}$$

$$\frac{dB_r}{dt} = G_r(s,\rho_r,\theta) - \mu_r B_r - \gamma_r B_r^2, \tag{49}$$

with the growth terms given in full by

$$G_l = \lambda_{sb}\,(1-\rho_r)\,\phi_{\text{load}}\,E_{\text{night}}\,B_l, \tag{50}$$

$$G_r = \lambda_{sb}\,\rho_r\,\phi_{\text{load}}\,E_{\text{night}}\,B_l, \tag{51}$$

where λ_{sb} is the growth coefficient, ϕ_{load} the phloem-loading frequency, and

$$E_{\text{night}} = \min(s, s_{\max})\Big[\lambda_g + (1-\lambda_g)\big(1-L+L\,(1-k_s)\big)\Big] \tag{52}$$

the sucrose actually available for structural synthesis, with λ_g the baseline fraction usable irrespective of the diel state. Three features of Eqs. (50)-(51) deserve comment, none of which is visible in a proportionality statement.

First, both fluxes scale with *leaf* biomass B_l, not with the biomass of the receiving organ: root growth is driven by the shoot's sucrose export, so G_r depends on B_r only through the loss terms of Eq. (49). Second, the same E_{night} appears here and in the phloem-loading cost C_{load} of Eq. (30), which is what couples growth to the sucrose it consumes; the bracket in Eq. (52) reduces to unity at night ($L = 0$) and to $\lambda_g + (1 - \lambda_g)(1 - k_s)$ by day, so daytime partitioning to starch withholds sucrose from growth in proportion to k_s. Third, $\min(s, s_{\max})$ saturates the supply, so growth is sucrose-limited only below $s_{\max}$ and thereafter set by λ_{sb}, ρ_r and B_l alone. The units of λ_{sb} follow from Eqs. (50)-(51) and are gFW (μmol sucrose)$^{-1}$ h^{-1} per unit ϕ_{load}; it is an empirical conversion from a carbohydrate flux to fresh-weight accumulation and not a carbon stoichiometry, which is why the biomass states are excluded from the carbon-closure budget of Section 3.11.

Here μ_l, μ_r are mortality rates and γ_l, γ_r self-competition coefficients. Calling the quadratic term "density-dependent" overstates its role in the simulations reported here, and the magnitudes should be seen. With $\gamma_l = \gamma_r = \mu_l = \mu_r = 5 \times 10^{-7}$ (in gFW^{-1} h^{-1} and h^{-1} respectively) and a terminal leaf biomass of 13.4 gFW, the self-competition loss is $\gamma_l B_l^2 = 9.0 \times 10^{-5}$ gFW h^{-1} against a growth flux $G_l + G_r = 3.1 \times 10^{-1}$ gFW h^{-1}: smaller by a factor of 3.5×10^3. The mortality loss is smaller still, 6.7×10^{-6} gFW h^{-1}. Over the horizons simulated here neither loss term shapes the trajectory: the biomass dynamics are set by G_l and G_r, and the quadratic term is best understood as a guarantee that the system cannot grow without bound rather than as an active mechanism. It becomes comparable to growth only at $B_l \sim \lambda_{sb}\phi_{\text{load}}E_{\text{night}}/\gamma_l \approx 4.6 \times 10^4$ gFW, five thousand times the biomass the 30-day runs reach. We say so explicitly because a reader who took "density-dependent self-competition" at face value would expect the saturating trajectory it usually implies, and would not find it in Figure 5.

The allocation state ρ_r is the last of the six dynamic states and completes the system. It increases with the nutrient investment the plant is currently making — which is where a larger root would pay — and decreases with nutrient contents that are already adequate and with cytosolic sucrose scarcity, which makes root construction unaffordable:

$$\frac{d\rho_r}{dt} = (1 - \rho_r)\Big[\alpha_N\,\varsigma + (1 - \varsigma)\,\alpha_P\Big] - \rho_r\left[\frac{N\,\varsigma}{N + N_{\min}} + \frac{P\,(1 - \varsigma)}{P + P_{\min}} + \frac{s_{\min}}{s_{\min} + s}\right], \tag{53}$$

where the stoichiometric signal

$$\varsigma = \frac{q^*}{q^* + N/(P + \epsilon)} \in (0, 1) \tag{54}$$

weights the nitrogen and phosphorus channels by which nutrient is currently limiting: $\varsigma \to 1$ when the internal N:P ratio falls below q^* (nitrogen limiting), and $\varsigma \to 0$ when it rises above it (phosphorus limiting). The same signal appears in Eqs. (45)-(47), so a single algebraic quantity governs how the nutrient investment and allocation states respond together to an imbalance. As with k_s and α_N, α_P, the factors $(1 - \rho_r)$ and ρ_r keep the state in $[0, 1]$ without clipping.

Equations (37), (42)-(47) and (53) shift investment toward the resource (carbon or nutrients) that most limits growth, in the spirit of functional-balance allocation theory (Thornley, 1972; Thornley and Johnson, 1990). This closes the coupling: carbon gain sets sucrose, sucrose supports nutrient uptake and growth, and the resulting nutrient status feeds back onto allocation. It does *not* feed back onto photosynthetic capacity: in the production Farquhar-Jarvis path $V_{c\max}$ and $J_{\max}$ are fixed parameters and the nutrient- and starch-saturation factors enter only the basic carbon-input sub-model M0 of Table 15; leaf biomass reaches the gas-exchange layer solely through the viability gate $B_l \geq B_{l,\min}$ (Fig. 1). Together with Eqs. (28), (30), (31), (48) and (49), they constitute the complete eleven-equation right-hand side of Eq. (1); total biomass $B_l + B_r$ is a derived quantity and not a twelfth state.

3.11. Computational model and simulation algorithm

The mathematical model of Sections 3.2-3.10 constitutes an initial-value problem for the slow state vector $X(t)$ (Eq. 1) in which the right-hand side embeds the fast algebraic gas-exchange subsystem S2-S4. The numerical model advances this system by an adaptive explicit Runge–Kutta method with local error control a Dormand–Prince `ODE45` embedded pair (Hairer, Nørsett and Wanner, 1993), not by a fixed-step scheme. The distinction matters for how the

fast-slow structure is realised. The gas-exchange subsystem is not solved once per accepted step: it is embedded in the derivative evaluation, so it is re-solved at every stage of every trial step, including the stages of steps the error controller subsequently rejects. Because the operating point is obtained in closed form (Eq. (22)) this is inexpensive and, more importantly, exact — there is no inner convergence tolerance to interact with the outer error control, and the derivative is a deterministic function of (X, t) as the embedded-pair error estimate assumes. The fast subsystem is therefore better described as an algebraic constitutive layer inside $f(\cdot)$ than as a nested iterative solve (Cellier, 1991).

The step size is controlled by the integrator and is not a user-set quantity. The configuration parameter `dt_hours` does not set the integration step: it supplies the initial step h_{init} offered to the controller, and it sets the interval on which an observer samples the trajectory for output. The step sequence actually taken is chosen by the local error estimate against the tolerance `solver_tolerance`, and it varies over the diel cycle, contracting at the light-dark transitions and expanding through the night. Reported quantities are consequently defined on the observer's output grid while the accuracy of the trajectory is governed by the tolerance; Section 4.4 separates the two and shows that only the tolerance behaves as a convergence parameter.

The overall step is given in Algorithm 2.

Algorithm 2 Derivative evaluation $f(X, t)$ and one accepted integration step. Lines 1–13 constitute the right-hand side and are executed once per Runge–Kutta stage, i.e. several times per trial step and also for trial steps that are rejected. Lines 14–17 are the integrator's step logic. The gas-exchange operating point is obtained in closed form, so no inner iteration appears; the iterative alternative of Algorithm 1 substitutes for line 8 only when explicitly selected for the verification study of Section 4.3.

Require: state $X(t)$, parameters $\boldsymbol{\theta}$, current step size h, tolerance τ
Ensure: accepted state $X(t + h_{\text{acc}})$ and next step size h'
1: **procedure** $f(X, t)$ {one derivative evaluation}
2: $h_d \leftarrow t \bmod 24$ {hour of day}
3: Evaluate photoperiod switch $L(h_d)$ and radiation $PAR(h_d)$ from Eqs. (4)-(5)
4: Read environmental forcing T, C_a, RH, ψ_w
5: Compute VPD from T, RH (Eqs. (12)-(14))
6: **if** $PAR > 0$ **and** $B_l \geq B_{l,\min}$ **then**
7: Compute J from PAR (Eq. (19)); evaluate Jarvis modifiers $f_{PAR}, f_{VPD}, f_T, f_{CO_2}, f_w$ and form g_s (Eq. (6))
8: $A_n \leftarrow$ larger admissible root of Eq. (22), per branch, taking $\min(A_c, A_j)$; if $g_c \leq 0$ apply Eq. (23)
9: $\Phi_c \leftarrow \max\big(0,\ A_n\big)$ scaled by Eq. (24) {$A_n < 0$ is not carried into the pools}
10: **else**
11: $\Phi_c \leftarrow 0$
12: **end if**
13: Compute algebraic quantities: starch degradation D_a (using the dusk-latched reserve $a^{\dagger}$), night efficiency E_{night}, sucrose costs, nutrient uptakes U_N, U_P, allocation signals
14: Assemble and **return** dX/dt from Eqs. (28)-(49)
15: **end procedure**
16: Form the seven Dormand–Prince stages by calling f at $(t + c_i h,\ X + h \sum_j a_{ij} k_j)$
17: Form the fifth- and fourth-order solutions and the local error estimate e
18: **if** $e \leq \tau$ **then**
19: Accept: $X \leftarrow X + h \sum_i b_i k_i$, $t \leftarrow t + h$; notify observers, which record output whenever t crosses the next sampling instant
20: **end if**
21: $h' \leftarrow h\ \min\big(f_{\max},\ \max\big(f_{\min},\ \sigma\,(\tau/e)^{1/5}\big)\big)$ {retry the step from t if rejected}

Three differences from the algorithm as given in the original submission should be noted, since they change what the reader is entitled to conclude from it. The daylight gate is $PAR > 0$ together with the leaf-biomass viability condition $B_l \geq B_{l,\min}$; the $PAR \geq 20$ μmol m^{-2} s^{-1} threshold that appeared in the submitted version has been removed from the implementation, because it made the carbon input discontinuous in a state the adaptive controller must integrate across (Section 3.5). The step is not a fixed Δt but an error-controlled trial step that may be rejected and retried, so the number of derivative evaluations per unit simulated time is a property of the trajectory rather than of the configuration.

And the output the reader sees is produced by an observer on a separate sampling grid, not by the integrator's own step sequence.

Two properties of the numerical model support its verification, in the sense of confirming that the simulator faithfully realises the mathematical model (Zeigler et al., 2000). First, the fast subsystem has a closed-form solution, Eq. (22), so the operating point is exact and no convergence question arises for the inner problem. This matters because the fixed-point alternative is *not* unconditionally convergent: the supply map $C_i^{(k+1)} = \max(0,\, C_a - A_n(C_i^{(k)})/g_c)$ has local slope proportional to g_c^{-1}, so it is contractive only while conductance is large enough, and it loses that property when the Jarvis modifiers drive g_s towards zero. Section 4.3 quantifies where the boundary lies and what the iteration returns beyond it. Second, the model admits conservation checks that are invariant to the discretisation and can be monitored at run time: the cytosol-vacuole exchange conserves total sucrose (Section 3.8), and over a closed day-night cycle the integrated carbon input $\int \Phi_c \, dt$ must balance the sum of respiration, storage change, phloem-loading and nutrient-uptake costs within the carbohydrate subsystem $\{a, s, s_v\}$. The biomass states are deliberately outside this budget: B_l and B_r are fresh weights governed by an empirical rate law whose coefficient λ_{sb} carries units of gFW^{-1} and is not a carbon-to-fresh-weight stoichiometry, so including them would compare quantities that are not commensurable. Residuals of the carbohydrate balance quantify the numerical error and provide a model-independent test of the implementation, complementing the comparison of simulated day-night trajectories against experimental data reported in Section 6.

4. Numerical Implementation

The model was implemented as a modular simulation engine written in C (ISO C99) and built with CMake. The implementation separates the computation of environmental forcing, gas-exchange variables, carbohydrate dynamics, nutrient uptake, and biomass growth into distinct translation units, so that each physiological component can be inspected and tested independently while the coupled flow from environmental forcing to growth response is preserved. This section describes how the state equations are advanced numerically and which measures were taken to make every reported result reproducible.

4.1. Simulation architecture and time integration

Each evaluation step follows the fast-slow structure of Section 3. Four computational blocks are executed in sequence. The first evaluates the photoperiod forcing and environmental inputs (PAR, temperature, ambient CO_2, relative humidity, vapour-pressure deficit and leaf water potential). The second computes the Farquhar-Jarvis gas-exchange layer, stomatal conductance, intercellular CO_2, Rubisco-limited and electron-transport-limited assimilation, and net assimilation in which the intercellular CO_2 concentration is obtained from the closed-form solution of Eq. (22). The iterative alternative (Algorithm 1; cap $k_{\max} = 1000$, tolerances $\varepsilon_{C_i} = \varepsilon_{A_n} = 10^{-4}$) is selectable at run time and is used only for the verification study of Section 4.3. The third block converts net assimilation into the model carbon-input term, and the fourth advances the slow states: starch, sucrose, nutrient pools, allocation variables and leaf and root biomass.

The eleven slow states are integrated by an adaptive explicit Runge-Kutta solver (a Dormand-Prince `ODE45` embedded pair) provided by a separate, reusable C ODE-solver library. The solver uses an absolute/relative error tolerance of 10^{-8}, and states are sampled for output at a fixed reporting interval of $\Delta t = 0.1$ h; the diel forcing $L(t)$ of Eq. (4) is evaluated at each stage from the continuous clock time. Two integration horizons appear in this paper, and which one applies depends on the driver rather than on the model. The validation, diel, ablation and two-dimensional sweep runners set a 30-day horizon in the driver itself; the convergence, carbon-balance, cross-validation and sensitivity drivers take the horizon from `config/default.config`, which specifies 35 days. Each reported result states its own horizon, and the choice is immaterial to the conclusions: a 35-day cross-check of the convergence study reproduces every conclusion, with RGR differing from the 30-day value only in the third significant figure (0.17140 against 0.17107 d^{-1} at 8 h). The day-night dynamics were in every case analysed after the transient had settled onto a repeating diel pattern. Identical solver settings (tolerance, reporting interval, horizon) were used for the 4, 6, 8 and 12 h photoperiods within each driver, so that differences in the outputs reflect photoperiod forcing and parameter variation rather than changes in numerical configuration. After each run the state trajectory was checked for numerical and physiological validity: non-negative sucrose, starch and biomass, absence of nonfinite samples, and zero photosynthetic carbon input during the night phase (Table 9).

4.2. Reproducibility and code availability

Every scenario is fully specified by a plain-text configuration file that lists the photoperiod length, the environmental forcing, the initial state and the complete parameter set; no parameter is set implicitly in code. The ablation experiments are driven by per-variant, per-photoperiod configuration files (`config/ablation/<variant>/<Nh>.config`) that inherit from a single base configuration and override only the calibrated per-photoperiod parameters (the growth rate λ_{sb} and the starch-synthesis rate λ_{sdi}), so that the basic, Farquhar and Farquhar-Jarvis variants differ only in the declared carbon-input pathway. Each reported table and figure is generated by a dedicated command-line runner (validation, ablation and sensitivity runners) from these configuration files, and the analysis and plotting scripts read the resulting CSV output, so that the path from raw simulation output to every figure is explicit.

The sensitivity analysis is likewise fully specified. Sobol' total-order indices are computed with the variance-based Saltelli estimator Saltelli, Annoni, Azzini, Campolongo, Ratto and Tarantola (2010) using 512 base samples and hence $512\,(d+2)$ model evaluations over the $d \leq 16$ analysed parameters. The two base matrices the estimator requires are drawn as two independent Latin-hypercube samples rather than from a split low-discrepancy sequence, for the reason set out in Section 5.5.1; the design used is recorded in every output file. Morris screening uses 30 trajectories at 6 levels. All stochastic sampling is seeded from a fixed value (seed 0), and the sample counts, seed and photoperiods are recorded as command-line arguments, so that the Sobol' and Morris results are reproducible. The simulation engine, the ODE-solver library, the configuration files and the analysis scripts are openly available in the project repository, allowing the complete validation, ablation and sensitivity workflow to be regenerated from source.

4.3. Numerical verification of the solver components

Verification asks whether the simulator solves the equations it claims to solve, and is logically prior to validation against data (Zeigler et al., 2000; Cellier, 1991). Because agreement with measurements at one operating point cannot establish correctness over a sampling domain, each numerical component was tested independently of any comparison with observations. The tests are implemented in a standalone driver (`tools/solver_verification.c`) that links the production model sources, so the code under test is the code that produces the results, and they write machine-readable output to `results/verification/`. Four tests are reported, and a separate convergence study of the discretisation itself follows in Section 4.4.

Residual test against the closed form. Over a PAR sweep spanning the full diel range, the operating point returned by each solver is substituted back into the coupled supply-demand system and the residual $\left|g_c(C_a - C_i) + R_d - A(C_i)\right|$ is recorded. This tests the solution, not the iteration count, and it is the sense in which Eq. (22) can be called exact.

Water-stress sweep. The leaf water potential ψ_w is stepped across the sampled range $[-1.5, -0.1]$ MPa at fixed forcing, and for each value the fraction of daylight solves in which the fixed-point iteration reaches its cap without satisfying Eq. (21) is recorded, together with the daily integrated carbon fixation under each solver. This locates the boundary of the region in which the iteration is contractive, and quantifies the error committed beyond it.

Parity test. At a forcing point inside the non-convergent region the iteration cap $k_{\max}$ is varied over consecutive integers and the returned A_n and C_i are recorded. A convergent iteration is insensitive to the cap; a periodic orbit is not, and the parity of $k_{\max}$ then determines which point of the cycle is returned. This test distinguishes slow convergence from non-convergence, which a residual or tolerance check alone cannot do.

Monte Carlo audit over the sampling domain. Because the sensitivity analysis evaluates the model at points drawn from the full parameter ranges of Appendix A, and not only at the calibrated operating point, 2×10^4 parameter vectors were drawn from those ranges and, for each, the fixed-point and analytic solvers compared. Reported are the fraction of non-convergent solves, the fraction returning $A_n < 0$ under light, the fraction returning $C_i > 2C_a$, and the distribution of $|A_n^{\mathrm{fp}} - A_n^{\mathrm{analytic}}|$. The third quantity requires a word of qualification. The only hard admissibility constraint on the operating point is $C_i \geq 0$; $C_i > C_a$ is not inadmissible but the physically correct supply solution wherever $A_n < 0$, as noted after Eq. (22). The $2C_a$ threshold is therefore reported as a *diagnostic* for iterations that have wandered far outside the range in which the coupled problem has a meaningful solution — at ψ_w values where the map is not contractive the iterates leave the physical range entirely — and not as a violation of a bound the model imposes. It is used here because it separates divergent from convergent behaviour cleanly on this domain, not because $2C_a$ has physiological

meaning. This audit measures the exposure of a published sensitivity study to a solver defect, which is the quantity that matters and is not visible from the calibrated run.

Two further checks are reported alongside these. The temperature modifier of Eq. (9) is evaluated on a 1°C grid from T_1 to T_h and tested for the two properties its normalisation is supposed to guarantee, $f_T(T_0) = 1$ and $\arg\max f_T = T_0$; the same driver records the conductance ratio between the corrected and the previous parameterisation at each temperature, which is the quantity that determines whether previously reported results are affected. Finally, carbon conservation is verified by integrating ten flux accumulators alongside the eleven state equations on the augmented system, using the same weights and the same step for both (`tools/carbon_balance.c`). This test uses a fixed-step classical `RK4` scheme rather than the adaptive production integrator, deliberately: sharing a fixed step between states and accumulators is what makes the residual a pure bookkeeping check, whereas an adaptive controller stepping on an error estimate of the states alone would leave a quadrature error in the accumulators that is indistinguishable from a missing flux. The model equations integrated are the production ones. Because the accumulators share the integrator, the closure residual is bounded by integrator arithmetic rather than by an independent quadrature, and any non-trivial residual indicates a missing or double-counted flux rather than discretisation error. The budget is closed over the carbohydrate subsystem $\{a, s, s_u\}$ for each photoperiod, and the fresh-weight biomass states are excluded for the dimensional reason given in Section 4; the size of the excluded term is quantified in Section 6 rather than left implicit.

4.4. Convergence study: tolerance, output interval and integrator

A reported simulation result is only meaningful if it is insensitive to the discretisation that produced it. The parameters of the numerical procedure were therefore varied independently around their production values, for all four photoperiods over the full 35-day horizon, and the effect on every reported quantity recorded (`tools/convergence_study.c`, output in `results/convergence/`). Identifying *which* parameter controls the discretisation error requires care under an adaptive integrator, and the original submission got it wrong; the study was consequently redesigned and rerun, and this section reports the corrected result.

Solver tolerance. Under the production configuration the eleven slow states are advanced by the adaptive `ODE45` backend, for which the configuration entry `dt_hours` is not the integration step: it is passed as the initial trial step, which the step-size controller immediately overrides, and as the interval at which the trajectory is sampled for output. The quantity that controls the discretisation error is therefore the solver tolerance, and it is the tolerance that was refined. With Δt held at its production value of 0.1 h, the tolerance was tightened from 10^{-4} to 10^{-12}. Relative to the tightest setting, RGR at the production tolerance 10^{-8} differs by 1.7×10^{-7} to 3.6×10^{-7} across the four photoperiods, end-of-day starch by at most 7.6×10^{-7}, and total biomass by at most 1.6×10^{-6}; the largest deviation in total sucrose is 1.0×10^{-5}. Every reported digit is therefore resolved at the production tolerance, sucrose included. The original submission restricted sucrose agreement to three significant figures on convergence grounds; that restriction rested on a 1.3×10^{-2} cytosolic-sucrose deviation which belongs to the fixed-step verification sweep and not to the production configuration, so it is not forced by the discretisation. We nevertheless retain three significant figures for sucrose throughout, on the independent ground that its absolute level is set by the saturated calibration of Section 6 rather than predicted; the convergence result removes one reason for the caution but not the operative one. Over the full swept range the tolerance moves RGR by 1.0×10^{-3} to 1.9×10^{-3}, so the coarsest setting would not be adequate. The observed exponent of the tolerance response, estimated by Richardson extrapolation on successive triples and referred to the actual refinement ratio of 100 between levels, has a median of 0.92 for RGR and 0.97 for end-of-day starch, with no negative estimates in 60 triples. This exponent is a tolerance response, not a step-size order of convergence: it has no formal target value, and in particular it must not be compared against the formal order of the embedded pair. It is reported because it establishes that the error falls monotonically and predictably as the tolerance is tightened.

Output interval. Because `dt_hours` sets both the initial step and the output interval, the two effects were separated by a second sweep in which the tolerance was held at the production 10^{-8} and Δt was varied from 0.4 h to 0.00625 h. Across that entire range RGR moves by at most 1.0×10^{-7} in relative terms at any photoperiod — three to five orders of magnitude less than the tolerance moves it. At the production point the ratio of the span attributable to the tolerance to the span attributable to Δt is 2.1×10^4 to 8.2×10^4 for RGR, and the tolerance is the dominant source for every one of the twenty (photoperiod, metric) combinations examined. The reported quantities are thus effectively independent of the reporting interval, and the accuracy statement for the published runs rests on the tolerance alone. A Richardson exponent computed on this sweep has a median of 0.46 with 40% of estimates negative, which is the expected signature

of output-grid noise at the 10^{-7} level rather than of a convergent refinement; it is reported here only to make clear that it is *not* an order of convergence and should not be read as one.

Fixed-step verification of the integrator. Fixed-step classical `RK4` was refined separately, as a verification of the integrator rather than as a characterisation of the production configuration. In that sweep Δt genuinely is the integration step, and halving it from 0.4 h to 0.00625 h reduces the deviation in RGR monotonically; at $\Delta t = 0.1$ h the deviation from the finest step is 1.9×10^{-4} to 4.3×10^{-4}. The observed order is close to *one* (median 0.90 for RGR, 0.99 for end-of-day starch), not the fourth order of the underlying scheme. This is expected and worth stating explicitly rather than presenting a formal order the model does not achieve: the forcing and several constitutive functions are only piecewise smooth in time — the light schedule has a derivative discontinuity at sunset, the starch degradation law switches branch at the day/night transition, the night-efficiency term clamps, and the Jarvis light modifier is discontinuous at q (Section 3.4) — so the global order is set by the kinks, not by the stage weights. Sixteen per cent of the estimates on this sweep are negative, indicating non-monotone pre-asymptotic behaviour at the coarsest steps. This sweep also carries the carbon-balance bookkeeping, since the ten flux accumulators can be advanced by the same fixed-step weights as the states; the relative closure residual stays between 6.6×10^{-16} and 8.4×10^{-14} across the whole sweep, at the arithmetic floor.

Choice of backend. Comparing the two backends at production settings quantifies what the choice of integrator is worth: at $\Delta t = 0.1$ h and tolerance 10^{-8}, fixed-step `RK4` differs from `ODE45` by 1.1×10^{-4} to 1.2×10^{-3} in RGR and by up to 1.1×10^{-2} in final biomass. This deviation is larger than the residual tolerance error of the production configuration, which is why the tools that override to fixed-step `RK4` (the sensitivity, one-at-a-time and two-dimensional sweep runners) are reported as such rather than treated as interchangeable with the production path.

Gas-exchange closure tolerance. The convergence criterion of the algebraic closure solve was tightened from 10^{-3} to 10^{-10}. Between the production tolerance 10^{-4} and the tightest setting, RGR changes by 1.7×10^{-8} to 4.9×10^{-8} in relative terms, i.e. roughly an order of magnitude below the residual error of the ODE tolerance at its production value and some four orders of magnitude below the error the ODE tolerance commits over its swept range; the closure tolerance is therefore not a limiting approximation. The mean iteration count of the legacy fixed-point solver rises from 9.2 to 21.7 over the same range while the analytic solver is unaffected by construction, and at $\psi_w = -0.6$ MPa no solve fails to converge at any tolerance.

Closure solver. At the calibrated water potential the analytic and fixed-point solvers agree to 10^{-7}–10^{-8} in every reported metric with zero non-convergence, which is the quantitative basis for the claim that the validation, diel and ablation results are unaffected by the defect of Section 6.2. Repeating the comparison at $\psi_w = -1.4$ MPa shows what the defect costs where the map is not contractive: 96–97% of solves reach the iteration cap (mean 965 iterations of a 1000 cap), and simulated RGR collapses from 0.0404 to 0.00032 d^{-1} at 4 h, a relative deviation of 0.99.

Integrator. The three integrators available in the solver library were compared on a problem with a known solution, $\dot{y} = -y$, $y(0) = 1$ over $t \in [0, 1]$, at tolerances 10^{-4}, 10^{-6} and 10^{-8}. The production integrator (Dormand–Prince `ODE45`) converges as expected, with relative error falling from 2.5×10^{-6} to 3.9×10^{-9} as the tolerance is tightened; fixed-step `RK4` returns 9.1×10^{-7} at every tolerance, as it must. The library's `RKF78` implementation, however, returns a relative error of 1.026×10^{-1} at 10^{-4} and 1.025×10^{-1} at 10^{-6} — i.e. *independent of tolerance* — and does not terminate at all at 10^{-8}. The decay rate it implies, 0.902380 at 10^{-6} (0.902330 at 10^{-4}), matches $1 - 82/840 = 0.9023810$ to five figures — the signature of a weight vector whose entries do not sum to unity, i.e. an inconsistent method rather than an inaccurate one. We report this because it is a defect of a third-party component that a user of that library would not otherwise detect from tolerance-refinement alone; it does not affect any result in this paper, since all reported runs use `ODE45`, and `RKF78` is not used anywhere in the production configuration.

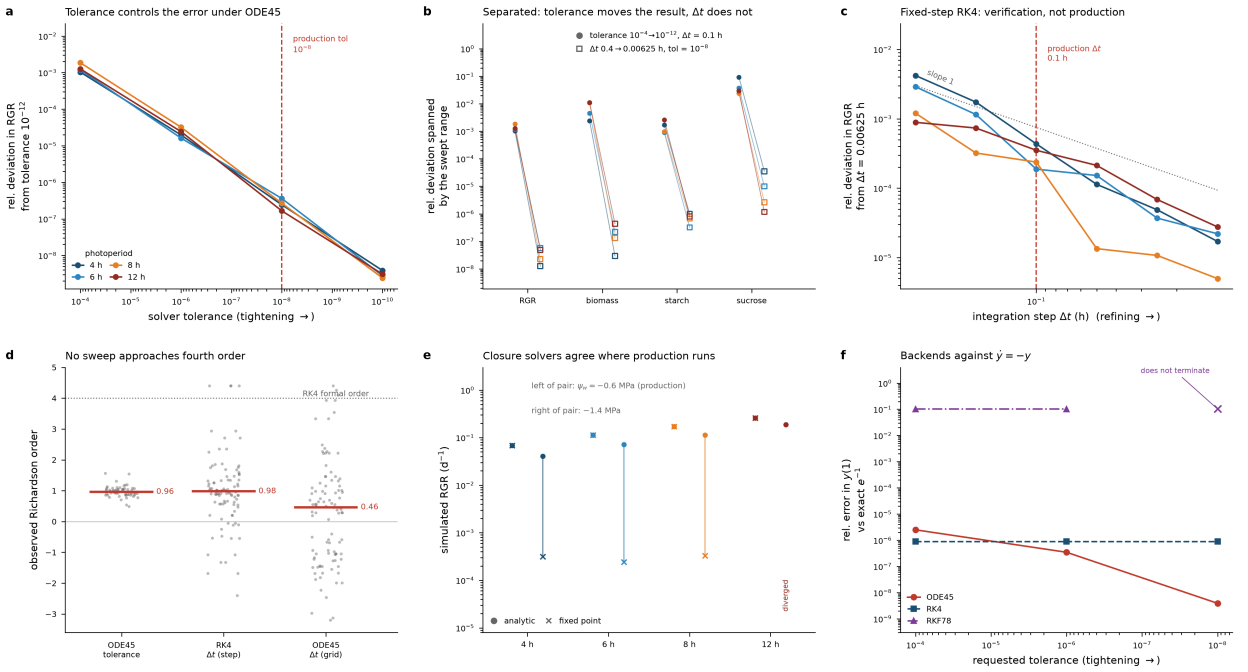


Figure 3: Convergence of the numerical procedure. (a) Relative deviation of RGR from the tightest tolerance (10^{-12}) against solver tolerance, per photoperiod, at the production reporting interval $\Delta t = 0.1$ h; the production tolerance 10^{-8} is marked. Under the adaptive `ODE45` backend used for all reported runs, the tolerance is the quantity that controls the discretisation error, so this is the production-relevant convergence test. (b) The two error sources separated: for each metric, the relative deviation spanned by the tolerance sweep at fixed Δt (filled) against that spanned by the Δt sweep at fixed tolerance (open). Because `dt_hours` is the initial step and the output interval rather than the integration step, the Δt span is an output-grid effect and is three to five orders of magnitude smaller. (c) Fixed-step `RK4` step refinement, shown as verification of the integrator and not as the production configuration; here Δt is the integration step and the error falls monotonically with a slope near one. (d) Observed Richardson exponent by sweep, with medians marked; the formal fourth order of the Runge–Kutta scheme is shown for reference and is approached by none of the three, since the forcing and several constitutive laws are only piecewise smooth in time. The `ODE45` Δt exponent is output-grid noise, not an order of convergence. (e) Analytic versus fixed-point closure at two leaf water potentials: indistinguishable at the calibrated $\psi_w = -0.6$ MPa with zero non-convergent solves, and a near-total collapse of simulated growth at -1.4 MPa, where 96–97% of solves reach the iteration cap; the 12 h fixed-point run there diverges outright and is marked as such rather than plotted as a small value. (f) Verification of the three library integrators against the exact solution of $\dot{y} = -y$: `ODE45` converges with tolerance, fixed-step `RK4` is tolerance-independent as expected, and `RKF78` returns a 10% error at both tolerances at which it terminates and does not terminate at 10^{-8}. All panels: 35-day integration, four photoperiods, the default configuration `config/default.config`.

5. Model Evaluation Protocol

The model was evaluated under controlled photoperiod scenarios using *Arabidopsis thaliana* targets drawn from published metabolite and growth measurements, principally the photoperiod series of Sulpice et al. (2014), which supplies the relative-growth-rate targets, together with the diel carbohydrate time courses of Annunziata, Apelt, Carillo, Krause, Feil, Mengin, Lauxmann, Köhl, Nikoloski, Stitt and Lunn (2017) and the photoperiod trends of Mengin, Pyl, Alexandre Moraes, Sulpice, Krohn, Encke and Stitt (2017). The evaluation was designed to test three complementary aspects of the proposed simulation workflow: the ability to reproduce photoperiod-dependent relative growth rate (RGR), the ability to generate physiologically plausible day-night carbohydrate dynamics, and the robustness of the coupled model with respect to parameter variation.

5.1. Photoperiod scenarios

Four photoperiod treatments were considered: 4 h, 6 h, 8 h, and 12 h of light per 24 h cycle (Table 4). These scenarios provide a controlled way to perturb the balance between daytime carbon assimilation and nighttime reserve mobilization. Each simulation was run for 30 days, and the reported carbohydrate dynamics were analysed after the transient response had reached a repeating diel pattern.

Two properties of this design must be stated explicitly, because both affect how the photoperiod results should be read.

First, photoperiod and daily photon dose are confounded. Holding $\mathrm{PAR}_{\max}$ fixed while lengthening the day increases DLI in proportion to day length, so the 12 h scenario receives three times the photon dose of the 4 h scenario (Table 4). A change in simulated RGR across scenarios therefore reflects the joint effect of day length and integrated irradiance, not day length alone. This matches the design of the reference experiments, in which irradiance was also held constant while photoperiod was varied, so the comparison to the observed RGR series is consistent; but it means that neither the observations nor the simulation isolate a pure photoperiod effect, and the ablation and sensitivity results in Sections 6 onward are interpreted accordingly. Mengin et al. (2017) separated the two factors experimentally at 6 and 12 h and found starch allocation to be sensitive to photoperiod but largely insensitive to irradiance, which supports attributing most of the observed response to day length; reproducing that factorial design in simulation is left to future work.

Second, the absolute irradiance level is higher than in the reference experiments. The reference photoperiod series was grown under approximately rectangular chamber lighting of order 150 $\mu\mathrm{mol\,m^{-2}\,s^{-1}}$, giving DLI values of roughly 2.2–6.5 $\mathrm{mol\,m^{-2}\,d^{-1}}$ across 4–12 h, whereas the half-sinusoidal schedule used here delivers 11.0–33.0 $\mathrm{mol\,m^{-2}\,d^{-1}}$ — a factor of 5.1 higher at matched day length (Table 5). The half-sinusoid was chosen to give continuous, differentiable forcing through dawn and dusk, which the adaptive integrator handles without step rejection at a light–dark discontinuity, and it is the natural profile for the outdoor conditions the model is ultimately intended for (Annunziata et al., 2017). The consequence is that the fitted carbon-input coefficient κ' (Eq. 25) absorbs this irradiance mismatch along with the leaf-to-rosette scaling: a model driven at five times the experimental photon dose must retain a correspondingly smaller fraction of the per-area assimilation to reproduce the observed growth. This is a recognised limitation of the present forcing specification rather than a physiological result, and it is one reason the per-photoperiod calibration in Section 6 cannot be read as parameter estimation.

5.2. RGR validation metric

The primary growth-level validation metric was relative growth rate:

$$RGR = \frac{\ln B(t_2) - \ln B(t_1)}{t_2 - t_1}, \tag{55}$$

where $B(t)$ denotes total plant biomass at time t. The simulated RGR was compared against the photoperiod-specific RGR targets reported for *Arabidopsis thaliana* rosettes by Sulpice et al. (2014) ($RGR_{\mathrm{ref}} = 0.068$, 0.114, 0.171 and 0.260 $\mathrm{d^{-1}}$ for the 4, 6, 8 and 12 h photoperiods, respectively). These four values are from Sulpice et al. (2014) and not, as the original submission stated, from Annunziata et al. (2017), whose photoperiod series does not report RGR at all four of these day lengths; the attribution is corrected here and the numerical targets are unchanged. The relative error was computed as:

$$E_{\mathrm{rel}} = 100\frac{\hat{RGR} - RGR_{\mathrm{ref}}}{RGR_{\mathrm{ref}}}, \tag{56}$$

where $\hat{RGR}$ is the simulated value and RGR_{ref} is the reference value.

The in-sample RGR comparison cannot be interpreted as a validation at all, and is reported as a calibration diagnostic. The calibration adjusts two coefficients per photoperiod — the growth coefficient λ_{sb} and the starch-synthesis rate λ_{sdi} — by nested bisection against exactly two observations per photoperiod, the published RGR and the published end-of-day starch content. The fitted model is therefore saturated: eight parameters, eight observations, zero degrees of freedom, and the residual that a bisection leaves behind is a convergence tolerance rather than a prediction error. To separate the two, the protocol adds an out-of-sample test in which a single photoperiod-independent parameter pair is fitted on the 8 and 12 h treatments and used to predict the 4 and 6 h treatments; the split is by photoperiod rather than at random, because interpolation within the photoperiod range is exactly the ability in question. Both regimes are

Table 4
Photoperiod validation scenarios. All non-photoperiod forcing is identical across scenarios and taken from `config/default.config`; only the light schedule changes. PAR follows a half-sinusoid $\mathrm{PAR}(t) = \mathrm{PAR}_{\max} \sin(\pi t/P)$ over the light phase, so with a photoperiod-independent $\mathrm{PAR}_{\max}$ the mean daytime irradiance is the same in every scenario ($2\,\mathrm{PAR}_{\max}/\pi = 764\ \mu\mathrm{mol\,m^{-2}\,s^{-1}}$) and the daily light integral (DLI) scales in proportion to day length. Photoperiod and photon dose are therefore deliberately confounded in this design; see the discussion of Table 5 below.

Sc.	Photoperiod	Dur.	$\mathrm{PAR}_{\max}$ μmol m^{-2} s^{-1}	DLI mol m^{-2} d^{-1}	Evaluation target
S1	4 h light / 20 h dark	30 d	1200	11.00	RGR, day–night carbohydrates
S2	6 h light / 18 h dark	30 d	1200	16.50	RGR, day–night carbohydrates
S3	8 h light / 16 h dark	30 d	1200	22.00	RGR, day–night carbohydrates
S4	12 h light / 12 h dark	30 d	1200	33.00	RGR, day–night carbohydrates

Shared forcing: $T = 25$ °C, RH = 60%, $C_a = 400$ μmol mol^{-1}, $\psi_w = -0.6$ MPa, SLA = 0.025 m^2 gFW^{-1}; initial state $B_l = B_r = 0.02$ gFW, sunrise at $t = 0$ h.

Table 5
Daily light integral of the simulated forcing compared with the approximate DLI of the reference growth-chamber experiments. Simulated values are the exact integral of the half-sinusoid, $\mathrm{DLI} = 2\,\mathrm{PAR}_{\max} P/\pi$ converted to mol m^{-2} d^{-1}; reference values assume rectangular lighting at 150 $\mu\mathrm{mol\,m^{-2}\,s^{-1}}$.

Photoperiod	Simulated DLI mol m^{-2} d^{-1}	Reference DLI mol m^{-2} d^{-1}	Ratio
4 h	11.00	2.16	5.1
6 h	16.50	3.24	5.1
8 h	22.00	4.32	5.1
12 h	33.00	6.48	5.1

reported in Section 6. The carbohydrate dynamics, ablation behaviour and sensitivity structure are used as evidence in their own right, not as a substitute for the missing degrees of freedom.

5.3. Carbohydrate validation metrics

Sucrose and starch dynamics were evaluated against the measured day-night carbohydrate profiles of Annunziata et al. (2017), which provide the primary reference for the shape and timing of the diel cycle at the 12 h photoperiod, supported by Mengin et al. (2017) at 6 and 12 h and by the photoperiod trends of Sulpice et al. (2014). No single published source supplies matched diel time courses at all four photoperiods studied here, so the comparison is against the 12 h reference curves for shape and against the photoperiod trend for the remaining day lengths. Because the available reference data were used primarily to evaluate the timing and shape of day-night dynamics, the comparison was performed using scale-adjusted error metrics. For each photoperiod, sucrose and starch were compared at matching reference time points.

Concretely, a single multiplicative scale α is fitted by least squares to each simulated trace before the error is evaluated, so that the metrics score the shape and timing of the diel cycle rather than the absolute level:

$$\alpha = \frac{\sum_{i=1}^{n} \hat{y}_i\, y_i}{\sum_{i=1}^{n} \hat{y}_i^{\,2}}, \qquad r_i = \alpha \hat{y}_i - y_i. \tag{57}$$

The following error metrics were then computed on the scaled residual r_i:

$$MAE = \frac{1}{n} \sum_{i=1}^{n} \left| r_i \right|, \tag{58}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n} r_i^2}, \tag{59}$$

$$nRMSE = \frac{RMSE}{|\bar{y}|}, \qquad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i. \tag{60}$$

Here, y_i denotes the reference value, $\hat{y}_i$ denotes the simulated value, n is the number of available reference points, and $\bar{y}$ is the mean reference level, which normalises the RMSE. MAE retains the units of the reference data (μmol gFW^{-1}) and is reported for the same fitted α, so the normalised and absolute errors describe one and the same fit. The three quantities are computed by `fit_scaled_nrmse` in `tools/validation_runner.c`. The predefined acceptance thresholds were $nRMSE < 0.50$ for sucrose and $nRMSE < 0.25$ for starch.

5.4. Ablation protocol

An ablation study was performed to isolate the contribution of the photosynthesis and gas-exchange components. Three model variants were compared using the same downstream growth framework (Table 6):

The three variants differ only in how the carbon input Φ_c is produced. Every downstream equation — the carbohydrate, nutrient and biomass dynamics of Sections 3.8 onward — is byte-identical across variants, and so is every parameter that appears in them. Explicitly:

- **M0 (Basic)** prescribes the carbon input as a saturating multiplicative form with no gas-exchange representation:

  $$\Phi_{c,M0} = L(t)\,\Lambda(a)\,\Phi_{c,\max}\,\min\left(\sigma_N,\,\sigma_P\right), \tag{61}$$

  where $L(t) \in \{0, 1\}$ is the light switch, $\Phi_{c,\max}$ a prescribed maximum rate (12.7 μmol gFW^{-1} h^{-1}), σ_N, σ_P the nutrient-saturation factors, and $\Lambda(a)$ the starch feedback on the photosynthetic rate,

  $$\Lambda(a) = \lambda_c + \left(1 - \lambda_c\right)\frac{a_{\max} - a}{a_{\max}}, \qquad a_{\max} = D_{a,\max}\,(24 - P), \tag{62}$$

  with λ_c the photoperiod-specific feedback coefficient of Table 2 and $D_{a,\max}$ the maximum starch degradation rate. This term is active in M0 only; in M1 and M2 the carbon input is set by the gas-exchange solution and Λ does not enter. Neither C_i, g_s, PAR intensity, temperature, humidity nor C_a appears: the carbon input responds to day length and internal state only, which is the formulation the coupled model is intended to replace.

- **M1 (Farquhar)** computes A_n from the full Farquhar–von Caemmerer–Berry biochemistry, but does not solve the supply–demand closure. Conductance is held at the fixed value $g_s = 0.1$ mol m^{-2} s^{-1} from `config/default.config`, so C_i still satisfies the same diffusion constraint $C_i = C_a - A_n/(g_s/1.6)$ and is obtained by the same closed-form solution, but with a g_s that does not respond to the environment:

  $$\Phi_{c,M1} = \kappa'\,3600\,\mathrm{SLA}\,A_n\left(C_i;\; g_s = \bar{g}_s\right), \qquad \bar{g}_s = 0.1. \tag{63}$$

  M1 therefore isolates the effect of biochemical demand alone.

- **M2 (Farquhar–Jarvis)** is the full model: g_s is computed by the Jarvis multiplicative response (Eq. (6)) from PAR, VPD, temperature, C_a and ψ_w, and the closure is solved for C_i and A_n simultaneously, giving Eq. (24).

The design point is that M0 → M1 adds biochemical demand, and M1 → M2 adds mechanistic stomatal supply. Because the fitted coefficients λ_{sb} and λ_{sdi} are held at the values obtained for M2 in every variant (see below), the comparison is not confounded by per-variant recalibration.

The purpose of this experiment was not only to compare reported RGR error, but also to test whether the proposed gas-exchange layer provides an environment-responsive carbon input that can be propagated into downstream carbohydrate and biomass dynamics.

Table 6
Ablation variants used to test the contribution of the gas-exchange layer.

Variant	Description
M0 - Basic	Tedone-style baseline with prescribed carbon input.
M1 - Farquhar	Baseline model with Farquhar C_3 biochemical photosynthesis and fixed conductance.
M2 - Farquhar-Jarvis	Farquhar photosynthesis coupled with Jarvis stomatal regulation.

5.5. Sensitivity analysis protocol

Sensitivity analysis was performed using three complementary approaches. First, one-at-a-time parameter sweeps were used to inspect the direction, saturation, and threshold behaviour of the RGR response. Second, Morris screening was used to identify parameters with large elementary effects and to distinguish approximately linear effects from nonlinear or interaction-driven effects. Third, Sobol' total-order indices were used to estimate the global contribution of each parameter to RGR variability, including interaction effects. The parameters examined were organised into functional groups spanning nutrient uptake, gas exchange, water and temperature status, respiration, and initial conditions (Table 7).

The sensitivity analysis focused on parameters related to nutrient uptake, water status, gas exchange, light response, respiration, and initial carbohydrate or biomass states. The main output analysed in this section was RGR. The admissible range, sampling scale (linear or logarithmic), functional role and literature basis of each of the $d = 16$ sampled parameters are given individually in Appendix A, which reproduces the machine-readable parameter specification used by the sensitivity driver. Table 8 summarises all evaluation outputs and the analysis stage in which each is used.

5.5.1. Sampling design and verification of the estimators

Because the sensitivity indices are themselves the output of a Monte Carlo estimator, the sampling design and the estimator were specified and tested before being applied to the model. Total-order Sobol' indices S_T and first-order indices S_1 were computed with the estimators of Sobol' (2001) in the form recommended by Saltelli et al. (2010), which requires two *independent* base sample matrices $\mathbf{A}$ and $\mathbf{B}$ of size $n \times d$ and evaluates the model on the d hybrid matrices $\mathbf{A}_{\mathbf{B}}^{(i)}$ obtained by substituting column i of $\mathbf{A}$ with column i of $\mathbf{B}$, for a total of $n(d+2)$ model runs. The independence requirement is not a formality: the estimator attributes variance to parameter i through the change produced by that single-column substitution, so any correlation between the corresponding columns of $\mathbf{A}$ and $\mathbf{B}$ biases the index towards zero. Splitting one low-discrepancy sequence of length $2n$ into a first and a second half does *not* produce independent matrices, and Section 6 reports what that construction does to the estimates.

The base matrices are therefore constructed by one of two designs, selected at run time according to the dimension of the problem. When $2d$ does not exceed the maximum dimension supported by the low-discrepancy sampler, a single Sobol' sequence in $2d$ dimensions is drawn and split by *column*, the construction specified by Saltelli et al. (2010). When it does — as here, where $d = 16$ and the sampler is limited to 16 dimensions — two independent Latin-hypercube samples are drawn instead, stratified per dimension and independent by construction (McKay, Beckman and Conover, 1979). The design actually used is written to a `base_design` column in every output file, so that no reported index is ambiguous about its own provenance.

The estimator implementation was verified against the Ishigami function (Ishigami and Homma, 1990),

$$f(x_1, x_2, x_3) = \sin x_1 + 7 \sin^2 x_2 + 0.1\, x_3^4 \sin x_1, \qquad x_i \sim \mathcal{U}(-\pi, \pi), \tag{64}$$

whose first- and total-order indices are known in closed form, so that the estimator can be tested rather than assumed correct. Convergence was assessed as the mean absolute error against the analytic indices over eight seeds, for n from 256 to 16384.

Uncertainty in the model indices is reported at two levels, because the two levels answer different questions. Bootstrap confidence intervals (10^3 resamples of the model evaluations) quantify sampling noise *given* a design and are blind to error contributed by the design itself; a replicate-design spread, obtained by repeating the whole analysis with ten independently generated base samples, quantifies the latter. Both are reported for every index.

Morris screening followed Morris (1991) with the elementary-effect measures μ^* and σ of Campolongo, Cariboni and Saltelli (2007), using r trajectories with a p-level grid and a fixed step $\Delta = p/[2(p-1)]$. Morris trajectory base

Table 7
Parameter groups used in the sensitivity analysis.

Group	Parameters	Interpretation
Nutrient uptake capacity	$I_{N,\max}$, $I_{P,\max}$	Potential N and P acquisition capacity
Soil nutrient content	N_s, P_s	External nutrient availability
Gas exchange and light	V_{cmax}, $J_{\max}$, C_a, RH, $\mathrm{PAR}_{\max}$	Photosynthetic and stomatal response
Water and temperature	ψ_w, T	Abiotic limitation of gas exchange and growth
Respiration and growth	r_m, λ_{sb}	Carbon loss and growth-rate scaling
Initial conditions	s_0, B_0	Initial carbohydrate and biomass state
Vacuolar exchange	k_{in}	Sucrose exchange with vacuolar compartment

points are drawn from an independent pseudo-random generator rather than from the low-discrepancy sampler, as the method specifies, so the screening results are not affected by the base-sample construction discussed above.

Table 8
Evaluation outputs and their role in the analysis.

Output	Used in	Purpose
RGR	Growth validation, ablation, sensitivity	Main growth-level response
Sucrose	Carbohydrate validation	Mobile carbon dynamics
Starch	Carbohydrate validation	Day-night reserve dynamics
Φ_c	Numerical checks	Photosynthetic carbon input consistency
s_{cyt}	Numerical checks	State validity and carbohydrate stability
Biomass	RGR computation and state checks	Growth state consistency
Sobol' S_T	Global sensitivity	Parameter importance including interactions
Morris μ^*, σ	Screening sensitivity	Effect size and interaction/nonlinearity indication

6. Results

The results are presented in five steps. First, the numerical and physiological validity of the coupled simulations is checked. Second, the calibrated model is evaluated against photoperiod-dependent RGR targets. Third, the internal sucrose and starch dynamics are analysed to verify whether the calibrated model produces plausible day-night carbohydrate behaviour. Fourth, an ablation study is used to isolate the effect of replacing the simplified carbon-input formulation with Farquhar and Farquhar-Jarvis gas-exchange models. Finally, global and local sensitivity analyses are used to identify the dominant sources of RGR variability.

6.1. Numerical and physiological consistency

Before analysing the validation and sensitivity results, the simulation output was checked for numerical and physiological consistency. The coupled model was required to preserve non-negative sucrose, starch, and biomass states, to produce no photosynthetic carbon input during the dark phase, and to avoid nonfinite samples during the 30-day simulation horizon.

All four photoperiod scenarios passed the consistency checks (Table 9). No negative sucrose, starch, or biomass values were observed, and no nonfinite values were detected. The photosynthetic carbon input was zero during the night phase in all scenarios, confirming that the imposed photoperiod forcing was propagated consistently into the carbon-input term. These results indicate that the gas-exchange layer and the downstream carbohydrate, nutrient, and biomass states remained numerically stable under the evaluated conditions.

6.2. Solver verification and carbon closure

The consistency checks above are necessary but weak: they test the trajectory that the calibrated configuration actually produces. The verification tests of Section 4.3 test the solver over the domain the sensitivity analysis samples, and they reveal a defect that the calibrated trajectory does not expose.

The closed form is exact; the fixed-point iteration is not unconditionally convergent. Substituted back into the coupled system, the analytic operating point of Eq. (22) leaves a residual of at most 6.5×10^{-14} across the PAR sweep, i.e. at the level of double-precision arithmetic. The fixed-point iteration matches it wherever it converges. It does not always converge. Table 10 steps the leaf water potential across the range sampled in the sensitivity analysis. Down to $\psi_w = -1.0$ MPa not a single daylight solve fails, and the two solvers agree on daily carbon fixation to 0.1%. Between -1.1 and -1.2 MPa the behaviour changes abruptly: the non-convergent fraction rises from 3.50% to 59.75%, and the daily fixation reported by the iterative solver falls 21.5% below the analytic value. By -1.4 MPa the iteration fails in 97.75% of daylight solves and reports 99.9% less carbon fixed than the exact solution. The transition is sharp, near $\psi_w \approx -1.15$ MPa, which is why it is invisible in a sweep of the calibrated configuration: all production runs are performed at $\psi_w = -0.6$ MPa, where the failure rate is exactly zero.

The failure is a period-two orbit, not slow convergence. The parity test identifies the mechanism. At a forcing point inside the failing region, varying the iteration cap over consecutive integers returns two alternating values and nothing in between: an even cap returns $A_n = -13.389$ μmol m^{-2} s^{-1} with $C_i = 862.6$ μmol mol^{-1}, an odd cap returns $A_n = +23.695$ with $C_i = 0$, and the analytic solution is $A_n = 7.704$ at every cap. The map has a stable two-cycle, so no increase in the iteration budget would help; and because the reported value depends on the parity of the cap, the

Table 9
Numerical and physiological consistency checks for the Farquhar-Jarvis model at the repeating day-night cycle. Minima are taken over all sampled steps of the 30-day run. Generated by `tools/validation_runner.c` into `results/validation/validation_summary.csv`.

PP	Min. s_{cyt}	Min. starch	Min. biomass	Night Φ_c	Nonfinite	Pass
4 h	0.298	0.225	0.040	0.0	0	✓
6 h	0.423	0.225	0.040	0.0	0	✓
8 h	0.569	0.225	0.040	0.0	0	✓
12 h	0.908	0.225	0.040	0.0	0	✓

Table 10
Fixed-point solver behaviour across the sampled range of leaf water potential, at fixed remaining forcing. Non-convergence is the fraction of daylight solves reaching the iteration cap without satisfying Eq. (21). 'Carbon deficit' is the shortfall of the daily integrated fixation reported by the iterative solver relative to the analytic solution of Eq. (22). All production runs use $\psi_w = -0.6$ MPa. Abridged from `results/verification/solver_water_stress.csv`.

ψ_w (MPa)	Solves	Non-conv. (%)	$\int A_n$ analytic (mol m^{-2})	Carbon deficit (%)
−0.20	2400	0.00	0.5671	0.11
−0.60	2400	0.00	0.5601	0.10
−1.00	2400	0.00	0.5269	0.09
−1.10	2400	3.50	0.5047	0.10
−1.20	2400	59.75	0.4678	21.47
−1.30	2400	87.67	0.4024	95.83
−1.40	2400	97.75	0.2755	99.92
−1.50	2400	99.25	0.0000	—

sign of the flux is determined by an implementation detail. Neither branch is individually detectable as impossible: the even-cap pair satisfies the supply relation Eq. (20) exactly at the local conductance ($g_c = 0.029$ mol m^{-2} s^{-1}), and $C_i > C_a$ is the correct supply solution for the net efflux it reports, so a bound check on C_i alone would have passed it. What is diagnostic is the pair taken together — two mutually inconsistent operating points on the same forcing, straddling zero assimilation, with the analytic root lying between them. Cap-parity sensitivity, not a range check on C_i, is the test that would have caught this, and it was not being applied.

Exposure of the sensitivity analysis. Over 2×10^4 parameter vectors drawn from the full ranges of Appendix A, the iterative solver fails to converge in 15.16% of cases, returns a negative net assimilation under light in 13.52%, and leaves the physically meaningful range with $C_i > 2C_a$ in 7.64% — the last a divergence diagnostic in the sense of Section 4.3, not a violated model bound. The discrepancy against the closed form has median zero — most of the domain is solved correctly — but a mean of 2.43 μmol m^{-2} s^{-1}, a 95th percentile of 18.72 and a maximum of 38.11, the last comparable in magnitude to the assimilation rate itself. A sixth of the design points were therefore being evaluated with a carbon input that was wrong, often with the wrong sign, and the affected points are not randomly located: they concentrate at low conductance, which is where the water-status and VPD dimensions act. This is the reason the sensitivity results were recomputed with the analytic solver, and it is the reason the local response diagnostics and the validation, diel and ablation results were not: those are all evaluated at $\psi_w = -0.6$ MPa, inside the region where the two solvers agree to 0.1%.

Corrected Jarvis temperature response. The temperature modifier as originally implemented satisfied neither property its normalisation was intended to guarantee: it returned $f_T(25°\text{C}) = 0.7071$ rather than unity, and its maximum lay at 32°C rather than at the stated optimum $T_0 = 25°$C. Two independent errors were responsible, one in each normalisation constant of Eq. (9). With both corrected, $f_T(25°\text{C}) = 1.000000$, the maximum lies at 25.0°C, and $f_T(5°\text{C}) = f_T(45°\text{C}) = 0$ exactly, so the clipping previously required is not. The correction is calibration-neutral: because the modifier enters multiplicatively and all calibration runs are performed at $T = 25°$C, setting the explicit $g_{s,\max}$ of Eq. (6) equal to the former value $f_T(25) = 1/\sqrt{2}$ reproduces the previous conductance at 25°C

exactly: across the PAR range the corrected and previous values agree to within 3×10^{-17} mol m^{-2} s^{-1}, a relative difference below 10^{-16}. No calibrated parameter changes. What changes is the response away from 25°C, where the corrected conductance is higher below the optimum and lower above it, by up to a factor of 1.32 at 10°C and 0.71 at 35°C. Results that vary temperature — the temperature–humidity and water-potential–temperature surfaces of Fig. 12 and the `gas_exchange.leaf_temperature` sensitivity dimension — are affected and are reported here with the corrected response; results at fixed 25°C are unchanged bit for bit. Separately, a guard that returned zero assimilation below $PAR = 20$ μmol m^{-2} s^{-1} was removed. It introduced a genuine discontinuity, since $f_{PAR}(20) = 0.0374 \neq 0$, and the terminal non-negativity constraint already yields zero below the light compensation point without a special case.

Carbon closure. Table 11 reports the carbohydrate budget over a 35-day integration at $\Delta t = 0.1$ h. Assimilated carbon, the starch and sucrose fluxes, degradation, maintenance respiration, phloem loading, transport surcharge and vacuolar exchange balance the change in the three carbohydrate pools to a relative residual between 1.5×10^{-15} and 1.3×10^{-14} — machine precision, as it must be, since the flux accumulators are integrated with the same fixed-step `RK4` weights and step as the states in this test (Section 4.3). The value of the test is not the magnitude of the residual but its sensitivity: a mis-signed or double-counted flux would appear immediately as a residual many orders of magnitude larger, and during development that is exactly how two bookkeeping errors were found.

The excluded term can now be quantified rather than asserted. The sucrose drawn from the cytosol to support growth is $\phi_{\mathrm{load}} E_{\mathrm{night}}$, while the carbon credited to the two biomass pools is $\lambda_{sb}\, \phi_{\mathrm{load}} E_{\mathrm{night}} B_l$, so the ratio of credited to withdrawn carbon is exactly $\lambda_{sb} B_l$. With $\lambda_{sb} = 0.00587$ gFW^{-1} this ratio reaches unity only at $B_l = 170.4$ gFW, four orders of magnitude above the seedling range 0.01–0.12 gFW simulated here, over which it spans 5.9×10^{-5} to 7.0×10^{-4}. The biomass equations are therefore an empirical rate law and not a carbon budget, and λ_{sb} is a rate coefficient with units gFW^{-1} rather than a carbon-to-fresh-weight yield. We state this explicitly because the alternative reading — that the model conserves carbon from assimilation through to biomass — is not supported and would be a straightforward misinterpretation of Table 11.

6.3. Photoperiod-dependent RGR validation

The calibrated Farquhar-Jarvis model with the vacuolar sucrose compartment reproduced the RGR targets of Sulpice et al. (2014) across all four photoperiod scenarios (Table 12, Fig. 5). The mean absolute error was 2.41×10^{-4}, the RMSE was 2.68×10^{-4}, and the mean absolute percentage error was 0.24%.

These residuals must not be read as predictive errors, and the earlier description of them as "calibration-consistent validation" understated the problem. The calibration procedure adjusts two parameters per photoperiod: the growth coefficient λ_{sb} is bisected until the simulated RGR equals the published RGR, inside an outer bisection that adjusts the starch-synthesis rate λ_{sdi} until the simulated end-of-day starch matches the published value. With four photoperiods this is eight fitted parameters against eight observations: the model is saturated and has zero degrees of freedom. The numbers in the fourth and fifth columns of Table 12 are therefore convergence residuals of the bisection, and re-running the calibration to tighter tolerance drives the RGR error to 0.0000% in every photoperiod. A mean absolute percentage error of 0.24% measures how well a root-finder converged, not how well the model predicts. Note also that the parameter identified in the original submission as the calibrated one — the vacuolar influx k_{in} — is not what the calibration routine adjusts; λ_{sb} and λ_{sdi} are.

To obtain a figure that does measure prediction, the photoperiod response was cross-validated out of sample. The design imposes a photoperiod-*independent* parameterisation, fits it to a subset of photoperiods, freezes it, and predicts the held-out ones. Two hypotheses were tested: a *constant* pair (λ_{sb}, λ_{sdi}) shared by all photoperiods (2 parameters), and a *linear* law $\lambda_{sb} = c_0 + c_1 P$ with a shared λ_{sdi} (3 parameters). Each was fitted by Nelder–Mead on seven training/test splits — the three two-versus-two partitions and the four leave-one-photoperiod-out cases — giving 14 cross-validated fits, plus the four saturated per-photoperiod reference fits: 18 fits in total. Table 13 reports all of them. Each regime therefore yields ten held-out predictions (two from each two-versus-two split, one from each leave-one-out split). All ten are admissible under the constant hypothesis; under the linear hypothesis the frozen parameters of the 4, 6 h training split make one predicted photoperiod inadmissible, so nine are scored. The aggregate held-out errors quoted below are means over those ten and nine predictions respectively, not over the seven splits: an unweighted mean over splits would give 26.32% and 6.31%, and we report the per-prediction figures so that the two-versus-two splits, which contribute two predictions each, are not under-weighted.

Three conclusions follow, and they point in different directions. First, the photoperiod dependence of growth in this model is carried substantially by the calibrated coefficients rather than by the coupled gas-exchange and carbohydrate

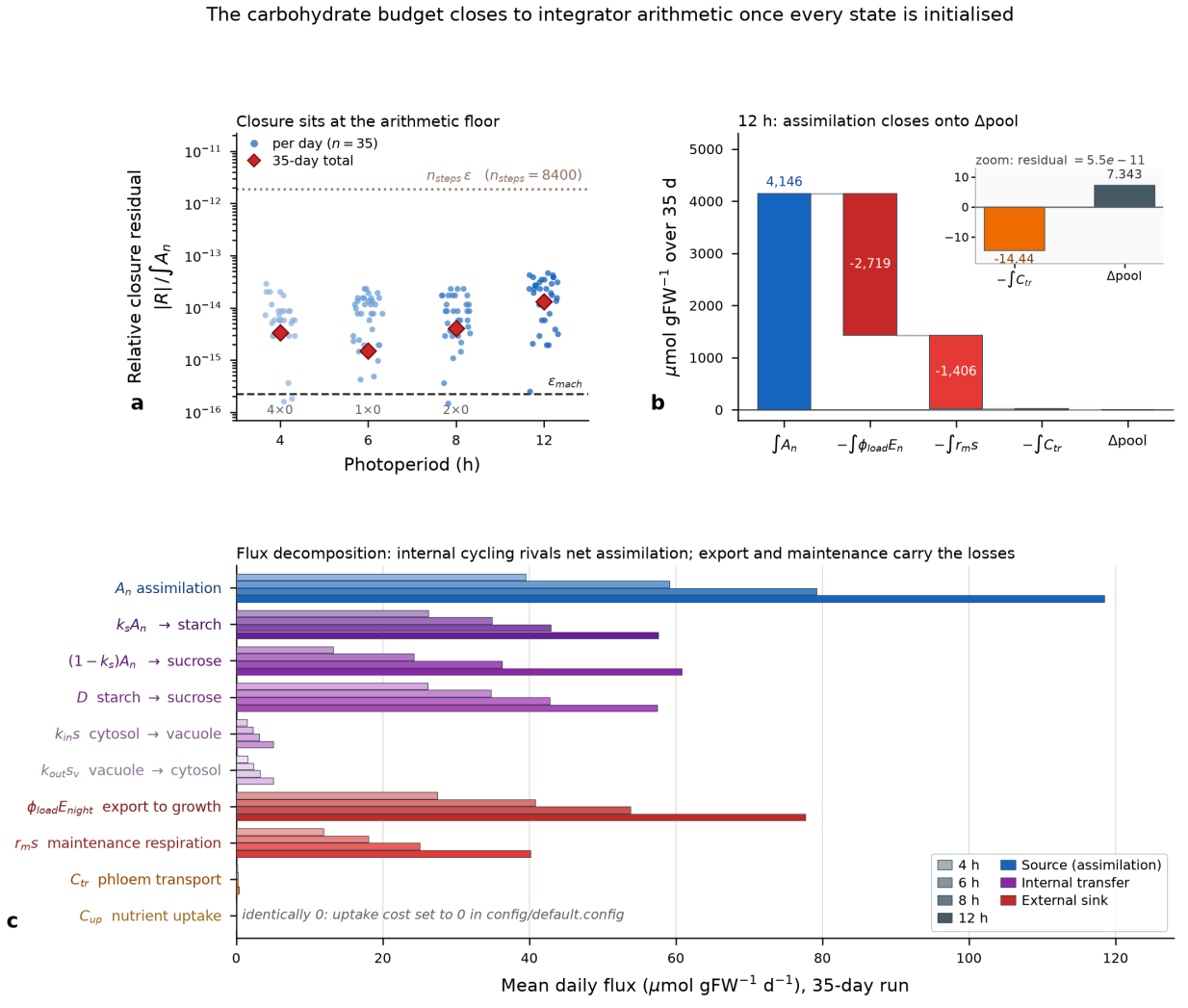


Figure 4: Closure of the carbohydrate budget. (a) Absolute relative closure residual against photoperiod on a logarithmic axis, per simulated day and for the 35-day total, with the double-precision machine epsilon and the accumulated-rounding bound $n_{\text{steps}}\epsilon$ shown for scale. The residual sits at the arithmetic floor at every photoperiod, so the closure is limited by floating-point representation rather than by any modelling approximation — the budget is an identity of the discretisation, not an approximate balance. (b) The 12 h budget as a waterfall: integrated assimilation against export to growth, maintenance respiration and phloem transport, closing onto the change in pool size, with the residual shown magnified in the inset. (c) Decomposition of the mean daily fluxes at all four photoperiods into assimilation, the two partitioning channels, starch remobilisation, the vacuolar exchange pair, export to growth, maintenance respiration and transport; nutrient uptake cost is identically zero because it is set to zero in the production configuration. Internal cycling is of the same order as net assimilation. In panels (a) and (c) the annotations `4×0`, `1×0` and `2×0` record how many of the per-day residuals at that photoperiod were exactly zero in double precision (4 of the 35 days at 4 h, 1 at 6 h, 2 at 8 h); such days are omitted from the logarithmic axis rather than plotted. This test integrates the flux accumulators on the augmented system with the same fixed-step `RK4` weights and step as the eleven state equations (Section 4.3), which is why the residual reaches machine precision instead of the 10^{-6}–10^{-8} typical of a budget reconstructed post hoc from sampled trajectories.

Table 11
Carbon closure over the carbohydrate subsystem $\{a, s, s_v\}$ for a 35-day integration at $\Delta t = 0.1$ h. Fluxes are time-integrated μmol gFW^{-1}. The residual is the closure error of the subsystem budget; the relative residual normalises it by integrated assimilation. Fresh-weight biomass is outside the budget by construction (see text). From `results/carbon_balance/carbon_balance.csv`.

PP (h)	$\int \Phi_c$	$\int$ to starch	$\int$ to sucrose	$\int D_a$	$\int C_{\text{resp}}$	$\int C_{\text{load}}$	Rel. residual
4	1381.7	917.5	464.3	915.1	416.7	961.1	-3.4×10^{-15}
6	2069.7	1221.5	848.2	1217.7	631.5	1429.9	-1.5×10^{-15}
8	2772.3	1502.1	1270.2	1496.9	876.8	1882.5	$+4.1 \times 10^{-15}$
12	4146.5	2017.5	2129.0	2010.1	1406.2	2718.5	$+1.3 \times 10^{-14}$

Table 12
RGR validation under photoperiod treatments. The Farquhar-Jarvis model with vacuolar sucrose compartment was calibrated per photoperiod.

Photoperiod	Literature RGR	Simulated RGR	Abs. error	Rel. error (%)
4 h	0.0680	0.068435	0.000435	0.640
6 h	0.1135	0.113679	0.000179	0.158
8 h	0.1708	0.170925	0.000125	0.073
12 h	0.2600	0.260224	0.000224	0.086

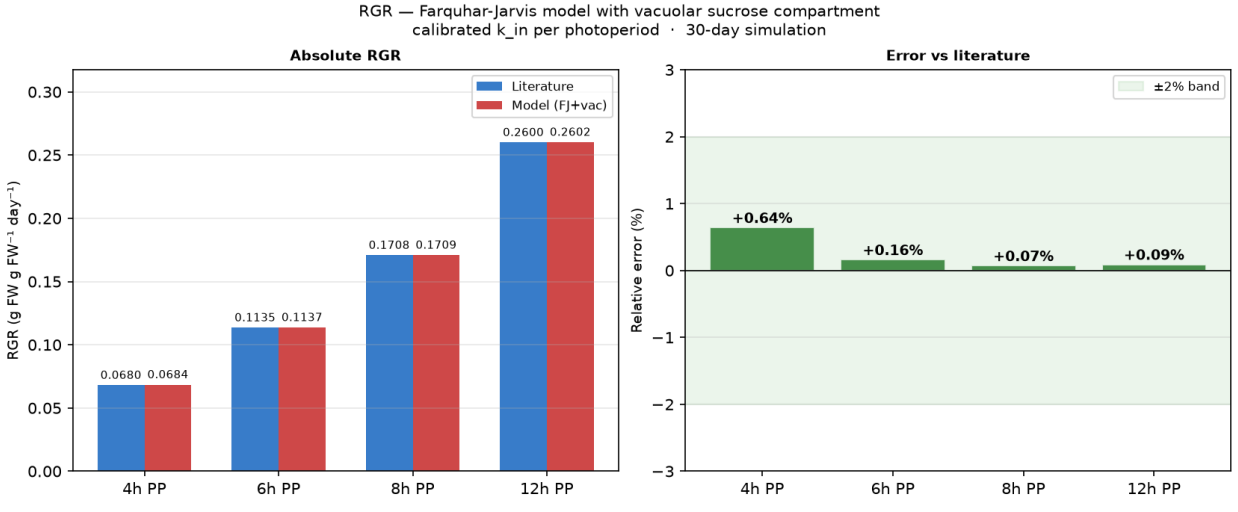


Figure 5: RGR validation of the Farquhar-Jarvis model with vacuolar sucrose compartment under four photoperiod treatments. The left panel compares absolute RGR values against the literature targets of Sulpice et al. (2014), while the right panel shows the relative error with respect to the ±2% acceptance band. The model overshoots at every photoperiod, by at most 0.64%; these are convergence residuals of the per-photoperiod calibration and not predictive errors, as the text below the figure sets out.

dynamics: if it were carried by the dynamics, a single parameter pair would extrapolate across photoperiod, and it does not — the constant hypothesis gives a mean held-out error of 28.49%. The model as calibrated is an interpolator over photoperiod, not a predictor of it. Second, the deficiency is localised rather than diffuse. Allowing λ_{sb} to vary linearly with photoperiod — one additional parameter — reduces the mean held-out error to 7.27% and the training error to 1.35%, so most of the photoperiod signal the coupled dynamics fail to generate is a smooth, almost affine trend in a single growth coefficient. That is a considerably more actionable diagnosis than a global statement of poor prediction, and it identifies exactly which mechanism a subsequent model version has to supply. Third, the saturated calibration is

not thereby useless — it is the right procedure for the question the ablation study asks, which is a comparison between model variants calibrated identically — but it cannot support a claim of predictive accuracy, and no such claim is made here. The held-out errors, not the in-sample residual, are the figures of merit for extrapolation across photoperiod, and reducing them is a modelling problem, not a calibration problem: it would require the photoperiod dependence now absorbed into λ_{sb} and λ_{sdi} to be generated by the state dynamics instead.

Three things follow from Table 13. First, the in-sample residual carries no information about predictive accuracy: it is 0.00% by construction, while the same model predicting a photoperiod it was not fitted to is wrong by 7 to 28% on average and by up to 87% in the worst split. Second, the comparison between the two cross-validated regimes locates where the photoperiod dependence lives. A single constant pair cannot represent it — 28.49% mean held-out error, and the fit is poor even on its own training photoperiods (12.93%) — whereas allowing λ_{sb} to vary linearly with photoperiod reduces held-out error to 7.27% with a near-exact training fit. The photoperiod response of growth in this model is therefore carried substantially by a smooth trend in one calibrated coefficient, not by the coupled dynamics. Third, the splits are not interchangeable: extrapolation beyond the training range is markedly worse than interpolation within it. Training on 8 and 12 h and predicting the short photoperiods gives the largest error of the leave-two-out splits in both regimes (35.91% and 16.70%), while training on 4 and 12 h and predicting the interior (8.54% and 4.52%) is the best. The leave-one-out results show the same ordering, with the interior photoperiods (6 and 8 h) predicted far more accurately than the endpoints.

One split deserves comment because it exposed a numerical defect rather than a modelling one. Fitting the linear law on the two shortest photoperiods gives a poorly determined slope, and the frozen parameters imply $\lambda_{sdi} = 0.636$ at 12 h, beyond the bracket used in calibration; at that value the adaptive integrator stalls rather than diverging, which is how the step-budget guard described in Section 7.3 came to be added. The affected prediction is reported as non-admissible rather than scored.

The internal carbohydrate trajectories and the ablation and sensitivity analyses, which do not depend on the RGR fit being predictive, are therefore the substantive evidence of model behaviour, and are reported next.

6.4. Diel sucrose and starch dynamics

The model produced stable repeating day-night carbohydrate cycles after the transient phase. Starch accumulated during the light phase and was depleted during the dark phase in all photoperiod scenarios (Fig. 6). This behaviour is consistent with the expected role of transient starch as a daytime carbon reserve and a nighttime carbon source. Sucrose also showed repeating diel behaviour, although the agreement with reference sucrose profiles was weaker than for starch at every photoperiod, and weakest under the short days (Table 14).

Starch nRMSE remained below the predefined threshold of 0.25 in all scenarios, with values between 0.128 and 0.192. This indicates that the model captured the timing and shape of starch accumulation and degradation across the evaluated photoperiods. Sucrose nRMSE also remained below its threshold of 0.50 in all four scenarios, between 0.284 and 0.390, so every photoperiod passes the carbohydrate criterion.

The residual sucrose error is not uniform across photoperiod, and it is largest where the day is shortest: 0.390 at 4 h and 0.382 at 6 h, against 0.315 at 8 h and 0.284 at 12 h. The ordering is monotone in day length. It tracks the fitted scale: α_{suc} is furthest from unity under short days (3.37 at 4 h against 1.13 at 12 h), so the cytosolic pool the model holds at 4 h is roughly a third of the measured level and, once that offset is divided out, the comparison is carried entirely by the shape of the rhythm. Where the model must be stretched most to meet the data it also matches its shape least well, which is the expected signature of a compartment whose level is set by a calibrated partition rather than by the reference series.

The short photoperiods are also the cases in which the transient starch pool carries the largest share of the diel carbon budget: with four hours of assimilation, night-time sucrose supply is set almost wholly by the rate at which starch is remobilised, so an error in the phase or curvature of starch degradation is transmitted more directly into the sucrose trace than under a 12 h day, where a much larger fraction of the sucrose requirement is met by concurrent assimilation. The two carbohydrate residuals are therefore not independent, and the short-day sucrose error is at least partly a symptom of the turnover law rather than a separate sucrose defect. We note this as the physically motivated reading of the ordering rather than as a demonstrated mechanism: the present comparison scores one photoperiod at a time and does not isolate the contribution of the degradation law to the sucrose residual.

These values differ from those of the first submission, and the reason is almost entirely a difference of definition rather than of model behaviour. The earlier table was not produced by the validation runner. It scored the total (cytosolic plus vacuolar) sucrose trace, used the unscaled residual $\hat{y}_i - y_i$, and normalised by the reference range $y_{\max} - y_{\min}$

Table 13
Out-of-sample validation of the photoperiod response: all 18 fits. The *saturated* regime is the manuscript's original procedure — one λ_{sb} and one λ_{sdi} per photoperiod against one RGR and one end-of-day starch observation per photoperiod (8 parameters, 8 observations, zero degrees of freedom) — so its RGR residual is a bisection tolerance and not an error. The two cross-validated regimes impose a photoperiod-*independent* law, fit it on the training photoperiods by Nelder–Mead, freeze it, and predict the held-out ones. Errors are |relative error| in RGR, averaged over the photoperiods in each role. "n.a." marks a predicted photoperiod at which the frozen parameters make the trajectory inadmissible (the adaptive step collapses; see Section 7.3), recorded rather than scored. The **all splits** rows are means over individual held-out predictions — ten for the constant regime, nine for the linear regime after the one inadmissible case — and not over the seven splits, so that the two-versus-two splits carry their two predictions each.

Regime	Split	Train MAPE	Test MAPE	Test worst	λ_{sdi}
Saturated (8 par., 8 obs.)	in-sample, all 4 PP	0.00	—	—	per PP
(in-sample end-of-day starch)		3.29	—	—	
Constant (2 par.)	train 8, 12 h	5.62	35.91	46.74	0.0742
	train 4, 6 h	8.48	56.19	87.42	0.5708
	train 4, 12 h	20.41	8.54	16.34	0.0655
	hold out 4 h	10.61	33.27	33.27	0.0897
	hold out 6 h	18.36	4.19	4.19	0.0767
	hold out 8 h	13.77	16.13	16.13	0.0755
	hold out 12 h	11.84	30.02	30.02	0.2890
	all splits	**12.93**	**28.49**	**87.42**	
Linear in P (3 par.)	train 8, 12 h	0.02	16.70	21.79	0.0727
	train 4, 6 h	0.01	4.60	4.60	0.6358 (1 n.a.)
	train 4, 12 h	0.01	4.52	7.33	0.0596
	hold out 4 h	2.88	6.39	6.39	0.0846
	hold out 6 h	3.30	0.23	0.23	0.0710
	hold out 8 h	0.79	6.96	6.96	0.0694
	hold out 12 h	1.09	4.76	4.76	0.3467
	all splits	**1.35**	**7.27**	**21.79**	

Table 14
Sugar and starch validation summary. Errors are computed on the scale-fitted residual of Section 5.3; MAE is in μmol gFW^{-1} and α is the fitted scale. The scored sucrose state is the cytosolic pool s_{cyt}, which is the pool the reference series is matched against by the runner; the total (cytosolic plus vacuolar) sucrose shown in Fig. 6 is a different quantity and is not scored here. Each photoperiod was evaluated using 13 reference points from `data/paper/sugars_starch_reference.csv`. Generated by `tools/validation_runner.c` into `results/validation/validation_summary.csv`; the superseded values of the first submission are retained in `results/validation/validation_summary_2026-07-02.csv`.

Photoperiod	Ref. points	α_{suc}	Sucrose nRMSE	Sucrose MAE	α_{sta}	Starch nRMSE	Starch MAE
4 h	13	3.365	0.390	0.800	1.036	0.154	1.981
6 h	13	2.211	0.382	0.713	1.216	0.175	3.894
8 h	13	1.971	0.315	0.683	1.052	0.192	4.453
12 h	13	1.130	0.284	0.592	0.923	0.128	3.313

— that is, the metric equations exactly as they were then written, which did not mention a fitted scale. The runner scores the cytosolic pool, fits the scale of Section 5.3, and normalises by the mean reference level; and it is the runner that applies the acceptance thresholds and emits the pass flags. Evaluating the earlier definition on the present trajectories returns 0.356, 0.266, 0.348 and 0.532 for sucrose and 0.090, 0.113, 0.102 and 0.073 for starch, against the published 0.370, 0.268, 0.345, 0.556 and 0.097, 0.117, 0.103, 0.071: the old numbers are recovered to within a few percent, so the two tables are two different measurements of one model rather than a change in the model. The residual few percent is the model drift, and it is bounded independently — re-running the runner against its own July

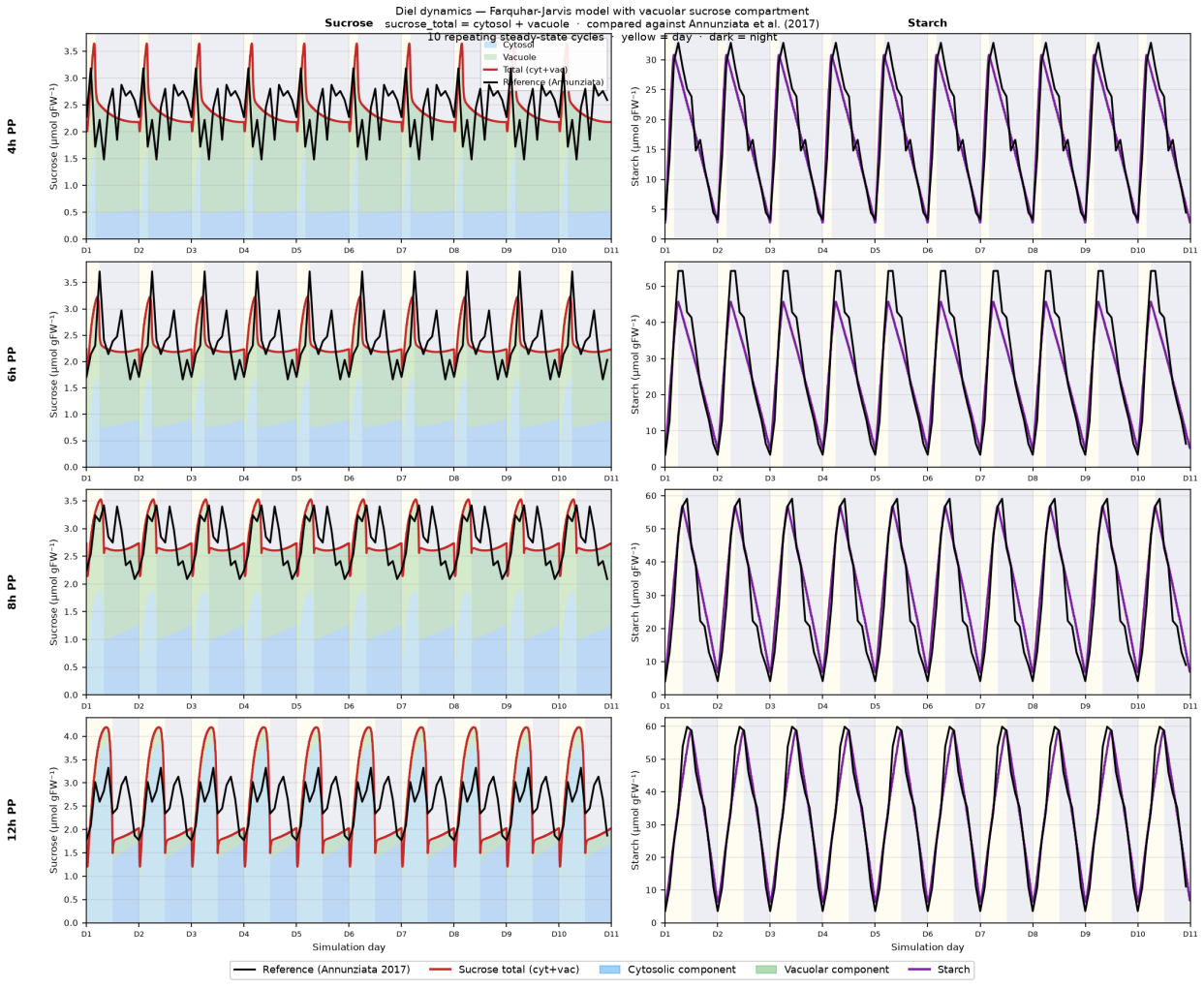


Figure 6: Diel sucrose and starch dynamics produced by the Farquhar-Jarvis model with the vacuolar sucrose compartment. Rows are the 4, 6, 8 and 12 h photoperiods; the left column shows cytosolic, vacuolar and total sucrose and the right column starch. Black curves are the measured profiles of Annunziata et al. (2017). Pale bands mark the light period and grey bands the dark period, over ten repeating steady-state cycles. Absolute levels carry the calibration of Section 6; the comparison is of shape and timing (Section 3.8).

output moves every carbohydrate metric by less than 4%, which is the combined effect of the analytic C_i solution, the Jarvis temperature-response correction and the removal of the PAR gate on these particular quantities. Both definitions evaluated on the present trajectories are tabulated in `results/validation/metric_definition_comparison.csv`. The definitions in Section 5.3 have been corrected to state what the code computes, and the table now reports runner output directly.

Two consequences should be stated plainly. First, the starch fit is looser than the first submission reported: nRMSE rises from a 0.071–0.117 band to 0.128–0.192, roughly 50 to 85% higher at every photoperiod, though still inside the 0.25 threshold. That increase is largely the change of denominator — for starch the reference range is about twice the reference mean, so range-normalisation reports a smaller number for the same residual — and we report the larger, mean-normalised value because it is the one the threshold is applied to in code. Second, the ordering of the sucrose residual across photoperiod is a property of the metric as much as of the model: under the earlier total-sucrose, range-normalised measure the 12 h case was the largest error, whereas under the scored cytosolic, mean-normalised measure it is the smallest. The amplitude overshoot visible in the 12 h total-sucrose trace of Fig. 6 is real and is discussed

below, but it is not a threshold failure under the criterion the model is tested against, and it is not the largest normalised carbohydrate error in the set.

6.5. Ablation results

The ablation study compared three carbon-input formulations: the basic Tedone-style formulation, the Farquhar photosynthesis variant, and the Farquhar-Jarvis gas-exchange variant. The basic formulation systematically underestimated RGR across all photoperiods. Replacing this simplified carbon input with Farquhar C_3 photosynthesis removed most of the systematic bias. Adding the Jarvis stomatal conductance layer produced similar reported RGR accuracy under calibrated conditions, while introducing explicit environmental responsiveness into the carbon-input pathway (Table 15, Fig. 7).

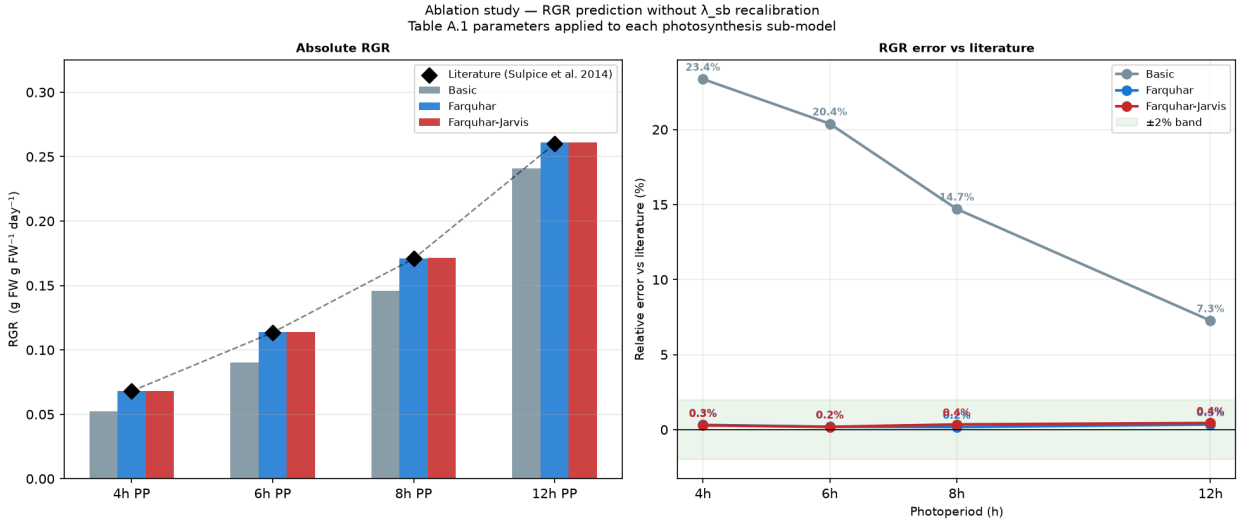


Figure 7: Ablation study comparing RGR predictions obtained with the basic carbon-input model, the Farquhar photosynthesis model, and the Farquhar-Jarvis gas-exchange model. The left panel gives absolute RGR against the literature targets, the right the relative error with the ±2% acceptance band; λ_{sb} is not recalibrated between sub-models, so all three are run with the same Table 3 values. Errors in the right panel are plotted as *magnitudes*: the basic model lies below the target at every photoperiod, which the body text reports with the corresponding negative sign. The basic model systematically underestimates RGR, while the Farquhar and Farquhar-Jarvis variants remain close to the literature targets.

The basic model produced errors ranging from −23.4% under the 4 h photoperiod to −7.3% under the 12 h photoperiod. This confirms that the simplified carbon-input formulation was not sufficient to reproduce the photoperiod-dependent RGR response. The Farquhar variant reduced the mean absolute percentage error to below 0.3%. The Farquhar-Jarvis variant did not substantially improve reported RGR relative to the Farquhar-only variant, because the RGR target was already closely matched. Its contribution is instead to expose the model to environmental drivers through stomatal regulation, which is evaluated further in the sensitivity analyses.

6.6. Global sensitivity of RGR

All Sobol' indices reported here were recomputed after the sampling-design defect described in Section 5.5.1 was corrected, and with the analytic gas-exchange solver of Eq. (22); they supersede the values obtained with the original design, and the ranking changes. The total-order analysis shows that RGR variability is dominated by the two nutrient uptake capacity parameters, maximum phosphorus uptake $I_{P,\max}$ and maximum nitrogen uptake $I_{N,\max}$, followed by the respiration frequency r_m and then by the photosynthetic capacities $V_{c\,\max}$ and $J_{\max}$, with leaf water potential ψ_w and ambient CO_2 close behind (Table 16, Fig. 8).

The full set of sixteen dimensions is listed rather than a top-ten subset, because the indices that are near zero are themselves a result: the initial conditions and both soil-content drivers contribute almost nothing to RGR variance over the ranges of Appendix A, and reporting only the dominant parameters would leave that unstated. All sixteen interaction

Table 15
Ablation study: simulated RGR under three photosynthesis sub-models without recalibration of the downstream growth-rate coefficient. Relative error with respect to the literature target is shown in parentheses.

Variant	4 h PP	6 h PP	8 h PP	12 h PP
Literature	0.0680	0.1135	0.1708	0.2600
M0 - Basic	0.0521 (−23.4%)	0.0904 (−20.4%)	0.1457 (−14.7%)	0.2411 (−7.3%)
M1 - Farquhar	0.0682 (+0.3%)	0.1137 (+0.2%)	0.1711 (+0.2%)	0.2609 (+0.3%)
M2 - Farquhar-Jarvis	0.0682 (+0.3%)	0.1137 (+0.2%)	0.1714 (+0.4%)	0.2612 (+0.5%)

Table 16
Dominant parameters ranked by mean Sobol' total-order index across photoperiods. The table provides a compact global-sensitivity summary for the RGR output.

	Parameter	4 h	6 h	8 h	12 h	$\bar{S}_T$	$\bar{S}_1$	$\bar{S}_T - \bar{S}_1$
1	$I_{P,\max}$	0.38	0.49	0.53	0.52	0.481	0.225	0.256
2	$I_{N,\max}$	0.46	0.41	0.40	0.25	0.378	0.093	0.285
3	r_m	0.46	0.37	0.34	0.22	0.348	0.028	0.320
4	$V_{c\max}$	0.21	0.21	0.22	0.31	0.238	0.039	0.199
5	C_a	0.24	0.17	0.18	0.26	0.211	0.042	0.169
6	$J_{\max}$	0.21	0.20	0.20	0.23	0.211	0.019	0.192
7	ψ_w	0.15	0.15	0.14	0.19	0.155	0.043	0.113
8	$\mathrm{PAR}_{\max}$	0.13	0.13	0.13	0.19	0.147	0.018	0.130
9	T	0.10	0.06	0.08	0.09	0.081	0.017	0.064
10	RH	0.09	0.03	0.06	0.07	0.063	0.022	0.041
11	P_{soil}	0.06	0.05	0.08	0.05	0.059	0.020	0.040
12	N_{soil}	0.06	0.04	0.04	0.03	0.042	0.011	0.032
13	λ_{sb}	0.05	0.02	0.05	0.04	0.040	0.023	0.017
14	k_{in}	0.00	0.02	0.01	0.01	0.010	0.002	0.008
15	$s(0)$	0.00	0.02	0.01	0.01	0.009	0.002	0.007
16	$B(0)$	0.00	0.00	0.00	0.00	0.001	0.001	0.001

shares are non-negative in the recomputed design, in contrast to the earlier estimates; the Saltelli estimator's total-order form is not constrained to dominate the first-order term at finite n, and indices are reported unclipped throughout, since clipping would misrepresent precision at this sample size.

The two nutrient uptake capacities remain dominant across the photoperiod range but respond to it in opposite directions: $I_{P,\max}$ rises with day length ($S_T = 0.38$ at 4 h to 0.52 at 12 h, peaking at 0.54 at 8 h), whereas $I_{N,\max}$ declines monotonically from 0.46 at 4 h to 0.25 at 12 h. The respiration frequency shows the same decline, from 0.46 to 0.22. The declining pair is consistent with a shift in what limits growth as the photoperiod lengthens: under short days the plant is carbon-limited, so the parameters governing the losses from and the demands on a scarce carbohydrate pool matter most, while under long days carbon is less scarce and the assimilation capacities $V_{c\max}$, $J_{\max}$ and the light and CO_2 supply gain influence — $V_{c\max}$ rises from 0.21 at 4 h to 0.31 at 12 h, $J_{\max}$ from 0.20 at 8 h to 0.23 at 12 h, and $\mathrm{PAR}_{\max}$ from 0.13 to 0.19. Leaf water potential acts in the opposite direction to $I_{N,\max}$, rising from 0.15 to 0.19, which is expected if its effect is mediated by stomatal limitation of an assimilation flux whose importance itself grows with day length. All values in this paragraph are the total-order indices of Table 16, i.e. the recomputed design of Section 6.7.

The most informative column of Table 16 is the last. For fifteen of the sixteen parameters — all but $B(0)$, whose indices are indistinguishable from zero — the interaction share $\bar{S}_T - \bar{S}_1$ exceeds the first-order index, and for the most extreme cases it does so by an order of magnitude: r_m has $\bar{S}_1 = 0.028$ against $\bar{S}_T = 0.348$, and $J_{\max}$ has $\bar{S}_1 = 0.019$ against $\bar{S}_T = 0.211$. Resolved by photoperiod rather than averaged over them, the interaction remainder exceeds the first-order index in 44 of the 52 influential parameter-photoperiod cells, and the margin is thinnest at the 4 h photoperiod (9 of 16 parameters) — the averaged column overstates the uniformity of the effect, and the per-photoperiod count is the honest statement of it. Little of the RGR variance is attributable to any parameter acting alone. This is the quantitative

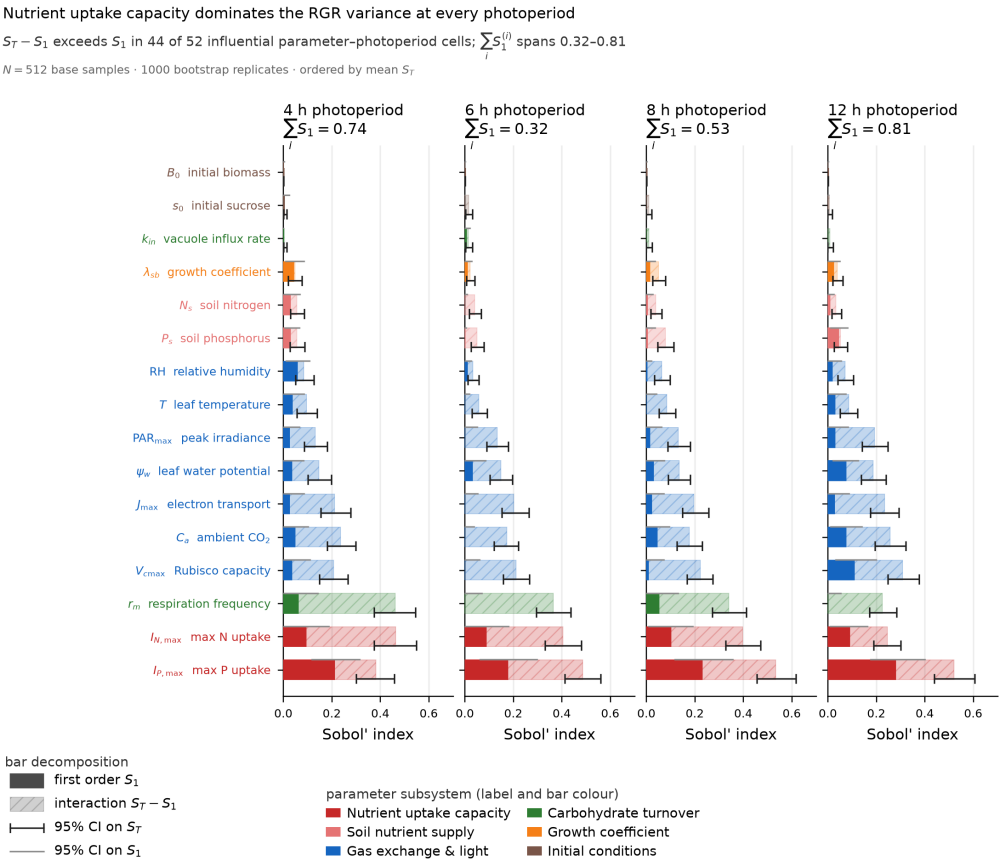


Figure 8: Sobol' variance decomposition of RGR under four photoperiod treatments. For each parameter the bar shows the first-order index S_1 (solid) and the interaction remainder $S_T - S_1$ (hatched), with bootstrap 95% confidence intervals from 1000 resamples of the base rows; parameters are ordered by mean total-order index across photoperiods. The interaction remainder exceeds the first-order index in 44 of the 52 influential parameter-photoperiod cells (9 of 16 parameters at 4 h, 16 of 16 at 6 and 8 h, 13 of 16 at 12 h), which is the central feature of the decomposition. Two reporting conventions should be noted. Indices are given unclipped: the Saltelli total-order estimator is not constrained to dominate S_1 at finite sample size, so where the estimator returns $S_1 > S_T$ within bootstrap noise the solid segment is drawn clamped at $\min(S_1, S_T)$ while the plotted intervals remain unclamped. The base design is two independent Latin-hypercube samples rather than a split low-discrepancy sequence, for the reason established in Section 6.7; the design actually used is recorded in every output file.

signature of the coupling the model is built around, and it has a methodological consequence worth stating: a one-at-a-time sensitivity analysis of this model would have found r_m and $J_{\max}$ nearly inert, and would have been wrong. The aggregate figures make the point compactly: summed over the sixteen dimensions, the first-order indices account for only 0.32 to 0.81 of the RGR variance depending on photoperiod, while the total-order indices sum to 2.36 to 2.59 against unity for a purely additive model. Between 19% and 68% of the variance is therefore unexplained by main effects, and is carried by interactions of second and higher order. The width of that range is itself informative: the main-effect share is smallest at the 6 h photoperiod, where the plant operates closest to the carbon-limited transition and the parameters act most strongly through one another.

The parameters associated with soil nutrient content, initial sucrose, initial biomass and vacuolar influx have low total-order indices over the analysed ranges. Two of these deserve comment rather than dismissal. The calibrated growth coefficient λ_{sb} has $\bar{S}_T = 0.040$, which is not a contradiction of the calibration result of Section 6.2: λ_{sb} sets the level of the growth response and the calibration exploits exactly that, but over the $\pm 70\%$ range sampled here it contributes little to the *variance* of RGR relative to parameters that act multiplicatively through the whole carbon chain. Sensitivity

and identifiability are different questions, and a low S_T does not license fixing a parameter that a calibration depends on. The vacuolar influx k_{in} has $\bar{S}_T = 0.010$, consistent with the equilibrium argument of Section 3.8: the vacuole redistributes sucrose in time without changing the integrated carbon available, so it shapes the diel trajectory much more than the 35-day growth integral used as the output metric here.

6.7. Verification of the sensitivity estimator

The indices above are only as trustworthy as the estimator that produced them, and in the originally submitted version they were not trustworthy. The Saltelli estimator requires two *independent* base samples, **A** and **B**. The implementation built them by drawing one Sobol' sequence of length $2n$ and splitting it into halves. Successive blocks of a low-discrepancy sequence are constructed to fill the gaps left by earlier blocks and are therefore strongly dependent: at $d = 16$ the column-wise correlation between the two halves reached 1.0000. The estimator then differences two nearly identical matrices, and the variance decomposition collapses.

Against the Ishigami function of Eq. (64), for which the indices are known in closed form ($S_1 = [0.3139, 0.4424, 0]$, $S_T = [0.5576, 0.4424, 0.2437]$), the published design does not converge at all: as n grows from 256 to 16384 the mean absolute error over eight seeds moves from 0.31 to 0.30 for S_1 and from 0.29 to 0.30 for S_T — no dependence on sample size — and S_1 for x_1 is estimated as exactly 0.000 against a true 0.3139. A correct column split of a single $n \times 2d$ sequence converges over the same range, MAE(S_1) falling from 0.034 to 0.0007; two independent Latin-hypercube samples converge at approximately $n^{-1/2}$, from 0.055 to 0.006. Merely reseeding the Sobol' sampler and calling it twice does *not* fix the problem: the error falls from 0.042 to 0.021 and then stops at that floor, because reseeding this sampler does not yield an independent set. Since the available sampler supports at most 16 dimensions and this problem has $d = 16$, the preferred $2d = 32$ column split is unavailable, and the adaptive design of Section 5.5.1 falls back to the Latin-hypercube construction; which design was used is recorded in a `base_design` column of every Sobol' output file.

The defect had already announced itself in our own convergence study, and was misread as noise. For the maximum nitrogen uptake capacity $I_{N,\max}$, S_T was 0.44 at $n = 512$, 0.15 at $n = 2048$, and 0.44 again at $n = 4096$, with a bootstrap 95% interval at $n = 2048$ of [0.128, 0.174] that excluded the $n = 4096$ estimate. Across ten replicate designs the spread at $n = 2048$ was *wider* than at $n = 512$, and the ratio of across-design standard deviation to within-design bootstrap standard error rose from 1.10 at $n = 512$ to 4.64 at $n = 2048$. An estimator whose variance grows with sample size is misspecified, not noisy, and that diagnostic is available without any external benchmark.

With the corrected design the across-design standard deviation of S_T falls as it should: averaged over the sixteen dimensions at the 12 h photoperiod it is 0.0347 at $n = 128$, 0.0177 at $n = 512$ and 0.0074 at $n = 2048$, close to the $n^{-1/2}$ rate. At $n = 512$ the across-design standard deviation and the within-design bootstrap standard error now agree, with a mean ratio of 0.88 (range 0.55 to 1.25) over the sixteen dimensions, against 1.10 rising to 4.64 before the fix. Both uncertainties are reported, because they measure different things: a bootstrap resamples within one fixed design and is blind to error contributed by the design itself, so the replicate-design spread is the quantity that would have caught this defect. The Morris screening required no recomputation — it draws trajectory base points from an independent pseudo-random generator, as Morris (1991) specifies, and never touches the affected sampler.

6.8. Morris screening and interaction behaviour

Morris screening supported the Sobol' ranking and provided additional evidence about nonlinear and interaction-driven effects. Parameters with high μ^* had large mean elementary effects, while parameters with high σ indicated nonlinear responses, interactions, or photoperiod-dependent behaviour (Fig. 10).

The nutrient uptake capacity parameters appear in the high-μ^*, high-σ region, indicating both strong influence and nonlinear or interaction-driven behaviour. Gas-exchange and light-related parameters also show substantial effects, particularly under longer photoperiods. The agreement with the Sobol' ranking is a meaningful check rather than a restatement, because the two methods share no sampling machinery: Morris draws its trajectory base points from an independent pseudo-random generator and was therefore unaffected by the design defect of Section 6.7, so it provides a ranking obtained without the corrected Saltelli estimator. Together the two analyses show that RGR is not controlled by a single fitted growth coefficient, but by the combined effect of nutrient uptake capacity, respiration, photosynthetic capacity, water status and photoperiod-dependent carbon availability.

6.9. Local response diagnostics

The one-at-a-time (OAT) parameter sweeps and parameter-pair response surfaces are reported as local diagnostics, not as a ranking. They cannot serve as one here: the interaction shares in Table 16 show that for fifteen of the sixteen

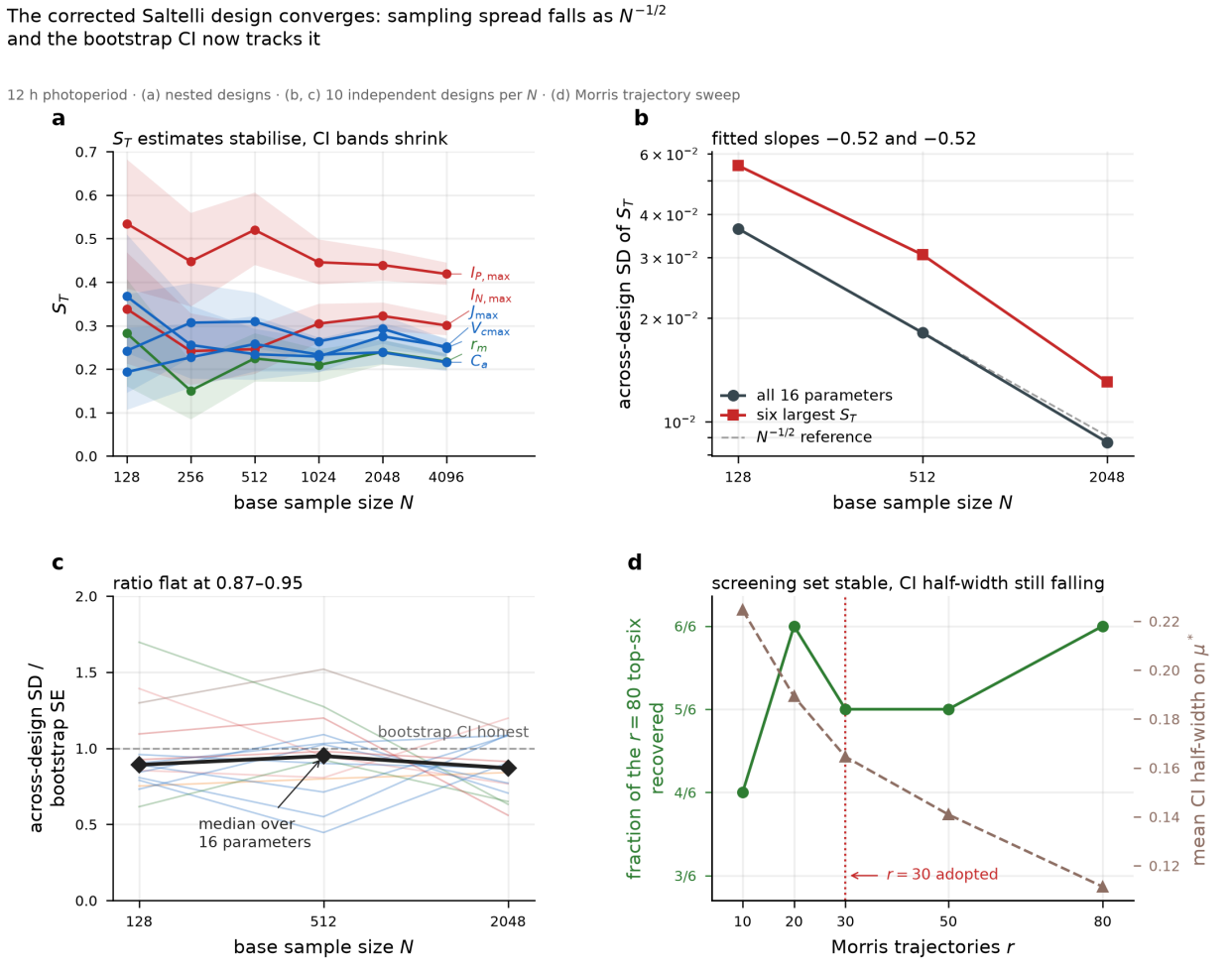


Figure 9: Convergence of the sensitivity estimators with the corrected base design, 12 h photoperiod. (a) Total-order indices S_T against base sample size N for the six leading dimensions over nested designs, with bootstrap 95% bands; the bands shrink steadily and the estimates settle to within 0.03 of their $N = 4096$ values from $N \approx 1024$ upward (to within 0.10 at $N = 512$). (b) Across-design standard deviation of S_T over ten independent designs per N, for all sixteen dimensions and for the six largest, on logarithmic axes with an $N^{-1/2}$ reference; the fitted slopes are −0.52 in both cases, in contrast to the pre-fix behaviour in which the spread grew with N. (c) Ratio of across-design standard deviation to within-design bootstrap standard error, per parameter and as a median; the ratio is flat at 0.87–0.95, so the bootstrap interval is now an honest measure of total estimator uncertainty, against a ratio rising to 4.64 before the fix. (d) Morris trajectory sweep: the fraction of the $r = 80$ top-six screening set recovered at each r (left axis) and the mean bootstrap confidence half-width on μ^* (right axis). The half-width falls monotonically with r (0.225, 0.189, 0.165, 0.141, 0.111 at $r = 10, 20, 30, 50, 80$), while the screened-in set is within one member of the $r = 80$ set from $r = 20$ upward and does not settle onto a single composition at any r tested (4, 6, 5, 5, 6 of six recovered). That is why $r = 30$ is adopted for screening only and no rank ordering is read from it: the leading μ^* values cluster within 0.34–0.51 at $r = 30$ and their relative order remains unstable, so Morris is used here to separate influential from negligible parameters and the variance decomposition is used for attribution.

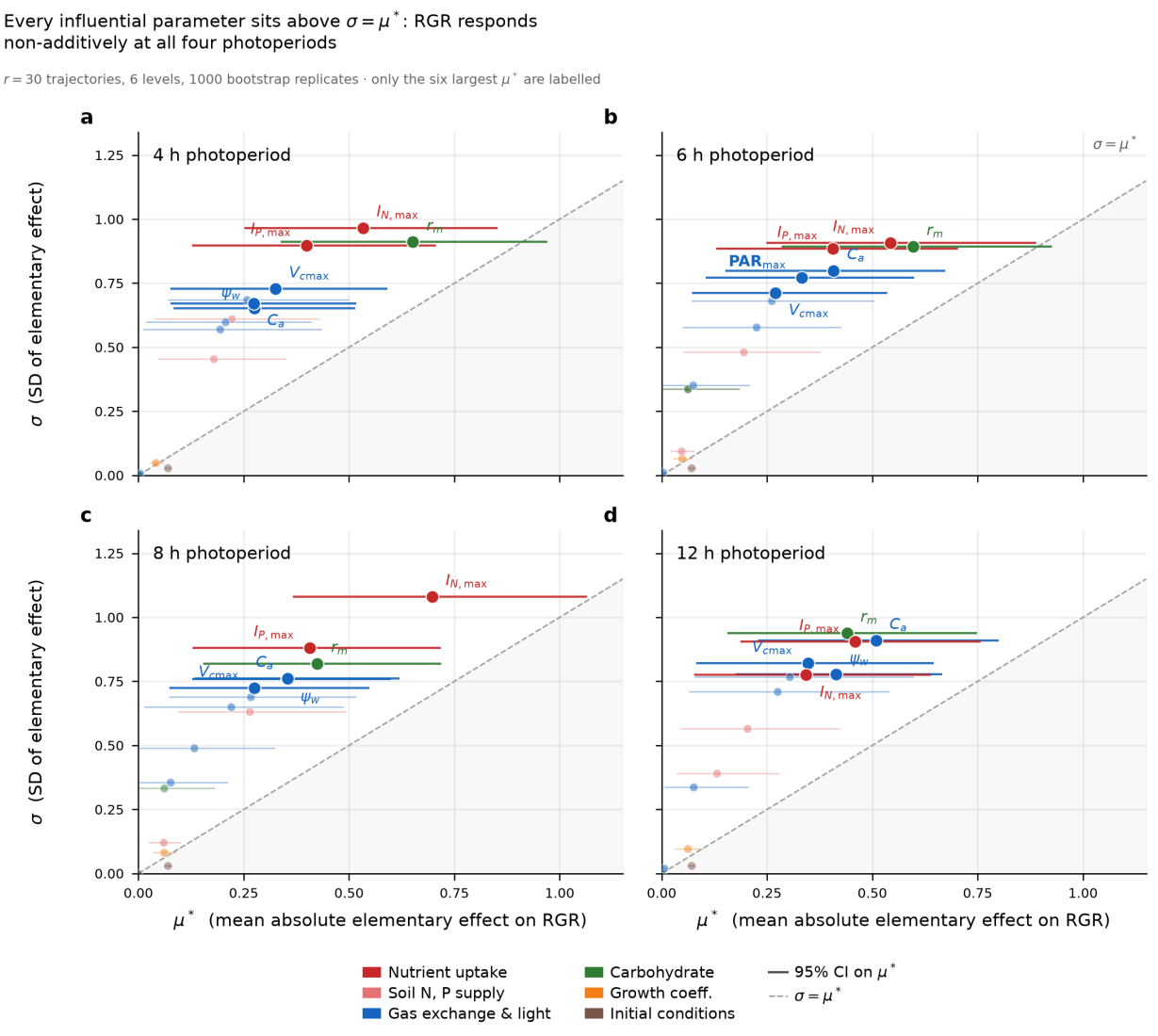


Figure 10: Morris screening of the RGR output at the 4, 6, 8 and 12 h photoperiods (panels a–d); $r = 30$ trajectories on 6 levels, with 95% bootstrap confidence bars on μ^* from 1000 replicates and colour by parameter subsystem. High μ^* indicates a large mean absolute elementary effect and high σ nonlinear or interaction-driven behaviour; the dashed diagonal is $\sigma = \mu^*$, and every parameter with appreciable μ^* lies above it at all four photoperiods. Only the six largest μ^* are labelled.

parameters most of the RGR variance arises from interactions, which an OAT sweep holding all other parameters at nominal values cannot see. Their value is in the shape of the response — where it saturates, where it changes sign, and how it depends on photoperiod — which the variance decomposition does not report.

The OAT sweeps (Fig. 11) showed saturating responses for nutrient uptake capacity, photosynthetic capacity, water potential, and irradiance-related parameters. Respiration frequency showed the expected negative effect on RGR: increasing r_m reduced growth. Environmental variables such as CO_2, temperature, relative humidity, and PAR_{max} produced photoperiod-dependent responses, supporting the interpretation that the Farquhar-Jarvis layer changes the model behaviour beyond a prescribed carbon-input formulation.

The two-parameter response surfaces (Fig. 12) confirm that interaction effects are strongest for pairs involving nutrient uptake, water status, temperature, CO_2 and photosynthetic capacity, and they add information the variance

RGR saturates in every gas-exchange parameter above its nominal value; only respiration changes it monotonically

Panels are the six largest mean S_T; every other parameter held nominal. Cells where the integration diverged are excluded, not plotted as zero growth — the nutrient-uptake panels lose their lower range entirely.

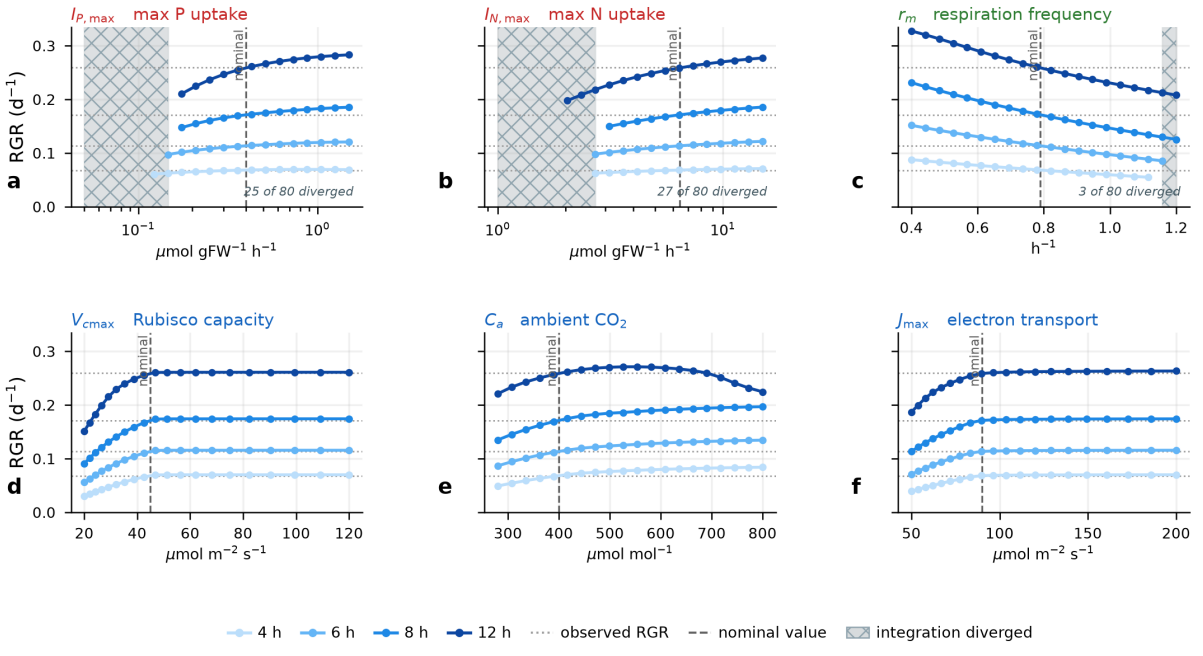


Figure 11: One-at-a-time parameter responses of relative growth rate at four photoperiods. Panels show the six parameters with the largest mean total-order Sobol' index S_T; every other parameter is held at its nominal value from `config/default.config` (dashed vertical line). Dotted horizontal lines mark the observed RGR of Sulpice et al. (2014) at the corresponding photoperiod. Grey hatched bands mark sub-ranges in which the integration diverged: in nutrient-limited parameter space the plant outgrows the soil supply, the nutrient content crosses zero and the affinity equation reaches its pole. These cells are excluded rather than plotted as zero growth — a distinction that matters, since all 55 zero-valued cells in the previously reported sweep were divergences of this kind and none was genuine starvation. Both nutrient-uptake capacities lose their lower range entirely. RGR saturates above the nominal value in the capacity parameters, but not in ambient CO_2 (panel e), which peaks at 0.2713 d^{-1} near 554 ppm and falls to 0.2246 d^{-1} by 800 ppm — a non-monotone response, not a plateau. Only respiration frequency r_m varies RGR monotonically across the whole admissible range.

decomposition cannot carry. The contours are not monotone ridges: the observed RGR is reached along an extended locus rather than at a point, so the calibration problem is ill-conditioned in exactly the directions the Sobol' interaction terms flag. Panel (c) also distinguishes two outcomes that a variance-based index cannot: the near-zero growth at $\psi_w = -1.5$ MPa is physiological stomatal closure, whereas the hatched regions in (a) and (b) are integration failures, excluded rather than scored. These surfaces are therefore consistent with the Sobol' and Morris results — the model response is dominated by the interaction of nutrient acquisition, gas exchange, environmental forcing and photoperiod-dependent carbon availability — while making the shape of that interaction legible.

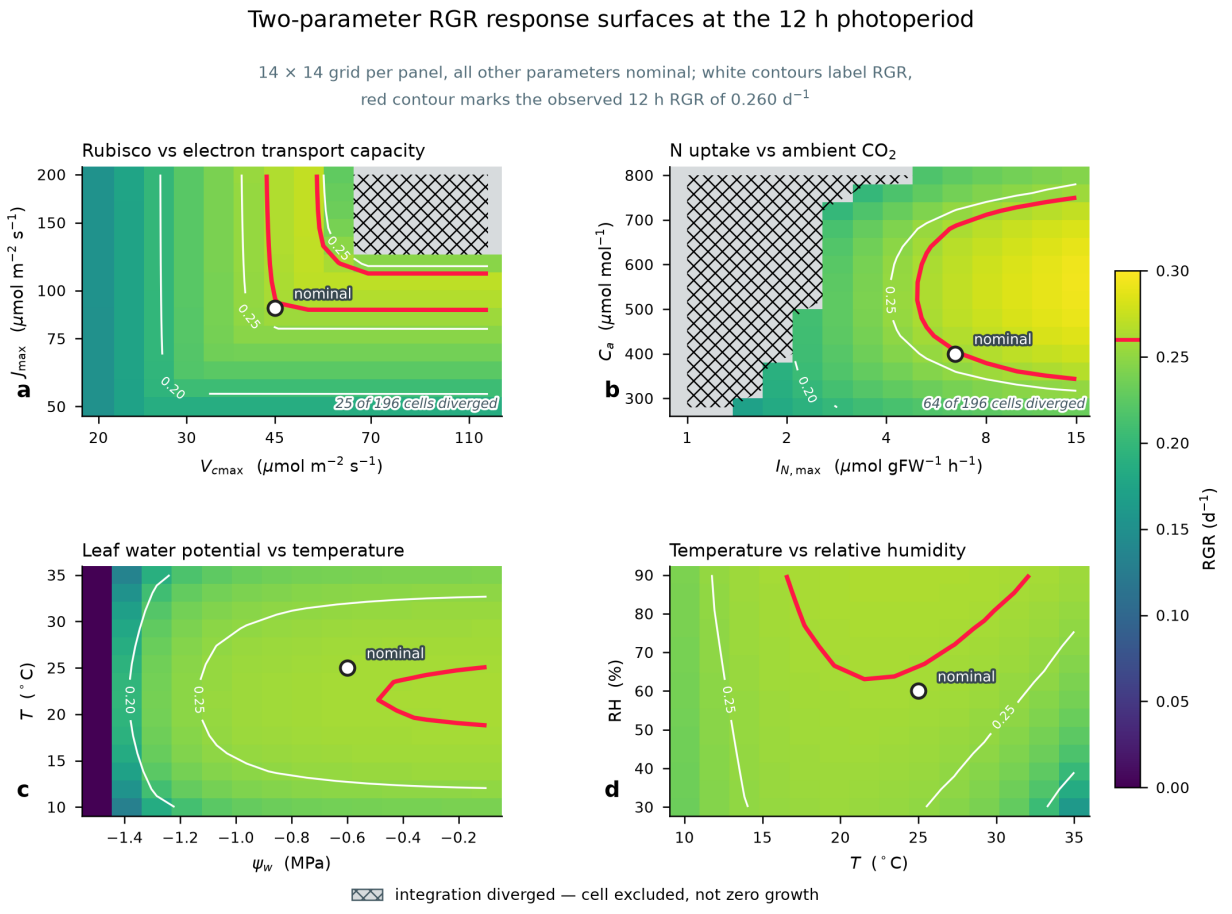


Figure 12: Two-parameter RGR response surfaces at the 12 h photoperiod, 14×14 grid per panel, all other parameters at their nominal values. White contours label RGR in d^{-1}; the red contour marks the observed 12 h value of 0.260 d^{-1} Sulpice et al. (2014) and the white-filled marker the nominal operating point. Grey hatched cells are excluded divergences, not zero growth. (a) Rubisco and electron-transport capacity co-limit assimilation, and the observed RGR is reached along an L-shaped locus that passes through the nominal point. (b) Nitrogen uptake and ambient CO_2 trade off; below $I_{N,\max} \approx 2$ μmol gFW^{-1} h^{-1} the run diverges over most of the CO_2 range. (c) The dark column at $\psi_w = -1.5$ MPa is genuine near-zero growth (RGR $\approx 10^{-4}$ d^{-1}), not a masked cell — stomatal closure, not integrator failure. (d) The temperature optimum shifts upward with relative humidity, from about 20°C at 30% to about 23°C at 90%, and the observed-RGR contour is open toward high humidity only: below about 60% relative humidity the observed value is not reached at any temperature.

7. Discussion

7.1. Principal findings and the role of the coupling

The substantive result of this work is not the closeness of the fitted RGR values but the behaviour of the coupled system, and the two must be separated clearly. The calibration adjusts two coefficients per photoperiod against two observations per photoperiod, so the reported mean absolute percentage error of 0.24% in RGR (Table 12) is the convergence residual of a bisection, not a predictive error; with eight parameters and eight observations the fit has no degrees of freedom, and tightening the tolerance drives the residual to zero. The out-of-sample test gives the figures that do carry information. Across seven training/test splits, a single photoperiod-independent parameter pair predicts held-out photoperiods with a mean absolute percentage error of 28.5%, rising to 87% in the worst split (Table 13). The

gap between 0% and 28.5% measures how much of the photoperiod response is carried by the calibrated coefficients rather than by the simulated dynamics, and it is large.

The second cross-validated hypothesis sharpens that statement into something more useful than a verdict. Allowing λ_{sb} to vary linearly with photoperiod — one additional parameter, still photoperiod-independent in the sense that no coefficient is refitted per treatment — reduces the mean held-out error from 28.5% to 7.3%. The missing photoperiod dependence is therefore not diffuse: it is close to an affine trend in the single coefficient that converts night-time phloem loading into biomass. That is a specific mechanistic gap. A plausible reading is that the model's night-time carbon economy is missing a photoperiod-dependent efficiency — longer nights impose a longer period of remobilisation at lower rate, which in the reference experiments is accompanied by adjustments in starch turnover that the fixed λ_{sb} cannot express. The splits also separate interpolation from extrapolation cleanly: predicting the interior photoperiods from the endpoints gives 4.5–8.5% error, whereas predicting the short photoperiods from the long ones gives 16.7–35.9%, so the model's photoperiod response degrades mainly outside the calibrated range rather than uniformly. We report it because the alternative — presenting a saturated fit as a validation — is a failure mode common enough in simulation studies to be worth naming, and because a reader cannot detect it from the reported errors alone.

What the model does reproduce without per-photoperiod adjustment is the internal carbohydrate behaviour. Transient starch accumulated in the light phase and was drawn down in the dark phase in all scenarios, with starch nRMSE between 0.128 and 0.192 (Table 14), with the observed near-linear nocturnal depletion arising from the structure of Eq. (29) rather than being fitted: the degradation rate is the degradable reserve latched at dusk divided by the night length, modulated by sucrose status, so depletion is paced to the night irrespective of how much starch was accumulated during the day. The correct statement is therefore that the shape is a consequence of the formulation and the photoperiod forcing, not of any parameter tuned to reproduce it. This is a weaker claim than a validated growth prediction, but it is a claim about dynamics rather than about a fitted level, and it is the sense in which the day-night carbon cycle is reproduced.

The physiological building blocks used here are Farquhar-von Caemmerer-Berry C_3 photosynthesis Farquhar et al. (1980), Jarvis-type stomatal conductance Jarvis (1976), and day-night starch turnover Scialdone et al. (2013); Stitt and Zeeman (2012). These are inherited from established work and are not presented as new physiology. The ablation study isolates what the coupling itself contributes. The simplified carbon-input formulation underestimated RGR systematically, from −23.4% at the 4 h photoperiod to −7.3% at 12 h (Table 15); replacing it with computed Farquhar assimilation removed almost all of that bias, reducing the mean absolute percentage error below 0.3%. Adding the Jarvis layer did not improve reported RGR further, because the target was already matched, but it changes the character of the model: the carbon input becomes a function of irradiance, vapour-pressure deficit, temperature, CO_2 and plant water status rather than a prescribed forcing. The local response diagnostics confirm this, with CO_2, temperature, humidity and PAR producing photoperiod-dependent RGR responses (Fig. 11, Fig. 12) that a prescribed-input model cannot generate. The value of the gas-exchange layer is therefore environmental responsiveness, which is exercised by the sensitivity analyses rather than by the calibrated RGR alone.

7.2. Dominant sensitivities and relation to existing models

The global sensitivity analysis shows that RGR variability is not controlled by a single fitted growth coefficient but by a combination of nutrient acquisition, respiration, photosynthetic capacity and water status. The maximum phosphorus and nitrogen uptake capacities $I_{P,\max}$ and $I_{N,\max}$ have the highest mean Sobol' total-order indices (0.48 and 0.38), followed by the respiration frequency r_m (0.35), the Rubisco and electron-transport capacities $V_{c\,\max}$ and $J_{\max}$ (0.24 and 0.21), ambient CO_2 (0.21) and leaf water potential ψ_w (0.16) (Table 16).

The dominant feature of the decomposition is not the ranking but the interaction structure. For fifteen of the sixteen parameters the photoperiod-averaged interaction share $\bar{S}_T - \bar{S}_1$ exceeds the first-order index (per photoperiod, in 44 of the 52 influential cells), and for r_m and $J_{\max}$ it exceeds it by roughly an order of magnitude ($\bar{S}_1 = 0.028$ and 0.019 against $\bar{S}_T = 0.348$ and 0.211). Summed over the sixteen dimensions, main effects account for 0.32 to 0.81 of the RGR variance while total-order indices sum to 2.36 to 2.59, so between 19% and 68% of the variance is carried by interactions, the share being largest at the shorter photoperiods. Little of the growth variance in this model is attributable to a parameter acting alone. That has a direct practical consequence: a one-at-a-time analysis — still the most common sensitivity method in the plant-simulation literature — would have reported r_m and $J_{\max}$ as nearly inert and would have been wrong about both. Variance-based methods are not a refinement here; they are the minimum required to see the model's behaviour at all.

The photoperiod dependence of the indices is interpretable in terms of which resource limits growth. Under short days the parameters governing losses from and demands on a scarce carbohydrate pool dominate: r_m falls from 0.46 at 4 h to 0.22 at 12 h, and $I_{N,\max}$ from 0.46 to 0.25. Under long days the supply-side parameters gain influence, with $V_{c\,\max}$ rising from 0.21 at 4 h to 0.31 at 12 h and $\mathrm{PAR}_{\max}$ from 0.13 to 0.19. Leaf water potential moves with the supply group, from 0.15 to 0.19, as expected if its effect is mediated by stomatal limitation of an assimilation flux whose importance itself grows with day length. This transition from demand-limited to supply-limited control is a property of the coupled system and is not available to a model with a prescribed carbon input, which is the substantive argument for computing assimilation rather than imposing it. The Morris screening places the nutrient-uptake capacities in the high-μ^*, high-σ region, agreeing with the Sobol' analysis on both magnitude and interaction, and providing an independent check — Morris uses a different sampling scheme and, as noted in Section 6.7, was unaffected by the sampling defect that required the Sobol' indices to be recomputed.

One result should be read with care. The calibrated growth coefficient λ_{sb} has a mean total-order index of 0.040, near the bottom of the ranking, while the cross-validation shows that the photoperiod response depends heavily on it. There is no contradiction: sensitivity indices measure contribution to output *variance* over a stated range, whereas the calibration exploits the parameter's control over the output *level*. A parameter can be a poor explanation of variance and still be the parameter a fit relies on. Reading low S_T as licence to fix a parameter, a common application of screening results, would be a mistake in exactly this case.

The model is positioned between two families reviewed in Section 2. Unlike crop simulators such as APSIM, DSSAT or WOFOST Holzworth et al. (2014); Jones et al. (2003); de Wit et al. (2019), which operate at field scale with prescribed or light-use-efficiency carbon input, the present model computes assimilation mechanistically and resolves the day-night carbohydrate cycle at the whole-plant scale. Unlike detailed functional-structural models de Reffye et al. (2020); Louarn and Song (2020), it deliberately omits explicit architecture in order to keep the coupled state vector small, transparent and reproducible. The intended contribution is not to compete with either family directly but to provide a documented, modular simulation core in which a mechanistic gas-exchange layer is transparently coupled to carbohydrate, nutrient and biomass dynamics, with all states, inputs, parameters and the numerical procedure stated explicitly.

7.3. What the verification exercise found, and what it generalises to

Four defects were found in the simulator during the preparation of this paper, none of them by comparison with data. They are reported here in full because their common structure is more transferable than any of the individual fixes: in each case the defect was invisible at the calibrated operating point, produced output that looked plausible, and would have been detected by a test that costs a small fraction of the effort already invested in validation.

The first was in the fast subsystem. The intercellular CO_2 concentration was obtained by fixed-point iteration between the Jarvis supply relation and the Farquhar demand relation, a construction used widely and usually without comment on its convergence. The map is contractive only while stomatal conductance is large enough; below roughly $\psi_w = -1.15$ MPa it acquires a stable period-two orbit, and the iteration then returns one of two alternating values selected by the parity of the iteration cap — -13.4 or $+23.7$ μmol m^{-2} s^{-1} at the tested point, against an exact value of 7.7 μmol m^{-2} s^{-1}. Increasing the iteration budget cannot help, and the sign of the carbon flux was determined by an implementation detail. The consequences were bounded, but only by accident: all calibrated runs use $\psi_w = -0.6$ MPa, where the failure rate is exactly zero, so the validation, diel and ablation results were unaffected. The sensitivity analysis was not, because it samples water potential down to -1.5 MPa; over the sampling domain the iteration failed in 15.2% of 2×10^4 draws and left the physically meaningful range with $C_i > 2C_a$ in 7.6%. The fix was to recognise that the coupled system is a quadratic with a closed-form solution (Eq. (22)) and to make it the default. Two lessons generalise. First, an iterative scheme embedded in a larger model should be tested for convergence across the domain the larger model samples, not the domain it is calibrated in; the two differ, and the calibrated point is systematically the benign one. Second — and this is the more useful half — the check that catches such a defect must be chosen with care. It is tempting to bound C_i by C_a, but that bound is false: $C_i > C_a$ is the correct supply solution whenever the leaf is a net source, so a range check on C_i would have rejected physical states and passed the pathological ones. What did discriminate was consistency between redundant computations of the same quantity: cap-parity sensitivity, and agreement between the iterative and closed-form solvers. A correctness check is only free if it is actually implied by the model, and it is worth verifying that it is before trusting its silence.

The second defect was in the stomatal submodel and had been present since the original implementation: the temperature modifier returned 0.707 rather than unity at its stated optimum, and peaked at 32°C rather than 25°C,

because both normalisation constants were wrong. The reason it survived calibration is instructive. The modifier enters multiplicatively and every calibration run was performed at a single fixed temperature, so its mis-normalisation was absorbed into the fitted parameters and produced no visible error. Because the model as written had no explicit maximum conductance $g_{s,\max}$ — its role having been absorbed into a parameter that simultaneously set the shape of the light response — there was no place where the misfit could show up as an implausible value. Introducing $g_{s,\max}$ explicitly both fixes the formulation and makes the correction exactly calibration-neutral at 25°C, so no reported number changes at the calibration temperature while the response away from it changes by up to a factor of 1.32. The general point is that a dimensionless modifier whose normalisation is never tested at its own reference condition is an untested assertion, and that conflating a scale parameter with a shape parameter removes the diagnostic that would have exposed it.

The third defect was in the analysis rather than the model, and is the one with the widest relevance. The Sobol' estimator requires two independent base samples; the implementation obtained them by splitting a single low-discrepancy sequence into halves. Successive blocks of such a sequence are constructed to be mutually complementary and are therefore strongly dependent — correlation up to 1.0000 at $d = 16$ — so the estimator differences two nearly identical matrices and the decomposition collapses. Against the Ishigami function, whose indices are known analytically, the design does not converge at any sample size, and returns exactly 0.000 for a first-order index whose true value is 0.314. The published indices were therefore wrong, and had to be recomputed. Two aspects of this deserve emphasis. First, the defect was visible in our own convergence study before any benchmark was run: the across-design spread of S_T *grew* with sample size, and the ratio of across-design spread to bootstrap standard error rose from 1.10 at $n = 512$ to 4.64 at $n = 2048$. An estimator whose variance increases with n is misspecified, and we initially read it as noise. Second, the bootstrap intervals reported in the original submission could not have caught it: resampling within a fixed design measures sampling noise *given* that design and is structurally blind to error contributed by the design itself. Both uncertainty measures are now reported. The general recommendation is concrete and cheap: a sensitivity estimator should be verified against a closed-form benchmark such as the Ishigami function before it is applied to a model whose true indices are unknown, and replicate designs should be run alongside bootstrap intervals, because only the former sees design-induced error.

A fourth defect surfaced only when the cross-validation of Section 6 pushed the parameters outside the calibrated bracket, and it is the one we would least have predicted. For some parameter combinations the adaptive step of the slow integrator collapses toward zero *without the state ever becoming non-finite*. The run then neither advances nor fails: it issues tens of millions of gas-exchange solves per second of wall-clock time — every one of them convergent, so the solver diagnostics report nothing wrong — while simulated time stands still. We measured 4.5×10^7 closure solves in twenty seconds with zero non-convergent cases, at t frozen. Bisecting the boundary places it at the 12 h photoperiod for $\lambda_{sdi} \in [0.5162, 0.5165]$ at $\lambda_{sb} = 6.7 \times 10^{-3}$; the other three photoperiods do not exhibit it anywhere in $\lambda_{sdi} \in [0.02, 1.0]$.

What makes this case instructive is that every guard already in place was structurally unable to see it. The non-finite-state guard cannot fire, because nothing is non-finite. A divergence test on the outputs cannot fire, because no outputs are produced. A wall-clock timeout in the calling tool would fire, but reports a hung process rather than an inadmissible parameter vector, which is useless to a batch runner that must classify thousands of them. The fix is a step budget: the integrator is given a hard cap on attempted steps (10^7, more than three orders of magnitude above what any admissible run requires at the production step size), and exhausting it stops the run through the solver's plugin interface and marks the result divergent. A stalled run then terminates in seconds and is classified exactly as a non-finite trajectory would be, and the batch caller records the parameter vector as inadmissible and continues. The general lesson is a corollary of the first defect: an adaptive integrator can fail by *not* producing a result, and that failure mode is invisible to every check formulated in terms of the values it produces. A budget on work done, not a test on output, is what detects it.

Taken together, these four cases suggest that the verification of a coupled simulator needs to be organised around the same fast-slow decomposition used to formulate it. The fast algebraic solve, the constitutive functions of the submodels, the conservation properties of the slow system and the estimators applied to the model's output are separate objects with separate failure modes, and each admits a test that does not require observational data: an exact solution where one exists, a normalisation condition at a reference point, a conservation residual, an analytic benchmark, a budget on work done. In this model those tests found four real defects and cost a small fraction of the effort spent on validation. That asymmetry, rather than any of the specific fixes, is the result we would expect to transfer to other coupled simulation studies.

7.4. Carbon accounting and the limits of what closes

The carbohydrate budget closes to machine precision — relative residuals between 1.5×10^{-15} and 1.3×10^{-14} across photoperiods (Table 11) — which addresses a natural concern about double counting in a model with several competing sinks on the same sucrose pool. That result should not be overstated. It holds over the three carbohydrate states $\{a, s, s_v\}$ and by construction: in this test the flux accumulators are integrated with the same fixed-step RK4 weights and the same step as the states, so the residual is bounded by integrator arithmetic rather than by an independent quadrature. What the test detects is a mis-signed or double-counted flux, and during development it detected two.

The budget does not extend to biomass, and the reason is worth stating plainly rather than leaving as a caveat. The ratio of carbon credited to the biomass pools to sucrose withdrawn from the cytosol is exactly $\lambda_{sb}B_l$, which reaches unity only at $B_l = 170$ gFW and spans 5.9×10^{-5} to 7.0×10^{-4} over the seedling range simulated here. The biomass equations are an empirical rate law and λ_{sb} is a rate coefficient with units gFW^{-1}, not a carbon-to-fresh-weight yield. The model is internally consistent as a growth model and is carbon-conserving within the carbohydrate subsystem; it is not carbon-conserving from assimilation through to biomass, and no claim to the contrary is made. Making it so would require an explicit carbon-to-fresh-weight conversion and would change every calibrated parameter, which is a reformulation rather than a correction and is left to future work. The present formulation is defensible provided the boundary of the conserved subsystem is declared, which is the practice we would recommend generally: report what closes, over which states, and quantify what is excluded rather than describing it qualitatively.

7.5. Limitations

Several limitations follow directly from these design choices and limit the conclusions.

1. **The RGR fit is saturated and the model interpolates rather than predicts over photoperiod.** Two coefficients per photoperiod are fitted against two observations per photoperiod, so the 0.24% agreement is a root-finder residual with no degrees of freedom, and no predictive claim follows from it. Over 14 cross-validated fits the held-out error is 28.5% with a single photoperiod-independent parameter pair, and 7.3% once λ_{sb} is allowed to vary linearly with photoperiod (Table 13). The latter figure localises the gap: what the coupled dynamics fail to generate is close to an affine trend in one growth coefficient. Closing it requires a mechanistic account of how photoperiod alters starch partitioning and the night-time growth response — a modelling problem, not a calibration problem — and is the most important item of future work.
2. **The reported errors are in-sample for every metric except the cross-validation.** Starch and sucrose nRMSE are computed on the same observations used in the fit, and no independent dataset, genotype or growth condition has been used. The parameter values are otherwise literature-derived and were not fitted, so the in-sample carbohydrate results are not vacuous, but they are not out-of-sample either.
3. **Carbon conservation is verified only within the carbohydrate subsystem.** It closes to machine precision over $\{a, s, s_v\}$ and does not extend through to biomass, for the dimensional reason set out in Section 7.4. The excluded term is quantified rather than assumed negligible, but the model is not a closed carbon budget from assimilation to dry matter.
4. **Sensitivity indices are conditional on the sampling ranges, which are heterogeneous in provenance.** Some ranges are published intervals and others are fractional perturbations around a nominal value (Appendix A); indices are not comparable across parameters whose ranges were set by different reasoning, and rankings would change under a different sampling domain. Sixteen of twenty candidate parameters are included, the cap being a limitation of the available sampler rather than a modelling decision.
5. **The four defects reported here were found by verification, which sets a floor rather than a ceiling on what remains.** The tests applied cover the algebraic solve, two constitutive functions, the carbohydrate conservation properties, the step-size and tolerance convergence of the integrator, and the sensitivity estimator. The nutrient and allocation submodels have not been subjected to comparable independent checks, and no claim is made that the simulator is now free of defects of the same class.
6. **Literature-based parameterisation.** Parameters were drawn from published sources rather than fitted to a single controlled experiment, and the available published validation data for whole-plant day-night dynamics are limited. Parameter uncertainty is therefore substantial, as the sensitivity analysis makes explicit.
7. **Sucrose under short days is the loosest carbohydrate fit, and only its shape is validated.** All four photoperiods satisfy the 0.50 sucrose criterion, but the residual is ordered in day length and is largest where the day is shortest (0.390 at 4 h and 0.382 at 6 h, against 0.284 at 12 h). The fitted scale is furthest from unity in

exactly those cases ($\alpha_{suc} = 3.37$ at 4 h), so the model holds a cytosolic pool well below the measured level and the metric validates the shape of the rhythm rather than its magnitude. Short days are also where the transient starch pool supplies most of the diel carbon budget, so an error in the degradation law is transmitted most directly into the sucrose trace there; the two carbohydrate residuals are not independent, and the present comparison cannot separate them.

8. **The carbohydrate errors are smaller under the metric the first submission printed than under the metric the code applies.** The table reported here is the runner output: cytosolic sucrose, normalised by the mean reference level. The first submission printed total sucrose normalised by the reference range, which for starch is about twice the mean and therefore yields a number roughly 50 to 85% smaller for the same residual (0.071–0.117 against 0.128–0.192). The starch cycle is captured within its 0.25 threshold either way, but with less margin than previously claimed, and the looser figure is the one the acceptance criterion is actually evaluated against.
9. **Simplified architecture and regulation.** Root architecture is lumped, and hormonal and developmental regulation are not represented; a full leaf energy balance is not solved, and leaf temperature is treated as an input rather than a computed quantity.
10. **Single validation species.** The model was evaluated only for *Arabidopsis thaliana* under photoperiod treatments; transfer to other species would require recalibration.
11. **Not a digital twin.** The model is an offline simulator. It does not include sensor assimilation, actuator feedback or online state correction, and no surrogate acceleration is claimed as part of the validated core.

8. Conclusion

This work reformulated a whole-plant, Tedone-based physiological framework as a coupled dynamical simulation model in which the photosynthetic carbon input is computed from a Farquhar-Jarvis gas-exchange layer rather than prescribed, and propagated consistently into day-night carbohydrate, nutrient and biomass dynamics. The model was documented as a modular simulator with explicitly declared states, inputs, parameters, outputs and numerical procedure, in line with the transparency expected of a reproducible simulation study. Under *Arabidopsis thaliana* photoperiod scenarios the model reproduces the day-night starch cycle within threshold and, in an ablation comparison, computing rather than prescribing the carbon input removes a systematic RGR bias of up to 23%. The RGR agreement itself is a saturated fit and is reported as such: held out across photoperiods over 14 cross-validated fits, the mean absolute percentage error is 28.5% with photoperiod-independent coefficients and 7.3% once one growth coefficient is allowed a linear photoperiod trend, which locates the model's weakness specifically in the mechanistic representation of the photoperiod response rather than in its calibration. Global sensitivity analysis shows that most of the growth variance is carried by parameter interactions — the interaction remainder exceeds the first-order index in 44 of the 52 influential parameter-photoperiod cells, and the mean total-order index exceeds the mean first-order index for 15 of the 16 parameters — and that control shifts from demand-side parameters under short days to supply-side parameters under long days, behaviour that a prescribed-input model cannot produce.

Equally central to the contribution is what the verification exercise found. Four defects were identified without reference to any observational data: a fixed-point gas-exchange solve that acquires a period-two orbit under water stress and returns a carbon flux whose sign depends on the iteration cap, a stomatal temperature modifier mis-normalised since the original implementation and hidden by calibration at a single temperature, a Sobol' sampling design whose two base samples were not independent, which invalidated the previously published indices, and an adaptive integrator step that collapses toward zero without the state ever becoming non-finite, so that the run neither advances nor fails. Each was found by a test that requires only an exact solution, a normalisation condition, a conservation residual, an analytic benchmark or a budget on work done, and each was invisible at the calibrated operating point. The transferable recommendation is to organise verification of a coupled simulator around its own fast-slow decomposition and to verify the analysis estimators, not only the model.

The remaining limitations define the work ahead: a mechanistic account of photoperiod-dependent partitioning to replace the per-photoperiod calibration, an out-of-sample evaluation on independent data, a carbon-to-fresh-weight conversion that would let the budget close through to biomass, a resolved leaf energy balance in place of prescribed leaf temperature, and in the longer term integration with sensor data and actuator feedback to extend the present offline simulator towards a digital-twin-enabled tool for controlled-environment plant growth.

A. Sensitivity-analysis parameter ranges

Table 17 reproduces the sampling specification used by both the Sobol' and the Morris analyses. It is not a summary written for the paper: the simulator reads this specification from a single machine-readable file (`config/sensitivity_params.csv`), one row per candidate parameter, and the table below is a transcription of that file. Each row carries the sampling bounds, the transformation under which the parameter is sampled (`linear` or `log`, the latter for parameters that range over more than one order of magnitude and whose uncertainty is naturally multiplicative), a role label distinguishing model parameters from environmental drivers, initial conditions and physical constants, and the provenance of the range. Sixteen rows are enabled and constitute the $d = 16$ dimensions analysed in Section 5.5.1; four are disabled, and each disabled row records why. The disabled rows are listed rather than omitted, because a parameter excluded from a sensitivity analysis without a stated reason is indistinguishable from one that was overlooked.

Two features of the specification deserve comment. First, the ranges are not uniform-percentage perturbations around the calibrated values: where a published range exists for the quantity the published range is used, and where the parameter is an artefact of this model's calibration — as λ_{sb} and the respiration frequency are — the range is an explicit fractional interval around the default and is labelled as such. Mixing the two kinds of range in one design is unavoidable, but it means the resulting indices must be read as sensitivity over the stated domain, not as an intrinsic property of the model. Second, the domain is wide enough to include operating points that the calibrated simulation never visits, in particular leaf water potentials down to -1.5 MPa; this is what made the solver audit of Section 4.3 necessary, and it is why the audit was run over the sampling domain rather than at the calibration point.

One exclusion reason requires updating relative to the originally submitted version, and is corrected here rather than quietly re-recorded. The provenance string for `photo.feedback_on_photosynthesis` justified its exclusion by the claim that a $PAR > 20$ guard prevented the reduced photosynthesis routine from ever being called. That guard has been removed (Section 4.3), so the original justification no longer holds. The parameter remains excluded, but for the correct reason: it parameterises the reduced carbon-input pathway used only in the `basic` ablation variant, and is therefore not a parameter of the coupled Farquhar-Jarvis model whose sensitivity is being analysed. Including it would place a dimension in the design that is inactive over the entire sampling domain, which inflates d and dilutes the total-order indices of the parameters that do act.

The exclusion of `carbohydrates.starch` is, at present, undocumented: the configuration file records a sampling range but no reason for disabling the row, and we do not reconstruct one after the fact. The dimension limit $d \leq 16$ imposed by the sampler forced some initial condition to be dropped and starch was the one dropped, but that is an inference from the design rather than a recorded decision.

Table 17
Sampling specification for the global sensitivity analysis, transcribed from `config/sensitivity_params.csv`. Bounds are given in the units of the corresponding symbol in Table 2. 'Scale' is the transformation under which the parameter is sampled. Enabled rows ($d = 16$) form the analysed design; disabled rows are retained with their exclusion reason. Provenance strings are reproduced as recorded in the configuration file.

Parameter	Role	Min	Max	Scale	Basis
Enabled: analysed dimensions ($d = 16$)					
`growth.lambda_sb`	model par.	0.002	0.010	linear	Calibrated; ±70% around default 0.00587 (Schulz et al. 2019)
`carbohydrates.respiration_frequency`	model par.	0.40	1.20	linear	Calibrated; ±50% around default 0.79
`nutrients.max_nitrogen_uptake`	model par.	1.0	15.0	log	Literature range for herbaceous crops (Barber 1995; Lambers et al. 2008)
`nutrients.max_phosphorus_uptake`	model par.	0.05	1.5	log	Literature range for herbaceous crops (Barber 1995)
`gas_exchange.max_rubisco_carboxylation_rate`	model par.	20.0	120.0	log	C_3 herbaceous range (Wullschleger 1993; Medlyn et al. 2002)
`gas_exchange.electron_transport_rate`	model par.	50.0	200.0	log	C_3 herbaceous range (Wullschleger 1993)
`carbohydrates.k_vacuole_in`	model par.	0.005	0.40	linear	Cytosol→vacuole exchange rate; covers the calibrated per-photoperiod range 0.011–0.230 h^{-1} (Lohaus 2007)
`nutrients.nitrogen_soil_content`	environment	3.0	30.0	linear	Low- to high-nutrient agricultural soil (Brady & Weil 2010)
`nutrients.phosphorus_soil_content`	environment	0.03	0.50	log	Low- to high-P agricultural soil (Brady & Weil 2010)
`gas_exchange.leaf_temperature`	environment	10.0	35.0	linear	Controlled-environment range; cold stress to heat stress
`gas_exchange.relative_humidity`	environment	30.0	90.0	linear	Arid to humid growing conditions
`gas_exchange.ambient_CO2_concentration`	environment	280.0	800.0	linear	Pre-industrial to RCP 8.5 projected 2100 (Meinshausen et al. 2011)
`gas_exchange.leaf_water_potential`	environment	−1.5	−0.1	linear	Severe water stress to turgid (Hsiao 1973; Jones 2014)
`photo.max_PAR`	environment	300.0	2000.0	linear	Shade canopy to full sun (McCree 1972)
`carbohydrates.sucrose`	initial cond.	0.80	3.00	linear	Observed range in spinach seedlings (% DW; Geiger & Servaites 1994)
`growth.total_biomass`	initial cond.	0.01	0.12	linear	Seedling (0.01 g FW) to established juvenile plant (0.12 g FW)
Disabled: excluded from the design, with reason					
`carbohydrates.lambda_sni`	model par.	0.04	0.18	linear	Excluded after screening: $\mu^* \approx 0$ throughout the range, and the Sobol' sampler in use is limited to 16 dimensions, so one screened-out parameter had to be dropped to keep $d \leq 16$
`photo.feedback_on_photosynthesis`	model par.	0.50	0.90	linear	Belongs to the reduced (non-Farquhar) carbon-input pathway, which is not exercised in the configuration analysed here; see the note below
`gas_exchange.oxygen_concentration`	phys. const.	210.0	210.0	fixed	Atmospheric O_2 held fixed; non-varying by construction
`carbohydrates.starch`	initial cond.	0.05	1.00	linear	Observed range in spinach seedlings (% DW; Geiger & Servaites 1994). The configuration file records the range but no exclusion reason; see the note below

CRediT authorship contribution statement

AUTHOR ONE: Conceptualization, Methodology, Software, Investigation, Writing – original draft. **AUTHOR TWO:** Conceptualization, Supervision, Writing – review & editing.

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